
ECE:4880 Group 21

Thermometer Quality Report

Run 2026-09-23T064945Z_a106867

Overall result: **PASS**

Profile: full

Group 21 | ECE:4880

Generated from frozen requirements, test definitions, and retained run evidence.

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1 Requirements traceability matrix

Requirement	Qualification traceability
PRJ-REQ-001	Status: PASS
SCRUM-399	Linked tests: DOC-01
PRJ-REQ-002	Status: PASS
SCRUM-400	Linked tests: SYS-01
PRJ-REQ-003	Status: PASS
SCRUM-401	Linked tests: SYS-01, DOC-01
PRJ-REQ-004	Status: PASS
SCRUM-402	Linked tests: DOC-01
SYS-HLR-100	Status: PASS
SCRUM-403	Linked tests: HW-01, SYS-01
SYS-HLR-110	Status: PASS
SCRUM-404	Linked tests: SYS-01
SYS-HLR-200	Status: PASS
SCRUM-405	Linked tests: HW-01, HW-02
SYS-HLR-210	Status: PASS
SCRUM-406	Linked tests: HW-03, SYS-02
SYS-HLR-220	Status: PASS
SCRUM-407	Linked tests: HW-03, SYS-03
SYS-HLR-300	Status: PASS
SCRUM-408	Linked tests: HW-01, HW-04, FW-01
SYS-HLR-310	Status: PASS
SCRUM-409	Linked tests: HW-04
SYS-HLR-320	Status: PASS
SCRUM-410	Linked tests: HW-04
SYS-HLR-400	Status: PASS
SCRUM-411	Linked tests: FW-04, FW-05
SYS-HLR-410	Status: PASS
SCRUM-412	Linked tests: HW-01, FW-04
SYS-HLR-420	Status: PASS
SCRUM-413	Linked tests: FW-02, FW-04, SYS-02
SYS-HLR-500	Status: PASS
SCRUM-414	Linked tests: BE-01, FE-02, SYS-01
SYS-HLR-510	Status: PASS
SCRUM-415	Linked tests: BLE-02, FE-07, SYS-04
SYS-HLR-520	Status: PASS
SCRUM-416	Linked tests: BE-03, SYS-03
SYS-HLR-600	Status: PASS
SCRUM-417	Linked tests: FW-03, BLE-03, BE-02, FE-03, SYS-03
SYS-HLR-700	Status: PASS
SCRUM-418	Linked tests: FE-06, ALR-01, ALR-03, SYS-05
SYS-HLR-321	Status: PASS
SCRUM-419	Linked tests: HW-04
SYS-HLR-322	Status: PASS
SCRUM-420	Linked tests: HW-04
HWE-MLR-200	Status: PASS
SCRUM-421	Linked tests: HW-01, HW-04
HWE-MLR-201	Status: PASS
SCRUM-422	Linked tests: HW-01, HW-02, HW-04
HWE-MLR-202	Status: PASS
SCRUM-423	Linked tests: HW-01, HW-02

Requirement	Qualification traceability
HWE-MLR-203	Status: PASS
SCRUM-424	Linked tests: HW-01, HW-02
HWE-MLR-204	Status: PASS
SCRUM-425	Linked tests: HW-01, HW-03
HWE-MLR-205	Status: PASS
SCRUM-426	Linked tests: HW-01, HW-03
HWE-MLR-206	Status: PASS
SCRUM-427	Linked tests: HW-01
HWE-MLR-209	Status: PASS
SCRUM-430	Linked tests: HW-01
HWE-MLR-210	Status: PASS
SCRUM-431	Linked tests: HW-01, HW-04
HWE-MLR-211	Status: PASS
SCRUM-432	Linked tests: HW-01, HW-03, HW-04, FW-04
HWE-MLR-212	Status: PASS
SCRUM-433	Linked tests: HW-01, HW-04
HWE-LLR-200	Status: PASS
SCRUM-434	Linked tests: HW-01
HWE-LLR-201	Status: PASS
SCRUM-435	Linked tests: HW-01, HW-02, HW-04
HWE-LLR-202	Status: PASS
SCRUM-436	Linked tests: HW-01, HW-02, FW-04
HWE-LLR-203	Status: PASS
SCRUM-437	Linked tests: HW-01, HW-02
HWE-LLR-204	Status: PASS
SCRUM-438	Linked tests: HW-01, HW-02
HWE-LLR-205	Status: PASS
SCRUM-439	Linked tests: HW-01, HW-02
HWE-LLR-206	Status: PASS
SCRUM-440	Linked tests: HW-01, HW-03
HWE-LLR-207	Status: PASS
SCRUM-441	Linked tests: HW-01, FW-04
HWE-LLR-208	Status: PASS
SCRUM-442	Linked tests: HW-01, HW-03, HW-04, FW-04
HWE-LLR-209	Status: PASS
SCRUM-443	Linked tests: HW-01, HW-04
HWE-LLR-210	Status: PASS
SCRUM-444	Linked tests: HW-01, FW-04
HWE-LLR-214	Status: PASS
SCRUM-448	Linked tests: HW-01
HWE-LLR-215	Status: PASS
SCRUM-449	Linked tests: HW-01
HWE-LLR-216	Status: PASS
SCRUM-450	Linked tests: HW-01
HWE-LLR-217	Status: PASS
SCRUM-451	Linked tests: HW-01
HWE-LLR-218	Status: PASS
SCRUM-452	Linked tests: HW-01, HW-04
HWE-LLR-219	Status: PASS
SCRUM-453	Linked tests: HW-01, HW-04, FW-04
HWE-LLR-220	Status: PASS
SCRUM-454	Linked tests: HW-01, HW-04
HWE-LLR-221	Status: PASS
SCRUM-455	Linked tests: HW-01

Requirement	Qualification traceability
SWE-EMB-MLR-300	Status: PASS
SCRUM-456	Linked tests: FW-01
SWE-EMB-MLR-301	Status: PASS
SCRUM-457	Linked tests: FW-02
SWE-EMB-MLR-302	Status: PASS
SCRUM-458	Linked tests: FW-02
SWE-EMB-MLR-303	Status: PASS
SCRUM-459	Linked tests: FW-02
SWE-EMB-MLR-304	Status: PASS
SCRUM-460	Linked tests: FW-02
SWE-EMB-MLR-305	Status: PASS
SCRUM-461	Linked tests: FW-02, FW-04
SWE-EMB-MLR-306	Status: PASS
SCRUM-462	Linked tests: FW-02, FW-03, FW-04
SWE-EMB-MLR-307	Status: PASS
SCRUM-463	Linked tests: FW-02, FW-04
SWE-EMB-MLR-308	Status: PASS
SCRUM-464	Linked tests: FW-02, FW-04
SWE-EMB-MLR-310	Status: PASS
SCRUM-466	Linked tests: FW-04
SWE-EMB-MLR-311	Status: PASS
SCRUM-467	Linked tests: FW-02, FW-03
SWE-EMB-MLR-312	Status: PASS
SCRUM-468	Linked tests: FW-02
SWE-EMB-MLR-313	Status: PASS
SCRUM-469	Linked tests: FW-02, FW-04, FW-05
SWE-EMB-LLR-300	Status: PASS
SCRUM-470	Linked tests: FW-01, FW-02, FW-03, FW-04
SWE-EMB-LLR-301	Status: PASS
SCRUM-471	Linked tests: FW-02
SWE-EMB-LLR-302	Status: PASS
SCRUM-472	Linked tests: FW-02
SWE-EMB-LLR-303	Status: PASS
SCRUM-473	Linked tests: FW-02
SWE-EMB-LLR-304	Status: PASS
SCRUM-474	Linked tests: FW-02, FW-03
SWE-EMB-LLR-305	Status: PASS
SCRUM-475	Linked tests: FW-02, FW-03
SWE-EMB-LLR-306	Status: PASS
SCRUM-476	Linked tests: FW-02, FW-03
SWE-EMB-LLR-307	Status: PASS
SCRUM-477	Linked tests: FW-03
SWE-EMB-LLR-308	Status: PASS
SCRUM-478	Linked tests: FW-02
SWE-EMB-LLR-309	Status: PASS
SCRUM-479	Linked tests: FW-02, FW-04
SWE-EMB-LLR-310	Status: PASS
SCRUM-480	Linked tests: FW-02
SWE-EMB-LLR-311	Status: PASS
SCRUM-481	Linked tests: FW-02, FW-04, SYS-02
SWE-EMB-LLR-312	Status: PASS
SCRUM-482	Linked tests: FW-02, FW-04
SWE-EMB-LLR-313	Status: PASS
SCRUM-483	Linked tests: FW-02, FW-04

Requirement	Qualification traceability
SWE-EMB-LLR-314	Status: PASS
SCRUM-484	Linked tests: FW-02, FW-03, FW-04
SWE-EMB-LLR-315	Status: PASS
SCRUM-485	Linked tests: FW-02, FW-03, FW-04
SWE-EMB-LLR-316	Status: PASS
SCRUM-486	Linked tests: FW-02, FW-04
SWE-EMB-LLR-317	Status: PASS
SCRUM-487	Linked tests: FW-04
SWE-EMB-LLR-318	Status: PASS
SCRUM-488	Linked tests: FW-02, FW-04
SWE-EMB-LLR-319	Status: PASS
SCRUM-489	Linked tests: FW-02, FW-04
SWE-EMB-LLR-323	Status: PASS
SCRUM-493	Linked tests: FW-01
SWE-EMB-LLR-324	Status: PASS
SCRUM-494	Linked tests: FW-02, FW-03
SWE-EMB-LLR-325	Status: PASS
SCRUM-495	Linked tests: FW-03
SWE-EMB-LLR-326	Status: PASS
SCRUM-496	Linked tests: FW-02
SWE-EMB-LLR-327	Status: PASS
SCRUM-497	Linked tests: FW-04
SWE-EMB-LLR-328	Status: PASS
SCRUM-498	Linked tests: FW-02
INT-MLR-400	Status: PASS
SCRUM-499	Linked tests: BLE-01
INT-MLR-401	Status: PASS
SCRUM-500	Linked tests: BLE-01
INT-MLR-402	Status: PASS
SCRUM-501	Linked tests: BLE-01
INT-MLR-403	Status: PASS
SCRUM-502	Linked tests: BLE-01
INT-MLR-404	Status: PASS
SCRUM-503	Linked tests: BLE-03
INT-MLR-405	Status: PASS
SCRUM-504	Linked tests: BLE-04
INT-MLR-406	Status: PASS
SCRUM-505	Linked tests: BLE-02
INT-MLR-407	Status: PASS
SCRUM-506	Linked tests: BLE-01
INT-LLR-400	Status: PASS
SCRUM-507	Linked tests: BLE-01
INT-LLR-401	Status: PASS
SCRUM-508	Linked tests: BLE-03
INT-LLR-402	Status: PASS
SCRUM-509	Linked tests: BLE-01
INT-LLR-403	Status: PASS
SCRUM-510	Linked tests: BLE-01
INT-LLR-404	Status: PASS
SCRUM-511	Linked tests: BLE-01
INT-LLR-405	Status: PASS
SCRUM-512	Linked tests: BLE-02
INT-LLR-406	Status: PASS
SCRUM-513	Linked tests: BLE-02

Requirement	Qualification traceability
INT-LLR-407	Status: PASS
SCRUM-514	Linked tests: BLE-03
INT-LLR-408	Status: PASS
SCRUM-515	Linked tests: BLE-03
INT-LLR-409	Status: PASS
SCRUM-516	Linked tests: BLE-03
INT-LLR-410	Status: PASS
SCRUM-517	Linked tests: BLE-03
INT-LLR-411	Status: PASS
SCRUM-518	Linked tests: BLE-03
INT-LLR-412	Status: PASS
SCRUM-519	Linked tests: BLE-01
INT-LLR-413	Status: PASS
SCRUM-520	Linked tests: BLE-02
INT-LLR-414	Status: PASS
SCRUM-521	Linked tests: BLE-02
INT-LLR-415	Status: PASS
SCRUM-522	Linked tests: BLE-04
SWE-CONN-MLR-500	Status: PASS
SCRUM-523	Linked tests: BE-01
SWE-CONN-MLR-501	Status: PASS
SCRUM-524	Linked tests: BE-01
SWE-CONN-MLR-502	Status: PASS
SCRUM-525	Linked tests: BE-01
SWE-CONN-MLR-503	Status: PASS
SCRUM-526	Linked tests: BE-01, BE-02, BE-03, BE-04
SWE-CONN-MLR-504	Status: PASS
SCRUM-527	Linked tests: BE-01
SWE-CONN-MLR-505	Status: PASS
SCRUM-528	Linked tests: BE-01
SWE-CONN-MLR-506	Status: PASS
SCRUM-529	Linked tests: BE-01, BE-03
SWE-DB-MLR-550	Status: PASS
SCRUM-530	Linked tests: DB-02
SWE-DB-MLR-551	Status: PASS
SCRUM-531	Linked tests: DB-02
SWE-DB-MLR-552	Status: PASS
SCRUM-532	Linked tests: DB-03
SWE-DB-MLR-553	Status: PASS
SCRUM-533	Linked tests: DB-03
SWE-DB-MLR-554	Status: PASS
SCRUM-534	Linked tests: DB-02
SWE-CONN-LLR-500	Status: PASS
SCRUM-535	Linked tests: BE-01
SWE-CONN-LLR-501	Status: PASS
SCRUM-536	Linked tests: BE-01, BE-02
SWE-CONN-LLR-502	Status: PASS
SCRUM-537	Linked tests: BE-01, BE-04
SWE-CONN-LLR-503	Status: PASS
SCRUM-538	Linked tests: BE-01, BE-03
SWE-CONN-LLR-504	Status: PASS
SCRUM-539	Linked tests: BE-01, BE-03
SWE-CONN-LLR-505	Status: PASS
SCRUM-540	Linked tests: BE-01, BE-02, BE-03

Requirement	Qualification traceability
SWE-CONN-LLR-506	Status: PASS
SCRUM-541	Linked tests: BE-01, BE-02
SWE-CONN-LLR-507	Status: PASS
SCRUM-542	Linked tests: BE-01, BE-02
SWE-CONN-LLR-508	Status: PASS
SCRUM-543	Linked tests: BE-01, BE-02
SWE-CONN-LLR-509	Status: PASS
SCRUM-544	Linked tests: BE-01, BE-02, BE-03
SWE-CONN-LLR-510	Status: PASS
SCRUM-545	Linked tests: BE-01, BE-04
SWE-CONN-LLR-511	Status: PASS
SCRUM-546	Linked tests: BE-01, BE-04
SWE-CONN-LLR-512	Status: PASS
SCRUM-547	Linked tests: BE-01, BE-04
SWE-CONN-LLR-513	Status: PASS
SCRUM-548	Linked tests: BE-01
SWE-CONN-LLR-514	Status: PASS
SCRUM-549	Linked tests: BE-01
SWE-DB-LLR-550	Status: PASS
SCRUM-550	Linked tests: DB-01
SWE-DB-LLR-551	Status: PASS
SCRUM-551	Linked tests: DB-02
SWE-DB-LLR-552	Status: PASS
SCRUM-552	Linked tests: DB-02
SWE-DB-LLR-553	Status: PASS
SCRUM-553	Linked tests: DB-02
SWE-DB-LLR-554	Status: PASS
SCRUM-554	Linked tests: DB-01
SWE-DB-LLR-555	Status: PASS
SCRUM-555	Linked tests: DB-01
SWE-DB-LLR-556	Status: PASS
SCRUM-556	Linked tests: DB-01
SWE-DB-LLR-557	Status: PASS
SCRUM-557	Linked tests: DB-03
SWE-DB-LLR-558	Status: PASS
SCRUM-558	Linked tests: DB-03
SWE-DB-LLR-559	Status: PASS
SCRUM-559	Linked tests: DB-01
SWE-DB-LLR-560	Status: PASS
SCRUM-560	Linked tests: DB-03
SWE-DB-LLR-561	Status: PASS
SCRUM-561	Linked tests: DB-01
SWE-WEB-MLR-600	Status: PASS
SCRUM-562	Linked tests: FE-07
SWE-WEB-MLR-601	Status: PASS
SCRUM-563	Linked tests: FE-01
SWE-WEB-MLR-602	Status: PASS
SCRUM-564	Linked tests: FE-05, FE-07
SWE-WEB-MLR-603	Status: PASS
SCRUM-565	Linked tests: FE-06
SWE-WEB-MLR-604	Status: PASS
SCRUM-566	Linked tests: FE-03
SWE-WEB-MLR-605	Status: PASS
SCRUM-567	Linked tests: FE-03, FE-05

Requirement	Qualification traceability
SWE-WEB-MLR-606	Status: PASS
SCRUM-568	Linked tests: FE-03
SWE-WEB-MLR-607	Status: PASS
SCRUM-569	Linked tests: FE-05, FE-07
SWE-WEB-MLR-608	Status: PASS
SCRUM-570	Linked tests: FE-03
SWE-WEB-MLR-610	Status: PASS
SCRUM-572	Linked tests: FE-03
SWE-WEB-LLR-600	Status: PASS
SCRUM-574	Linked tests: FE-01, FE-03
SWE-WEB-LLR-601	Status: PASS
SCRUM-575	Linked tests: FE-01, FE-03
SWE-WEB-LLR-602	Status: PASS
SCRUM-576	Linked tests: FE-05, FE-07
SWE-WEB-LLR-603	Status: PASS
SCRUM-577	Linked tests: FE-05, FE-07
SWE-WEB-LLR-604	Status: PASS
SCRUM-578	Linked tests: FE-02, FE-03
SWE-WEB-LLR-605	Status: PASS
SCRUM-579	Linked tests: FE-01
SWE-WEB-LLR-606	Status: PASS
SCRUM-580	Linked tests: FE-02, FE-03, FE-04, FE-06, FE-07
SWE-WEB-LLR-607	Status: PASS
SCRUM-581	Linked tests: FE-05
SWE-WEB-LLR-608	Status: PASS
SCRUM-582	Linked tests: FE-03, FE-05, FE-06, FE-07
SWE-WEB-LLR-609	Status: PASS
SCRUM-583	Linked tests: FE-05, FE-06, FE-07
SWE-WEB-LLR-610	Status: PASS
SCRUM-584	Linked tests: FE-05, FE-06
SWE-WEB-LLR-611	Status: PASS
SCRUM-585	Linked tests: FE-03, FE-05, FE-06
SWE-WEB-LLR-612	Status: PASS
SCRUM-586	Linked tests: FE-05, FE-06
SWE-WEB-LLR-613	Status: PASS
SCRUM-587	Linked tests: FE-03, FE-07
SWE-WEB-LLR-614	Status: PASS
SCRUM-588	Linked tests: FE-02, FE-03
SWE-WEB-LLR-615	Status: PASS
SCRUM-589	Linked tests: FE-03
SWE-WEB-LLR-616	Status: PASS
SCRUM-590	Linked tests: FE-03
SWE-WEB-LLR-617	Status: PASS
SCRUM-591	Linked tests: FE-02, FE-03, FE-06
SWE-WEB-LLR-618	Status: PASS
SCRUM-592	Linked tests: FE-02
SWE-WEB-LLR-619	Status: PASS
SCRUM-593	Linked tests: FE-02
SWE-WEB-LLR-620	Status: PASS
SCRUM-594	Linked tests: FE-03
SWE-WEB-LLR-621	Status: PASS
SCRUM-595	Linked tests: FE-03
SWE-WEB-LLR-622	Status: PASS
SCRUM-596	Linked tests: FE-05, FE-07

Requirement	Qualification traceability
SWE-WEB-LLR-623	Status: PASS
SCRUM-597	Linked tests: FE-07
SWE-WEB-LLR-624	Status: PASS
SCRUM-598	Linked tests: FE-03
SWE-WEB-LLR-625	Status: PASS
SCRUM-599	Linked tests: FE-03
SWE-WEB-LLR-626	Status: PASS
SCRUM-600	Linked tests: FE-03
SWE-ALR-MLR-700	Status: PASS
SCRUM-603	Linked tests: ALR-01, ALR-03
SWE-ALR-MLR-701	Status: PASS
SCRUM-604	Linked tests: ALR-01
SWE-ALR-MLR-702	Status: PASS
SCRUM-605	Linked tests: ALR-01
SWE-ALR-MLR-703	Status: PASS
SCRUM-606	Linked tests: ALR-03
SWE-ALR-MLR-704	Status: PASS
SCRUM-607	Linked tests: ALR-01
SWE-ALR-MLR-705	Status: PASS
SCRUM-608	Linked tests: ALR-02
SWE-SEC-MLR-750	Status: PASS
SCRUM-609	Linked tests: FE-05
SWE-ALR-LLR-700	Status: PASS
SCRUM-610	Linked tests: ALR-01, ALR-03
SWE-ALR-LLR-701	Status: PASS
SCRUM-611	Linked tests: ALR-01
SWE-ALR-LLR-702	Status: PASS
SCRUM-612	Linked tests: ALR-01
SWE-ALR-LLR-703	Status: PASS
SCRUM-613	Linked tests: ALR-01
SWE-ALR-LLR-704	Status: PASS
SCRUM-614	Linked tests: ALR-01
SWE-ALR-LLR-705	Status: PASS
SCRUM-615	Linked tests: ALR-01
SWE-ALR-LLR-706	Status: PASS
SCRUM-616	Linked tests: ALR-01, ALR-03
SWE-ALR-LLR-707	Status: PASS
SCRUM-617	Linked tests: ALR-01, ALR-03
SWE-ALR-LLR-708	Status: PASS
SCRUM-618	Linked tests: ALR-03
SWE-ALR-LLR-709	Status: PASS
SCRUM-619	Linked tests: ALR-01
SWE-ALR-LLR-710	Status: PASS
SCRUM-620	Linked tests: ALR-01
SWE-ALR-LLR-711	Status: PASS
SCRUM-621	Linked tests: ALR-01, ALR-02
SWE-ALR-LLR-712	Status: PASS
SCRUM-622	Linked tests: ALR-01, ALR-02
SWE-ALR-LLR-713	Status: PASS
SCRUM-623	Linked tests: ALR-01, ALR-02
SWE-SEC-LLR-750	Status: PASS
SCRUM-625	Linked tests: FE-05
SWE-SEC-LLR-751	Status: PASS
SCRUM-626	Linked tests: FE-05

Requirement	Qualification traceability
SWE-SEC-LLR-752	Status: PASS
SCRUM-627	Linked tests: FE-05
SWE-SEC-LLR-753	Status: PASS
SCRUM-628	Linked tests: FE-05
SWE-SEC-LLR-754	Status: PASS
SCRUM-629	Linked tests: FE-05
HWE-MLR-213	Status: PASS
SCRUM-634	Linked tests: HW-01, FW-04
HWE-MLR-214	Status: PASS
SCRUM-635	Linked tests: HW-01, FW-04
HWE-MLR-215	Status: PASS
SCRUM-636	Linked tests: HW-01, HW-02, HW-03
HWE-MLR-216	Status: PASS
SCRUM-637	Linked tests: HW-01, HW-04
HWE-LLR-222	Status: PASS
SCRUM-638	Linked tests: HW-01, FW-04
HWE-LLR-223	Status: PASS
SCRUM-639	Linked tests: HW-01, HW-02
HWE-LLR-224	Status: PASS
SCRUM-640	Linked tests: HW-01, HW-04
HWE-LLR-225	Status: PASS
SCRUM-641	Linked tests: HW-01, HW-04
SWE-CONN-MLR-507	Status: PASS
SCRUM-642	Linked tests: BE-01, BE-02
SWE-CONN-MLR-508	Status: PASS
SCRUM-643	Linked tests: BE-03
SWE-CONN-LLR-515	Status: PASS
SCRUM-644	Linked tests: BE-01, BE-02, SYS-03
SWE-CONN-LLR-516	Status: PASS
SCRUM-645	Linked tests: BE-01, BE-03, SYS-03
INT-LLR-416	Status: PASS
SCRUM-646	Linked tests: BLE-02, SYS-04
SWE-WEB-MLR-613	Status: PASS
SCRUM-647	Linked tests: FE-02
SWE-WEB-MLR-614	Status: PASS
SCRUM-648	Linked tests: FE-03
SWE-WEB-LLR-630	Status: PASS
SCRUM-649	Linked tests: FE-02, FE-03, FE-06, FE-07
SWE-WEB-LLR-631	Status: PASS
SCRUM-650	Linked tests: FE-03
SWE-WEB-LLR-632	Status: PASS
SCRUM-651	Linked tests: FE-03
SWE-WEB-LLR-633	Status: PASS
SCRUM-652	Linked tests: FE-03
PRJ-REQ-005	Status: PASS
SCRUM-653	Linked tests: HW-01

2 Requirements

Only applicable requirements are included. Each page records the current Jira description, its derivation, relationships, passing basis, and linked tests.

PRJ-REQ-001 — Design Documentation

Result PASS
Level / component Project / System
Jira [SCRUM-399 — open requirement](#)

Current Jira description

The project shall produce design documentation that describes the construction and operation of the temperature measurement system.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: None — top-level project or stakeholder requirement.

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **DOC-01 — Requirement Coverage Integrity:** PASS. Fails on unmapped/unknown requirements, missing tests, invalid physical claims, or mishandled TBD/conflict requirements.

PRJ-REQ-002 — Functional Prototype

Result PASS
Level / component Project / System
Jira [SCRUM-400 — open requirement](#)

Current Jira description

The project shall produce a prototype demonstrating that the proposed design functions.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: None — top-level project or stakeholder requirement.

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **SYS-01 — Complete Data Path:** PASS. Verifies both physical sensor values/status through ESP32, BLE, backend, MySQL, server, and browser.

PRJ-REQ-003 — Requirements Compliance

Result PASS
Level / component Project / System
Jira [SCRUM-401 — open requirement](#)

Current Jira description

The final system shall satisfy the applicable stakeholder requirements or document an approved stakeholder change for any deviation.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: None — top-level project or stakeholder requirement.

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **SYS-01 — Complete Data Path:** PASS. Verifies both physical sensor values/status through ESP32, BLE, backend, MySQL, server, and browser.
- **DOC-01 — Requirement Coverage Integrity:** PASS. Fails on unmapped/unknown requirements, missing tests, invalid physical claims, or mishandled TBD/conflict requirements.

PRJ-REQ-004 — Requirements Clarification

Result PASS
Level / component Project / System
Jira [SCRUM-402 — open requirement](#)

Current Jira description

Ambiguous or conflicting stakeholder requirements shall be clarified before the affected design is finalized.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: None — top-level project or stakeholder requirement.

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **DOC-01 — Requirement Coverage Integrity:** PASS. Fails on unmapped/unknown requirements, missing tests, invalid physical claims, or mishandled TBD/conflict requirements.

SYS-HLR-100 — System Composition

Result PASS
Level / component HLR / System
Jira [SCRUM-403 — open requirement](#)

Current Jira description

The system shall consist of a computer, two temperature-sensor assemblies, a battery-operated third-box assembly, and support for delivery of email alerts to one or more receiving addresses. SMS/text-message delivery is out of scope.

Revision note

Edited requirement. Notification scope was narrowed to email only; SMS delivery is not part of this release. The description above is the newest frozen version used by this run.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: None — top-level project or stakeholder requirement.

Children: [HWE-MLR-205](#), [HWE-MLR-206](#), [SWE-EMB-MLR-300](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **SYS-01 — Complete Data Path:** PASS. Verifies both physical sensor values/status through ESP32, BLE, backend, MySQL, server, and browser.

SYS-HLR-110 — Internet-Accessible Temperature Data

Result PASS
Level / component HLR / System
Jira [SCRUM-404 — open requirement](#)

Current Jira description

Temperature information produced by the third-box thermometer shall be accessible through an internet-connected computer while the required system components are operational.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: None — top-level project or stakeholder requirement.

Children: INT-MLR-400, SWE-CONN-MLR-500, SWE-DB-MLR-550, SWE-WEB-MLR-600

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **SYS-01 — Complete Data Path:** PASS. Verifies both physical sensor values/status through ESP32, BLE, backend, MySQL, server, and browser.

SYS-HLR-200 — Physical and Mechanical Robustness

Result PASS
Level / component HLR / System
Jira [SCRUM-405 — open requirement](#)

Current Jira description

The sensor assemblies, third box, cables, controls, and connectors shall satisfy the stakeholder-specified physical construction, handling, drop, and connection requirements.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: None — top-level project or stakeholder requirement.

Children: [HWE-MLR-200](#), [HWE-MLR-201](#), [HWE-MLR-202](#), [HWE-MLR-203](#), [HWE-MLR-209](#), [HWE-MLR-210](#), [HWE-MLR-215](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-02 — Mechanical Robustness:** PASS. Automatically captures pre/post full-system smoke evidence around guided drop, upside-down, connected-cable, handling, and immersion actions.

SYS-HLR-210 — Sensor Hot-Plug Recovery

Result PASS
Level / component HLR / System
Jira [SCRUM-406](#) — open requirement

Current Jira description

The system shall tolerate temperature-sensor disconnection and reconnection without requiring user intervention to restore normal operation.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: None — top-level project or stakeholder requirement.

Children: [HWE-MLR-204](#), [SWE-EMB-MLR-304](#), [SWE-EMB-MLR-312](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-03 — Power Switch and Hot-Plug:** PASS. Guides sensor hot-plug and power cycling while capturing independent operation, no-reset behavior, loss of current data, and recovery.
- **SYS-02 — Full-System Sensor Disconnect and Recovery:** PASS. Disconnects each real sensor in both ON and OFF display states and verifies other-sensor continuity, local/remote fault, graph gap, automatic recovery, and preservation of the pre-disconnection state.

SYS-HLR-220 — Third-Box Power Control

Result PASS
Level / component HLR / System
Jira [SCRUM-407 — open requirement](#)

Current Jira description

The third-box power switch shall control availability of local thermometer functionality and current remotely accessible temperature data.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: None — top-level project or stakeholder requirement.

Children: [HWE-MLR-205](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-03 — Power Switch and Hot-Plug:** PASS. Guides sensor hot-plug and power cycling while capturing independent operation, no-reset behavior, loss of current data, and recovery.
- **SYS-03 — Startup, Power-On, and History Recovery:** PASS. Separately measures box-OFF power recovery and 300-second history recovery after a host-radio outage while the box remains powered.

SYS-HLR-300 — Temperature Design Range

Result PASS
Level / component HLR / System
Jira [SCRUM-408 — open requirement](#)

Current Jira description

The system design shall support temperatures from at least -10 deg C through +63 deg C.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: None — top-level project or stakeholder requirement.

Children: [HWE-MLR-211](#), [HWE-MLR-212](#), [SWE-EMB-MLR-301](#), [SWE-EMB-MLR-310](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-04 — Temperature Accuracy and Thermal Response:** PASS. Captures room, ice-water, hand-heating, and soldering-iron response measurements for both sensors.
- **FW-01 — Firmware Build, Startup, and Configuration:** PASS. Builds or statically validates real and simulated firmware configuration, protocol generation, app_main initialization, Celsius, and boot identity.

SYS-HLR-310 — Thermal Response

Result PASS
Level / component HLR / System
Jira [SCRUM-409](#) — open requirement

Current Jira description

The temperature-measurement system shall visibly respond to sensor heating under the stakeholder-specified hand and soldering-iron tests.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: None — top-level project or stakeholder requirement.

Children: [SWE-EMB-MLR-301](#), [HWE-MLR-216](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-04 — Temperature Accuracy and Thermal Response:** PASS. Captures room, ice-water, hand-heating, and soldering-iron response measurements for both sensors.

SYS-HLR-320 — Temperature Accuracy

Result PASS
Level / component HLR / System
Jira [SCRUM-410 — open requirement](#)

Current Jira description

The thermometer shall satisfy the specified room-temperature and ice-water measurement tolerances.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: None — top-level project or stakeholder requirement.

Children: [SYS-HLR-321](#), [SYS-HLR-322](#), [HWE-MLR-212](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-04 — Temperature Accuracy and Thermal Response:** PASS. Captures room, ice-water, hand-heating, and soldering-iron response measurements for both sensors.

SYS-HLR-400 — Local Temperature Display and Controls

Result PASS
Level / component HLR / System
Jira [SCRUM-411 — open requirement](#)

Current Jira description

When powered ON, the third box shall display temperature information according to the independently controlled display state of Sensor 1 and Sensor 2.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: None — top-level project or stakeholder requirement.

Children: [HWE-MLR-211](#), [SWE-EMB-MLR-303](#), [SWE-EMB-MLR-305](#), [SWE-EMB-MLR-306](#), [SWE-EMB-MLR-310](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.
- **FW-05 — Shared Local/Remote Display State:** PASS. Confirms local and remote commands update one authoritative display state and malformed commands do not change it.

SYS-HLR-410 — Local Display Performance

Result PASS
Level / component HLR / System
Jira [SCRUM-412 — open requirement](#)

Current Jira description

The third-box display shall present correct temperature information with the required response time and indoor readability.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: None — top-level project or stakeholder requirement.

Children: [SWE-EMB-MLR-307](#), [HWE-MLR-213](#), [HWE-MLR-214](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

SYS-HLR-420 — Local Sensor Fault Indication

Result PASS
Level / component HLR / System
Jira [SCRUM-413](#) — open requirement

Current Jira description

The third box shall notify the local user when a temperature sensor is disconnected or otherwise malfunctioning.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: None — top-level project or stakeholder requirement.

Children: [HWE-MLR-211](#), [SWE-EMB-MLR-303](#), [SWE-EMB-MLR-304](#), [SWE-EMB-MLR-308](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.
- **SYS-02 — Full-System Sensor Disconnect and Recovery:** PASS. Disconnects each real sensor in both ON and OFF display states and verifies other-sensor continuity, local/remote fault, graph gap, automatic recovery, and preservation of the pre-disconnection state.

SYS-HLR-500 — Remote Real-Time Display

Result PASS
Level / component HLR / System
Jira [SCRUM-414 — open requirement](#)

Current Jira description

When the computer software and third box are operational, the system shall provide remotely accessible real-time temperature/status information for both temperature sensors.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: None — top-level project or stakeholder requirement.

Children: [SWE-EMB-MLR-311](#), [INT-MLR-400](#), [INT-MLR-401](#), [INT-MLR-402](#), [INT-MLR-407](#), [SWE-CONN-MLR-500](#), [SWE-CONN-MLR-501](#), [SWE-CONN-MLR-502](#), [SWE-CONN-MLR-504](#), [SWE-DB-MLR-551](#), [SWE-DB-MLR-554](#), [SWE-WEB-MLR-600](#), [SWE-WEB-MLR-601](#), [SWE-WEB-MLR-602](#), [SWE-WEB-MLR-604](#), [SWE-WEB-MLR-605](#), [SWE-WEB-MLR-606](#), [SWE-WEB-MLR-613](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.
- **FE-02 — Live Readout and Stale-Data Behavior:** PASS. Exercises valid/change/disconnected/box-off scenarios, 1-second refresh, stale-number removal, exact unavailable text, and prominent values.
- **SYS-01 — Complete Data Path:** PASS. Verifies both physical sensor values/status through ESP32, BLE, backend, MySQL, server, and browser.

SYS-HLR-510 — Remote Third-Box Display Control

Result PASS
Level / component HLR / System
Jira [SCRUM-415 — open requirement](#)

Current Jira description

The computer user shall be able to remotely enable or disable the local third-box temperature display for each temperature sensor, with a response in less than one second.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: None — top-level project or stakeholder requirement.

Children: [SWE-EMB-MLR-313](#), [INT-MLR-403](#), [INT-MLR-406](#), [INT-MLR-407](#), [SWE-CONN-MLR-505](#), [SWE-DB-MLR-552](#), [SWE-DB-MLR-553](#), [SWE-WEB-MLR-602](#), [SWE-WEB-MLR-607](#), [SWE-SEC-MLR-750](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-02 — Authenticated Display Control:** PASS. Confirms authenticated SET_DISPLAY and rejection of unauthorized/malformed state-changing requests.
- **FE-07 — Remote Display Control UI:** PASS. Tests independent controls, pending/completed/failed states, authoritative returned state, and no optimistic success.
- **SYS-04 — Remote Display Response Latency:** PASS. Measures 10 ON/OFF changes per sensor from user action to physical LCD render; every sample must be below 1000 ms.

SYS-HLR-520 — Startup and Recovery

Result PASS
Level / component HLR / System
Jira [SCRUM-416](#) — open requirement

Current Jira description

If the computer software is already running while the third box is OFF, required temperature information shall become available within 10 seconds after the third box is turned ON.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: None — top-level project or stakeholder requirement.

Children: INT-MLR-405, SWE-CONN-MLR-503, SWE-CONN-MLR-506, SWE-CONN-MLR-508

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-03 — Power-On Recovery:** PASS. Measures fake-device appearance through discovery, connection, current poll, and publication against 10 seconds.
- **SYS-03 — Startup, Power-On, and History Recovery:** PASS. Separately measures box-OFF power recovery and 300-second history recovery after a host-radio outage while the box remains powered.

SYS-HLR-600 — Historical Temperature Graph

Result PASS
Level / component HLR / System
Jira [SCRUM-417 — open requirement](#)

Current Jira description

The computer interface shall provide a chart-recorder-style graphical representation of the most recent 300 seconds of temperature data with the specified units, axes, missing-data behavior, and scrolling behavior.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: None — top-level project or stakeholder requirement.

Children: SWE-EMB-MLR-302, INT-MLR-404, INT-MLR-407, SWE-CONN-MLR-503, SWE-CONN-MLR-504, SWE-DB-MLR-550, SWE-DB-MLR-551, SWE-DB-MLR-554, SWE-WEB-MLR-604, SWE-WEB-MLR-605, SWE-WEB-MLR-606, SWE-WEB-MLR-608, SWE-WEB-MLR-610, SWE-CONN-MLR-507, SWE-WEB-MLR-614

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-03 — History, Average, and Atomic Snapshot:** PASS. Verifies 1 Hz logical sampling, 300-record wrap/order, record fields, averages, synchronization, and boot identity.
- **BLE-03 — History Transfer and MTU Adaptation:** PASS. Validates metadata-first chunks, boundaries, retries, complete reconstruction, small MTU behavior, and negotiated real MTU.
- **BE-02 — History Synchronization:** PASS. Runs deterministic 300-record metadata/chunk/retry/reconstruction/upsert/no-duplicate and time-budget tests.
- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.
- **SYS-03 — Startup, Power-On, and History Recovery:** PASS. Separately measures box-OFF power recovery and 300-second history recovery after a host-radio outage while the box remains powered.

SYS-HLR-700 — Temperature Alerting

Result PASS
Level / component HLR / System
Jira [SCRUM-418 — open requirement](#)

Current Jira description

When the computer and third box are ON, the system shall send configurable email notifications when real-time temperature exceeds a configured maximum or falls below a configured minimum. SMS/text-message delivery is out of scope.

Revision note

Edited requirement. Notification scope was narrowed to email only; SMS delivery is not part of this release. The description above is the newest frozen version used by this run.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: None — top-level project or stakeholder requirement.

Children: SWE-DB-MLR-552, SWE-DB-MLR-554, SWE-WEB-MLR-602, SWE-WEB-MLR-603, SWE-ALR-MLR-700, SWE-ALR-MLR-701, SWE-ALR-MLR-702, SWE-ALR-MLR-703, SWE-ALR-MLR-704, SWE-ALR-MLR-705, SWE-SEC-MLR-750

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-06 — Email Alert Configuration and Delivery:** PASS. Checks email-only recipients/rules, email validation, configurable thresholds/messages, Gmail-compatible SMTP delivery, and persistence schema.
- **ALR-01 — Alert Rule Matrix:** PASS. Table-drives HIGH/LOW/normal/missing/disconnected/invalid, both sensors/average, multiple recipients, and independent thresholds.
- **ALR-03 — Notification Provider Adapters and Failure Isolation:** PASS. Checks email destinations/messages, external SMTP credentials, and failure isolation from subsequent polling.
- **SYS-05 — End-to-End Notifications:** PASS. Configures actual UI rules, triggers controlled crossings, and verifies storage, evaluation, episode transition, provider, destination, and message.

SYS-HLR-321 — Room-Temperature Accuracy

Result PASS
Level / component HLR / System
Jira [SCRUM-419 — open requirement](#)

Current Jira description

At laboratory room temperature, the thermometer output shall be approximately 22 deg C +/- 4 deg C.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-320](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-04 — Temperature Accuracy and Thermal Response:** PASS. Captures room, ice-water, hand-heating, and soldering-iron response measurements for both sensors.

SYS-HLR-322 — Ice-Water Accuracy

Result PASS
Level / component HLR / System
Jira [SCRUM-420 — open requirement](#)

Current Jira description

When a sensor is placed in a water-ice mixture, the thermometer output shall be 0 deg C +/- 2 deg C.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-320](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-04 — Temperature Accuracy and Thermal Response:** PASS. Captures room, ice-water, hand-heating, and soldering-iron response measurements for both sensors.

HWE-MLR-200 — Sensor Quantity and Cable Length

Result PASS
Level / component MLR / Third Box Hardware
Jira [SCRUM-421 — open requirement](#)

Current Jira description

The hardware shall provide two temperature sensors, each at the end of a 1.0 +/- 0.1 meter cable.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-200](#)

Children: [HWE-LLR-200](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-04 — Temperature Accuracy and Thermal Response:** PASS. Captures room, ice-water, hand-heating, and soldering-iron response measurements for both sensors.

HWE-MLR-201 — Sensor Mechanical Robustness

Result PASS
Level / component MLR / Third Box Hardware
Jira [SCRUM-422 — open requirement](#)

Current Jira description

The sensor assemblies shall tolerate ordinary handling and placement in ice water without damage.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-200](#)

Children: [HWE-LLR-201](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-02 — Mechanical Robustness:** PASS. Automatically captures pre/post full-system smoke evidence around guided drop, upside-down, connected-cable, handling, and immersion actions.
- **HW-04 — Temperature Accuracy and Thermal Response:** PASS. Captures room, ice-water, hand-heating, and soldering-iron response measurements for both sensors.

HWE-MLR-202 — Third-Box Enclosure and Drop Survival

Result PASS
Level / component MLR / Third Box Hardware
Jira [SCRUM-423 — open requirement](#)

Current Jira description

The third box shall be enclosed and shall remain functional after the specified workbench-to-floor drop and upside-down operation.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-200](#)

Children: [HWE-LLR-202](#), [HWE-LLR-203](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-02 — Mechanical Robustness:** PASS. Automatically captures pre/post full-system smoke evidence around guided drop, upside-down, connected-cable, handling, and immersion actions.

HWE-MLR-203 — External Connector Robustness

Result PASS
Level / component MLR / Third Box Hardware
Jira [SCRUM-424 — open requirement](#)

Current Jira description

External cable connectors shall be securely mounted, casual-user connectable, and shall not break during the specified connected-cable drop.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-200](#)

Children: [HWE-LLR-204](#), [HWE-LLR-205](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-02 — Mechanical Robustness:** PASS. Automatically captures pre/post full-system smoke evidence around guided drop, upside-down, connected-cable, handling, and immersion actions.

HWE-MLR-204 — Hot-Plug Electrical Protection

Result PASS
Level / component MLR / Third Box Hardware
Jira [SCRUM-425 — open requirement](#)

Current Jira description

The sensor electrical interface shall permit powered connection and disconnection without damaging the sensor or third-box electronics.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-210](#)

Children: [HWE-LLR-206](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-03 — Power Switch and Hot-Plug:** PASS. Guides sensor hot-plug and power cycling while capturing independent operation, no-reset behavior, loss of current data, and recovery.

HWE-MLR-205 — Battery and Power Switch

Result PASS
Level / component MLR / Third Box Hardware
Jira [SCRUM-426 — open requirement](#)

Current Jira description

The third box shall contain a battery and physical ON/OFF power switch and shall operate as a standalone battery-powered thermometer.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-100](#), [SYS-HLR-220](#)

Children: [HWE-LLR-207](#), [HWE-LLR-208](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-03 — Power Switch and Hot-Plug:** PASS. Guides sensor hot-plug and power cycling while capturing independent operation, no-reset behavior, loss of current data, and recovery.

HWE-MLR-206 — ESP32 Processing Platform

Result PASS
Level / component MLR / Third Box Hardware
Jira [SCRUM-427 — open requirement](#)

Current Jira description

The third box shall be designed around an ESP32 microcontroller capable of running the required firmware and BLE interface.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-100](#)

Children: [HWE-LLR-209](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.

HWE-MLR-209 — USB-C Sensor Connections

Result PASS
Level / component MLR / Third Box Hardware
Jira [SCRUM-430 — open requirement](#)

Current Jira description

Each sensor shall connect to the third box through a USB-C termination, and Sensor 1 and Sensor 2 ports shall be visually distinguishable.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-200](#)

Children: [HWE-LLR-214](#), [HWE-LLR-215](#), [HWE-LLR-216](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.

HWE-MLR-210 — USB-C Misconnection Protection

Result PASS
Level / component MLR / Third Box Hardware
Jira [SCRUM-431 — open requirement](#)

Current Jira description

The custom sensor USB-C interface shall be electrically arranged or protected so that foreseeable accidental connection of a standard USB-C cable or power source does not damage the third box or sensor.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-200](#)

Children: [HWE-LLR-217](#), [HWE-LLR-218](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-04 — Temperature Accuracy and Thermal Response:** PASS. Captures room, ice-water, hand-heating, and soldering-iron response measurements for both sensors.

HWE-MLR-211 — Display Character Capability

Result PASS
Level / component MLR / Third Box Hardware
Jira [SCRUM-432 — open requirement](#)

Current Jira description

The LCD hardware shall support all characters required to display sensor labels, OFF/DISCONNECTED/error indications, signed temperatures across the required range, and the average value.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: SYS-HLR-300, SYS-HLR-400, SYS-HLR-420

Children: HWE-LLR-219

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-03 — Power Switch and Hot-Plug:** PASS. Guides sensor hot-plug and power cycling while capturing independent operation, no-reset behavior, loss of current data, and recovery.
- **HW-04 — Temperature Accuracy and Thermal Response:** PASS. Captures room, ice-water, hand-heating, and soldering-iron response measurements for both sensors.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

HWE-MLR-212 — Sensor/Signal-Chain Capability

Result PASS
Level / component MLR / Third Box Hardware
Jira [SCRUM-433 — open requirement](#)

Current Jira description

The selected sensors and associated electrical interface shall support the required temperature range and sufficient accuracy to meet the end-to-end accuracy requirements.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-300](#), [SYS-HLR-320](#)

Children: [HWE-LLR-220](#), [HWE-LLR-221](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-04 — Temperature Accuracy and Thermal Response:** PASS. Captures room, ice-water, hand-heating, and soldering-iron response measurements for both sensors.

HWE-LLR-200 — Cable Length Verification

Result PASS
Level / component LLR / Third Box Hardware
Jira [SCRUM-434 — open requirement](#)

Current Jira description

Each completed sensor cable assembly shall measure between 0.9 m and 1.1 m from the third-box connector interface to the sensing element.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [HWE-MLR-200](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.

HWE-LLR-201 — Wet Sensor Construction

Result PASS
Level / component LLR / Third Box Hardware
Jira [SCRUM-435 — open requirement](#)

Current Jira description

The sensor head and exposed cable region intended for immersion shall use insulation/encapsulation suitable for temporary immersion in an ice-water mixture.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [HWE-MLR-201](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-02 — Mechanical Robustness:** PASS. Automatically captures pre/post full-system smoke evidence around guided drop, upside-down, connected-cable, handling, and immersion actions.
- **HW-04 — Temperature Accuracy and Thermal Response:** PASS. Captures room, ice-water, hand-heating, and soldering-iron response measurements for both sensors.

HWE-LLR-202 — Internal Retention

Result PASS
Level / component LLR / Third Box Hardware
Jira [SCRUM-436](#) — open requirement

Current Jira description

The ESP32 board/module, LCD, battery, connectors, and other internal assemblies shall be mechanically retained so they do not become loose during the required drop test.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [HWE-MLR-202](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-02 — Mechanical Robustness:** PASS. Automatically captures pre/post full-system smoke evidence around guided drop, upside-down, connected-cable, handling, and immersion actions.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

HWE-LLR-203 — Orientation-Independent Mounting

Result PASS
Level / component LLR / Third Box Hardware
Jira [SCRUM-437 — open requirement](#)

Current Jira description

Internal wiring and mounted parts shall not depend on gravity or enclosure orientation for electrical continuity.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [HWE-MLR-202](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-02 — Mechanical Robustness:** PASS. Automatically captures pre/post full-system smoke evidence around guided drop, upside-down, connected-cable, handling, and immersion actions.

HWE-LLR-204 — Panel Connector Support

Result PASS
Level / component LLR / Third Box Hardware
Jira [SCRUM-438 — open requirement](#)

Current Jira description

Each external cable connector shall be panel-mounted or mechanically supported independently of PCB solder joints.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [HWE-MLR-203](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-02 — Mechanical Robustness:** PASS. Automatically captures pre/post full-system smoke evidence around guided drop, upside-down, connected-cable, handling, and immersion actions.

HWE-LLR-205 — Cable Strain Relief

Result PASS
Level / component LLR / Third Box Hardware
Jira [SCRUM-439 — open requirement](#)

Current Jira description

Sensor cable assemblies shall include strain relief at the probe end and connector end.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [HWE-MLR-203](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-02 — Mechanical Robustness:** PASS. Automatically captures pre/post full-system smoke evidence around guided drop, upside-down, connected-cable, handling, and immersion actions.

HWE-LLR-206 — Hot-Plug Input Protection

Result PASS
Level / component LLR / Third Box Hardware
Jira [SCRUM-440 — open requirement](#)

Current Jira description

Sensor input circuitry shall tolerate insertion/removal transients without requiring a third-box reset.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [HWE-MLR-204](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-03 — Power Switch and Hot-Plug:** PASS. Guides sensor hot-plug and power cycling while capturing independent operation, no-reset behavior, loss of current data, and recovery.

HWE-LLR-207 — Battery Supply Capacity

Result PASS
Level / component LLR / Third Box Hardware
Jira [SCRUM-441 — open requirement](#)

Current Jira description

The battery and regulator path shall supply the simultaneous peak current required by the ESP32, BLE radio, LCD/backlight, and both sensor interfaces.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [HWE-MLR-205](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

HWE-LLR-208 — Physical OFF State

Result PASS
Level / component LLR / Third Box Hardware
Jira [SCRUM-442 — open requirement](#)

Current Jira description

The power switch shall remove or disable operating power such that the ESP32/LCD cannot continue providing current temperature data when the switch is OFF.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [HWE-MLR-205](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-03 — Power Switch and Hot-Plug:** PASS. Guides sensor hot-plug and power cycling while capturing independent operation, no-reset behavior, loss of current data, and recovery.
- **HW-04 — Temperature Accuracy and Thermal Response:** PASS. Captures room, ice-water, hand-heating, and soldering-iron response measurements for both sensors.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

HWE-LLR-209 — ESP32 BLE Capability

Result PASS
Level / component LLR / Third Box Hardware
Jira [SCRUM-443 — open requirement](#)

Current Jira description

The selected ESP32 device/module shall include BLE capability supported by the selected ESP-IDF version.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [HWE-MLR-206](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-04 — Temperature Accuracy and Thermal Response:** PASS. Captures room, ice-water, hand-heating, and soldering-iron response measurements for both sensors.

HWE-LLR-210 — LCD Electrical Interface

Result PASS
Level / component LLR / Third Box Hardware
Jira [SCRUM-444 — open requirement](#)

Current Jira description

The character LCD shall be electrically interfaced to the ESP32 using the selected supported digital interface and shall provide enough character positions to render the required local states.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: None — top-level project or stakeholder requirement.

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

HWE-LLR-214 — Sensor 1 USB-C Port

Result PASS
Level / component LLR / Third Box Hardware
Jira [SCRUM-448 — open requirement](#)

Current Jira description

Sensor 1 shall terminate in a dedicated USB-C receptacle/plug pair assigned to the Sensor 1 electrical interface.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [HWE-MLR-209](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.

HWE-LLR-215 — Sensor 2 USB-C Port

Result PASS
Level / component LLR / Third Box Hardware
Jira [SCRUM-449 — open requirement](#)

Current Jira description

Sensor 2 shall terminate in a dedicated USB-C receptacle/plug pair assigned to the Sensor 2 electrical interface.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [HWE-MLR-209](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.

HWE-LLR-216 — Port Identification

Result PASS
Level / component LLR / Third Box Hardware
Jira [SCRUM-450 — open requirement](#)

Current Jira description

The enclosure shall use permanent labels, symbols, or other visual differentiation to identify Sensor 1 and Sensor 2 ports.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [HWE-MLR-209](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.

HWE-LLR-217 — USB-C VBUS Protection

Result PASS
Level / component LLR / Third Box Hardware
Jira [SCRUM-451 — open requirement](#)

Current Jira description

The custom sensor connector design shall prevent an accidental standard USB-C VBUS connection from applying damaging voltage to sensor-signal pins or the ESP32.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [HWE-MLR-210](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.

HWE-LLR-218 — USB-C Signal Isolation

Result PASS
Level / component LLR / Third Box Hardware
Jira [SCRUM-452 — open requirement](#)

Current Jira description

Custom sensor signals shall not be exposed on standard USB-C pins in a way that can damage a standard USB host/device during foreseeable accidental connection.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [HWE-MLR-210](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-04 — Temperature Accuracy and Thermal Response:** PASS. Captures room, ice-water, hand-heating, and soldering-iron response measurements for both sensors.

HWE-LLR-219 — LCD Signed Temperature Support

Result PASS
Level / component LLR / Third Box Hardware
Jira [SCRUM-453 — open requirement](#)

Current Jira description

The physical display size shall permit rendering at least a sensor identifier/state and a signed temperature value within -10 to +63 deg C.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [HWE-MLR-211](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-04 — Temperature Accuracy and Thermal Response:** PASS. Captures room, ice-water, hand-heating, and soldering-iron response measurements for both sensors.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

HWE-LLR-220 — Sensor Range Selection

Result PASS
Level / component LLR / Third Box Hardware
Jira [SCRUM-454 — open requirement](#)

Current Jira description

The selected sensor datasheet shall specify an operating/measurement range that includes -10 through +63 deg C.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [HWE-MLR-212](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-04 — Temperature Accuracy and Thermal Response:** PASS. Captures room, ice-water, hand-heating, and soldering-iron response measurements for both sensors.

HWE-LLR-221 — Accuracy Budget

Result PASS
Level / component LLR / Third Box Hardware
Jira [SCRUM-455 — open requirement](#)

Current Jira description

The hardware error budget at 0 deg C and approximately 22 deg C shall leave sufficient margin for software conversion/rounding while meeting the stakeholder tolerances.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [HWE-MLR-212](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **[HW-01 — Physical Configuration and Design Inspection](#):** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.

SWE-EMB-MLR-300 — ESP-IDF C Firmware

Result PASS
Level / component MLR / ESP32 Firmware
Jira [SCRUM-456](#) — open requirement

Current Jira description

Third-box firmware shall be implemented for the ESP32 using ESP-IDF and the C programming language.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-100](#)

Children: [SWE-EMB-LLR-300](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-01 — Firmware Build, Startup, and Configuration:** PASS. Builds or statically validates real and simulated firmware configuration, protocol generation, app_main initialization, Celsius, and boot identity.

SWE-EMB-MLR-301 — Continuous Sensor Acquisition

Result PASS
Level / component MLR / ESP32 Firmware
Jira [SCRUM-457 — open requirement](#)

Current Jira description

The third-box firmware shall continuously acquire readings/status from both temperature sensors while powered ON.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-300](#), [SYS-HLR-310](#)

Children: [SWE-EMB-LLR-301](#), [SWE-EMB-LLR-302](#), [SWE-EMB-LLR-303](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.

SWE-EMB-MLR-302 — Two 300-Reading History Buffers

Result PASS
Level / component MLR / ESP32 Firmware
Jira [SCRUM-458 — open requirement](#)

Current Jira description

The third box shall maintain separate rolling buffers containing the last 300 one-second records for Sensor 1 and Sensor 2.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-600](#)

Children: [SWE-EMB-LLR-304](#), [SWE-EMB-LLR-305](#), [SWE-EMB-LLR-306](#), [SWE-EMB-LLR-307](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.

SWE-EMB-MLR-303 — Sensor State Model

Result PASS
Level / component MLR / ESP32 Firmware
Jira [SCRUM-459](#) — open requirement

Current Jira description

The third box shall expose three user-visible states per sensor: ON, OFF, and DISCONNECTED.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-400](#), [SYS-HLR-420](#)

Children: [SWE-EMB-LLR-308](#), [SWE-EMB-LLR-309](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.

SWE-EMB-MLR-304 — Disconnect Transition Behavior

Result PASS
Level / component MLR / ESP32 Firmware
Jira [SCRUM-460](#) — open requirement

Current Jira description

A disconnected/faulted sensor shall be reported as DISCONNECTED while the fault exists and shall return to the OFF display state after a valid reconnection unless the user subsequently enables it.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-210](#), [SYS-HLR-420](#)

Children: [SWE-EMB-LLR-310](#), [SWE-EMB-LLR-311](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.

SWE-EMB-MLR-305 — Independent Local Display Control

Result PASS
Level / component MLR / ESP32 Firmware
Jira [SCRUM-461 — open requirement](#)

Current Jira description

The firmware shall independently control whether Sensor 1 and Sensor 2 temperatures are shown on the LCD.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-400](#)

Children: [SWE-EMB-LLR-312](#), [SWE-EMB-LLR-313](#), [SWE-EMB-LLR-319](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

SWE-EMB-MLR-306 — Local Average Display

Result PASS
Level / component MLR / ESP32 Firmware
Jira [SCRUM-462 — open requirement](#)

Current Jira description

When both sensor display states are ON and both readings are valid, the LCD shall display Sensor 1, Sensor 2, and the average of the two readings.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-400](#)

Children: [SWE-EMB-LLR-314](#), [SWE-EMB-LLR-315](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.
- **FW-03 — History, Average, and Atomic Snapshot:** PASS. Verifies 1 Hz logical sampling, 300-record wrap/order, record fields, averages, synchronization, and boot identity.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

SWE-EMB-MLR-307 — Local Display Response Time

Result PASS
Level / component MLR / ESP32 Firmware
Jira [SCRUM-463](#) — open requirement

Current Jira description

The firmware shall update the LCD to reflect a local sensor-display control action within approximately 20 ms.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-410](#)

Children: [SWE-EMB-LLR-316](#), [SWE-EMB-LLR-317](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

SWE-EMB-MLR-308 — Local Fault Display

Result PASS
Level / component MLR / ESP32 Firmware
Jira [SCRUM-464 — open requirement](#)

Current Jira description

The firmware shall render a clear local fault/disconnected indication instead of a valid temperature for an unavailable sensor.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-420](#)

Children: [SWE-EMB-LLR-318](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

SWE-EMB-MLR-310 — Local Celsius Presentation

Result PASS
Level / component MLR / ESP32 Firmware
Jira [SCRUM-466 — open requirement](#)

Current Jira description

The third-box LCD shall present temperature values in degrees Celsius and support signed values across the required system range.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-300](#), [SYS-HLR-400](#)

Children: [SWE-EMB-LLR-323](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

SWE-EMB-MLR-311 — Atomic Current Snapshot

Result PASS
Level / component MLR / ESP32 Firmware
Jira [SCRUM-467 — open requirement](#)

Current Jira description

The firmware shall provide an internally consistent current snapshot containing both sensor values, sensor states, display-enable states, average validity/value, and sample identity for a computer request.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-500](#)

Children: [SWE-EMB-LLR-324](#), [SWE-EMB-LLR-325](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.
- **FW-03 — History, Average, and Atomic Snapshot:** PASS. Verifies 1 Hz logical sampling, 300-record wrap/order, record fields, averages, synchronization, and boot identity.

SWE-EMB-MLR-312 — Automatic Sensor Recovery

Result PASS
Level / component MLR / ESP32 Firmware
Jira [SCRUM-468 — open requirement](#)

Current Jira description

Sensor acquisition shall automatically resume after reconnection without restarting the ESP32 firmware.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-210](#)

Children: [SWE-EMB-LLR-326](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.

SWE-EMB-MLR-313 — Remote Display Command Handling

Result PASS
Level / component MLR / ESP32 Firmware
Jira [SCRUM-469](#) — open requirement

Current Jira description

The firmware shall accept validated remote commands that change the same sensor display-enable state controlled by the physical sensor buttons.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-510](#)

Children: [SWE-EMB-LLR-327](#), [SWE-EMB-LLR-328](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.
- **FW-05 — Shared Local/Remote Display State:** PASS. Confirms local and remote commands update one authoritative display state and malformed commands do not change it.

SWE-EMB-LLR-300 — Firmware Entry Point

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-470 — open requirement](#)

Current Jira description

The ESP-IDF application shall initialize sensor drivers, LCD/backlight drivers, input handlers, shared state, history buffers, and the BLE service from `app_main()`.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-300](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-01 — Firmware Build, Startup, and Configuration:** PASS. Builds or statically validates real and simulated firmware configuration, protocol generation, `app_main` initialization, Celsius, and boot identity.
- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.
- **FW-03 — History, Average, and Atomic Snapshot:** PASS. Verifies 1 Hz logical sampling, 300-record wrap/order, record fields, averages, synchronization, and boot identity.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

SWE-EMB-LLR-301 — Per-Sensor Runtime Record

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-471 — open requirement](#)

Current Jira description

Firmware shall maintain one runtime record per sensor containing latest temperature in Celsius, validity/connection state, display-enabled state, and the latest sample sequence number.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-301](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.

SWE-EMB-LLR-302 — Continuous Acquisition Loop

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-472 — open requirement](#)

Current Jira description

The sensor acquisition task shall begin a new read for each sensor whenever the previous read cycle is complete and shall update the shared runtime record only after validating the result.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-301](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.

SWE-EMB-LLR-303 — Independent Sensor Reads

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-473 — open requirement](#)

Current Jira description

A read failure on one sensor shall not skip or block acquisition of the other sensor.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-301](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.

SWE-EMB-LLR-304 — One-Second History Timer

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-474 — open requirement](#)

Current Jira description

A periodic timer/task shall snapshot each sensor runtime record into its history buffer once every 1000 ms.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-302](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.
- **FW-03 — History, Average, and Atomic Snapshot:** PASS. Verifies 1 Hz logical sampling, 300-record wrap/order, record fields, averages, synchronization, and boot identity.

SWE-EMB-LLR-305 — Fixed-Size Circular Buffers

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-475 — open requirement](#)

Current Jira description

Each sensor history buffer shall be implemented as a fixed-capacity circular buffer of 300 records; adding record 301 shall overwrite the oldest record.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-302](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.
- **FW-03 — History, Average, and Atomic Snapshot:** PASS. Verifies 1 Hz logical sampling, 300-record wrap/order, record fields, averages, synchronization, and boot identity.

SWE-EMB-LLR-306 — History Record Contents

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-476 — open requirement](#)

Current Jira description

Each stored history record shall contain at least the sample sequence number, temperature value when valid, and a sensor status value sufficient to distinguish valid and missing/disconnected data.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-302](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.
- **FW-03 — History, Average, and Atomic Snapshot:** PASS. Verifies 1 Hz logical sampling, 300-record wrap/order, record fields, averages, synchronization, and boot identity.

SWE-EMB-LLR-307 — History Ordering

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-477 — open requirement](#)

Current Jira description

History retrieval code shall expose records in chronological order from oldest available to newest available regardless of circular-buffer wrap position.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-302](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-03 — History, Average, and Atomic Snapshot:** PASS. Verifies 1 Hz logical sampling, 300-record wrap/order, record fields, averages, synchronization, and boot identity.

SWE-EMB-LLR-308 — Internal State Separation

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-478 — open requirement](#)

Current Jira description

Firmware shall maintain sensor connection/health separately from the user display-enable flag and shall derive the externally visible ON/OFF/DISCONNECTED state from those values.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-303](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.

SWE-EMB-LLR-309 — Visible State Mapping

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-479 — open requirement](#)

Current Jira description

A valid connected sensor with display_enabled=true shall report ON; a valid connected sensor with display_enabled=false shall report OFF; an invalid/disconnected sensor shall report DISCONNECTED.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-303](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

SWE-EMB-LLR-310 — Fault Entry

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-480 — open requirement](#)

Current Jira description

After the configured sensor-driver fault criterion is met, firmware shall set the sensor connection/health state to DISCONNECTED and invalidate its current temperature.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-304](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.

SWE-EMB-LLR-311 — Reconnection State

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-481](#) — open requirement

Current Jira description

When a DISCONNECTED sensor produces valid readings again and meets the configured recovery criterion, firmware shall mark it connected and restore the sensor's display-enabled state from immediately before the disconnection. A sensor that was ON shall return ON; a sensor that was OFF shall remain OFF. This state retention shall be implemented locally in the third-box firmware and shall not depend on a computer connection. Verification: exercise ON-disconnect-reconnect and OFF-disconnect-reconnect cases for each sensor while the computer is absent or disconnected, and confirm the restored local state matches the pre-disconnection state.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-304](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.
- **SYS-02 — Full-System Sensor Disconnect and Recovery:** PASS. Disconnects each real sensor in both ON and OFF display states and verifies other-sensor continuity, local/remote fault, graph gap, automatic recovery, and preservation of the pre-disconnection state.

SWE-EMB-LLR-312 — Physical Button Toggle

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-482 — open requirement](#)

Current Jira description

A debounced press of Sensor 1 or Sensor 2 button shall toggle that sensor display_enabled flag only when the sensor is connected.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-305](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

SWE-EMB-LLR-313 — Disconnected Button Behavior

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-483](#) — open requirement

Current Jira description

Pressing a sensor display button while that sensor is DISCONNECTED shall not cause a valid temperature to be displayed or change the reported DISCONNECTED state.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-305](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

SWE-EMB-LLR-314 — Average Calculation

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-484 — open requirement](#)

Current Jira description

The firmware shall calculate the displayed average as $(\text{sensor1_c} + \text{sensor2_c}) / 2$ using sufficient intermediate precision to avoid overflow or unnecessary rounding error.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-306](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.
- **FW-03 — History, Average, and Atomic Snapshot:** PASS. Verifies 1 Hz logical sampling, 300-record wrap/order, record fields, averages, synchronization, and boot identity.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

SWE-EMB-LLR-315 — Average Validity

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-485](#) — open requirement

Current Jira description

The local average shall be valid/displayed only when both sensors are connected, valid, and their display_enabled flags are true.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-306](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.
- **FW-03 — History, Average, and Atomic Snapshot:** PASS. Verifies 1 Hz logical sampling, 300-record wrap/order, record fields, averages, synchronization, and boot identity.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

SWE-EMB-LLR-316 — Cached Display Data

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-486 — open requirement](#)

Current Jira description

Local button handling shall render from the latest cached sensor values rather than waiting for a new sensor conversion.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-307](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

SWE-EMB-LLR-317 — Button-to-Render Scheduling

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-487 — open requirement](#)

Current Jira description

The local input handler shall signal the LCD update path immediately after a valid button-state change so that software scheduling does not intentionally delay the update beyond 20 ms.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-307](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

SWE-EMB-LLR-318 — Disconnected LCD Text

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-488 — open requirement](#)

Current Jira description

For a disconnected/faulted sensor, the LCD rendering layer shall display a DISCONNECTED or equivalent clear error label instead of a numeric temperature.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-308](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

SWE-EMB-LLR-319 — Off LCD Text

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-489](#) — open requirement

Current Jira description

For a connected sensor with display_enabled=false, the LCD rendering layer shall display Sensor 1 OFF or Sensor 2 OFF, as applicable.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-305](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

SWE-EMB-LLR-323 — Canonical Temperature Unit

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-493 — open requirement](#)

Current Jira description

All third-box internal temperature calculations and BLE payload temperatures shall use Celsius as the canonical unit.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-310](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-01 — Firmware Build, Startup, and Configuration:** PASS. Builds or statically validates real and simulated firmware configuration, protocol generation, app_main initialization, Celsius, and boot identity.

SWE-EMB-LLR-324 — Snapshot Locking

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-494 — open requirement](#)

Current Jira description

The current-data response shall copy both sensor runtime records under a synchronization mechanism so one response cannot mix values from different update instants.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-311](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.
- **FW-03 — History, Average, and Atomic Snapshot:** PASS. Verifies 1 Hz logical sampling, 300-record wrap/order, record fields, averages, synchronization, and boot identity.

SWE-EMB-LLR-325 — Boot Identifier

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-495 — open requirement](#)

Current Jira description

Firmware shall generate or increment a boot/session identifier at startup and include it with current/history metadata so computer software can distinguish samples across third-box restarts.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-311](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-03 — History, Average, and Atomic Snapshot:** PASS. Verifies 1 Hz logical sampling, 300-record wrap/order, record fields, averages, synchronization, and boot identity.

SWE-EMB-LLR-326 — Recovery Without Restart

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-496 — open requirement](#)

Current Jira description

A successful sensor read after a disconnected period shall restore acquisition status without restarting the acquisition task, BLE service, or whole device.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-312](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.

SWE-EMB-LLR-327 — Remote/Local Shared State

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-497 — open requirement](#)

Current Jira description

Remote display commands and physical buttons shall update the same display_enabled fields and invoke the same LCD rendering path.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-313](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

SWE-EMB-LLR-328 — Remote Command Validation

Result PASS
Level / component LLR / ESP32 Firmware
Jira [SCRUM-498 — open requirement](#)

Current Jira description

Firmware shall reject remote display-control requests containing an invalid sensor identifier or unsupported state value without changing local state.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-EMB-MLR-313](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FW-02 — Sensor Acquisition, State Machine, and Recovery:** PASS. Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.

INT-MLR-400 — BLE Transport

Result PASS
Level / component MLR / BLE Interface
Jira [SCRUM-499](#) — open requirement

Current Jira description

The third box and computer connection component shall communicate using Bluetooth Low Energy.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-110](#), [SYS-HLR-500](#)

Children: [INT-LLR-400](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-01 — GATT and Protocol Contract:** PASS. Checks UUIDs, layouts, request IDs, GET_CURRENT, statuses/version rejection, authentication framing, and requester-only behavior.

INT-MLR-401 — Request/Response Application Protocol

Result PASS
Level / component MLR / BLE Interface
Jira [SCRUM-500 — open requirement](#)

Current Jira description

The third box shall transmit application data only in response to a computer-initiated BLE request; unsolicited application telemetry shall not be required.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-500](#)

Children: [INT-LLR-401](#), [INT-LLR-402](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-01 — GATT and Protocol Contract:** PASS. Checks UUIDs, layouts, request IDs, GET_CURRENT, statuses/version rejection, authentication framing, and requester-only behavior.

INT-MLR-402 — Current Data Transaction

Result PASS
Level / component MLR / BLE Interface
Jira [SCRUM-501](#) — open requirement

Current Jira description

The BLE interface shall support a request that returns the current state and value of both sensors and the current local display state.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-500](#)

Children: [INT-LLR-404](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-01 — GATT and Protocol Contract:** PASS. Checks UUIDs, layouts, request IDs, GET_CURRENT, statuses/version rejection, authentication framing, and requester-only behavior.

INT-MLR-403 — Remote Display Control Transaction

Result PASS
Level / component MLR / BLE Interface
Jira [SCRUM-502 — open requirement](#)

Current Jira description

The BLE interface shall support validated remote commands to change Sensor 1 and Sensor 2 local display-enable states and shall expose the resulting state.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-510](#)

Children: [INT-LLR-405](#), [INT-LLR-406](#), [INT-LLR-416](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-01 — GATT and Protocol Contract:** PASS. Checks UUIDs, layouts, request IDs, GET_CURRENT, statuses/version rejection, authentication framing, and requester-only behavior.

INT-MLR-404 — Historical Data Transaction

Result PASS
Level / component MLR / BLE Interface
Jira [SCRUM-503](#) — open requirement

Current Jira description

The BLE interface shall permit retrieval of both 300-record sensor history buffers from the third box.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-600](#)

Children: [INT-LLR-407](#), [INT-LLR-408](#), [INT-LLR-409](#), [INT-LLR-410](#), [INT-LLR-411](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-03 — History Transfer and MTU Adaptation:** PASS. Validates metadata-first chunks, boundaries, retries, complete reconstruction, small MTU behavior, and negotiated real MTU.

INT-MLR-405 — Connection Recovery

Result PASS
Level / component MLR / BLE Interface
Jira [SCRUM-504 — open requirement](#)

Current Jira description

The BLE protocol and computer client shall support automatic reconnection/recovery without requiring application restart.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-520](#)

Children: [INT-LLR-415](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-04 — BLE Link-Loss Recovery:** PASS. Forces physical link loss and verifies automatic rediscovery/reconnection without web restart.

INT-MLR-406 — BLE Access Control

Result PASS
Level / component MLR / BLE Interface
Jira [SCRUM-505 — open requirement](#)

Current Jira description

The BLE interface shall use an authentication/bonding mechanism sufficient to prevent an arbitrary nearby BLE client from issuing third-box control commands.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-510](#)

Children: [INT-LLR-414](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-02 — Authenticated Display Control:** PASS. Confirms authenticated SET_DISPLAY and rejection of unauthorized/malformed state-changing requests.

INT-MLR-407 — Protocol Versioning and Error Reporting

Result PASS
Level / component MLR / BLE Interface
Jira [SCRUM-506](#) — open requirement

Current Jira description

The BLE application protocol shall provide protocol-version identification and explicit success/error status for commands.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-500](#), [SYS-HLR-510](#), [SYS-HLR-600](#)

Children: [INT-LLR-403](#), [INT-LLR-412](#), [INT-LLR-413](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-01 — GATT and Protocol Contract:** PASS. Checks UUIDs, layouts, request IDs, GET_CURRENT, statuses/version rejection, authentication framing, and requester-only behavior.

INT-LLR-400 — Custom GATT Service

Result PASS
Level / component LLR / BLE Interface
Jira [SCRUM-507 — open requirement](#)

Current Jira description

The ESP32 shall expose one custom 128-bit GATT service dedicated to thermometer application communication; the computer connection process shall identify compatible devices by this service UUID.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [INT-MLR-400](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-01 — GATT and Protocol Contract:** PASS. Checks UUIDs, layouts, request IDs, GET_CURRENT, statuses/version rejection, authentication framing, and requester-only behavior.

INT-LLR-401 — No Application Notifications

Result PASS
Level / component LLR / BLE Interface
Jira [SCRUM-508 — open requirement](#)

Current Jira description

Application temperature and history data shall be transmitted only in response to computer-initiated GATT transactions. The third box shall not transmit application temperature or history data using unsolicited GATT notifications or indications.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [INT-MLR-401](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-03 — History Transfer and MTU Adaptation:** PASS. Validates metadata-first chunks, boundaries, retries, complete reconstruction, small MTU behavior, and negotiated real MTU.

INT-LLR-402 — Request Identifier

Result PASS
Level / component LLR / BLE Interface
Jira [SCRUM-509 — open requirement](#)

Current Jira description

Each application request shall carry a request identifier, and each response shall echo that identifier so the client can match responses to requests.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [INT-MLR-401](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-01 — GATT and Protocol Contract:** PASS. Checks UUIDs, layouts, request IDs, GET_CURRENT, statuses/version rejection, authentication framing, and requester-only behavior.

INT-LLR-403 — Protocol Header

Result PASS
Level / component LLR / BLE Interface
Jira [SCRUM-510 — open requirement](#)

Current Jira description

Each application message shall contain at least protocol version, opcode, request identifier, payload length, and response status when applicable. Optionally, the application messages may include parity bits

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [INT-MLR-407](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-01 — GATT and Protocol Contract:** PASS. Checks UUIDs, layouts, request IDs, GET_CURRENT, statuses/version rejection, authentication framing, and requester-only behavior.

INT-LLR-404 — GET_CURRENT Opcode

Result PASS
Level / component LLR / BLE Interface
Jira [SCRUM-511 — open requirement](#)

Current Jira description

The protocol shall define a GET_CURRENT request whose response includes boot identifier, newest sample sequence, Sensor 1 temperature/status/display state and Sensor 2 temperature/status/display state.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [INT-MLR-402](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-01 — GATT and Protocol Contract:** PASS. Checks UUIDs, layouts, request IDs, GET_CURRENT, statuses/version rejection, authentication framing, and requester-only behavior.

INT-LLR-405 — SET_DISPLAY Opcode

Result PASS
Level / component LLR / BLE Interface
Jira [SCRUM-512 — open requirement](#)

Current Jira description

The protocol shall define a SET_DISPLAY request containing sensor identifier and desired display-enabled Boolean value.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [INT-MLR-403](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-02 — Authenticated Display Control:** PASS. Confirms authenticated SET_DISPLAY and rejection of unauthorized/malformed state-changing requests.

INT-LLR-406 — SET_DISPLAY Response

Result PASS
Level / component LLR / BLE Interface
Jira [SCRUM-513 — open requirement](#)

Current Jira description

The SET_DISPLAY response shall return success/error and the resulting externally visible sensor state.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [INT-MLR-403](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-02 — Authenticated Display Control:** PASS. Confirms authenticated SET_DISPLAY and rejection of unauthorized/malformed state-changing requests.

INT-LLR-407 — GET_HISTORY_META Opcode

Result PASS
Level / component LLR / BLE Interface
Jira [SCRUM-514 — open requirement](#)

Current Jira description

The protocol shall define a GET_HISTORY_META request returning boot identifier, newest sequence, and available record count for each sensor buffer.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [INT-MLR-404](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-03 — History Transfer and MTU Adaptation:** PASS. Validates metadata-first chunks, boundaries, retries, complete reconstruction, small MTU behavior, and negotiated real MTU.

INT-LLR-408 — GET_HISTORY_CHUNK Opcode

Result PASS
Level / component LLR / BLE Interface
Jira [SCRUM-515 — open requirement](#)

Current Jira description

The protocol shall define a GET_HISTORY_CHUNK request containing sensor identifier, chronological starting offset/sequence, and requested record count.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [INT-MLR-404](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-03 — History Transfer and MTU Adaptation:** PASS. Validates metadata-first chunks, boundaries, retries, complete reconstruction, small MTU behavior, and negotiated real MTU.

INT-LLR-409 — History Chunk Response

Result PASS
Level / component LLR / BLE Interface
Jira [SCRUM-516 — open requirement](#)

Current Jira description

A history-chunk response shall contain only complete history records and shall identify the sensor, starting record position/sequence, and number of records returned.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [INT-MLR-404](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **[BLE-03 — History Transfer and MTU Adaptation](#):** PASS. Validates metadata-first chunks, boundaries, retries, complete reconstruction, small MTU behavior, and negotiated real MTU.

INT-LLR-410 — History Chunk Size

Result PASS
Level / component LLR / BLE Interface
Jira [SCRUM-517 — open requirement](#)

Current Jira description

The history retrieval implementation shall segment the 300-record history into BLE-safe chunks and shall not assume the entire history fits in one ATT payload.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [INT-MLR-404](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-03 — History Transfer and MTU Adaptation:** PASS. Validates metadata-first chunks, boundaries, retries, complete reconstruction, small MTU behavior, and negotiated real MTU.

INT-LLR-411 — MTU Negotiation

Result PASS
Level / component LLR / BLE Interface
Jira [SCRUM-518 — open requirement](#)

Current Jira description

The computer BLE client shall request a larger ATT MTU when supported. If a smaller MTU is negotiated, the history transfer shall remain functional by reducing the number of complete history records included in each chunk so that each response fits within the negotiated MTU.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [INT-MLR-404](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-03 — History Transfer and MTU Adaptation:** PASS. Validates metadata-first chunks, boundaries, retries, complete reconstruction, small MTU behavior, and negotiated real MTU.

INT-LLR-412 — Status Codes

Result PASS
Level / component LLR / BLE Interface
Jira [SCRUM-519 — open requirement](#)

Current Jira description

The protocol shall define at least SUCCESS, INVALID_COMMAND, INVALID_SENSOR, INVALID_VALUE, NOT_AVAILABLE, and INTERNAL_ERROR response statuses.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [INT-MLR-407](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-01 — GATT and Protocol Contract:** PASS. Checks UUIDs, layouts, request IDs, GET_CURRENT, statuses/version rejection, authentication framing, and requester-only behavior.

INT-LLR-413 — Unsupported Version Handling

Result PASS
Level / component LLR / BLE Interface
Jira [SCRUM-520 — open requirement](#)

Current Jira description

A receiver shall reject an application message using an unsupported protocol version without applying state-changing commands.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [INT-MLR-407](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-02 — Authenticated Display Control:** PASS. Confirms authenticated SET_DISPLAY and rejection of unauthorized/malformed state-changing requests.

INT-LLR-414 — BLE Pairing/Bonding

Result PASS
Level / component LLR / BLE Interface
Jira [SCRUM-521 — open requirement](#)

Current Jira description

The ESP32 and computer client shall use BLE pairing/bonding with authenticated/encrypted access for state-changing characteristics/commands.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [INT-MLR-406](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-02 — Authenticated Display Control:** PASS. Confirms authenticated SET_DISPLAY and rejection of unauthorized/malformed state-changing requests.

INT-LLR-415 — Reconnect Discovery

Result PASS
Level / component LLR / BLE Interface
Jira [SCRUM-522 — open requirement](#)

Current Jira description

After BLE link loss, the computer client shall resume scanning/connection attempts automatically and shall not require a user to restart the web application.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [INT-MLR-405](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-04 — BLE Link-Loss Recovery:** PASS. Forces physical link loss and verifies automatic rediscovery/reconnection without web restart.

SWE-CONN-MLR-500 — Three-Component Computer Architecture

Result PASS
Level / component MLR / Computer BLE Connection Process
Jira [SCRUM-523 — open requirement](#)

Current Jira description

The computer application shall consist of a MySQL database, a web application/server, and a third-box connection component.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-110](#), [SYS-HLR-500](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.

SWE-CONN-MLR-501 — Separate BLE Connection Process

Result PASS
Level / component MLR / Computer BLE Connection Process
Jira [SCRUM-524 — open requirement](#)

Current Jira description

The third-box BLE connection component shall run in separately (I.E: via different operating-system process or thread) from the web UI/server.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-500](#)

Children: [SWE-CONN-LLR-500](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.

SWE-CONN-MLR-502 — One-Hertz Third-Box Polling

Result PASS
Level / component MLR / Computer BLE Connection Process
Jira [SCRUM-525 — open requirement](#)

Current Jira description

The third-box connection process shall request current data from the third box at a nominal 1 Hz rate and publish each result to MySQL.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-500](#)

Children: [SWE-CONN-LLR-501](#), [SWE-CONN-LLR-502](#), [SWE-CONN-LLR-514](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.

SWE-CONN-MLR-503 — Startup History Synchronization

Result PASS
Level / component MLR / Computer BLE Connection Process
Jira [SCRUM-526](#) — open requirement

Current Jira description

On application startup or BLE reconnection, the connection process shall retrieve available third-box history and persist it so the most recent 300 seconds can be displayed within the stakeholder time limit.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: SYS-HLR-600, SYS-HLR-520

Children: SWE-CONN-LLR-505, SWE-CONN-LLR-506, SWE-CONN-LLR-507, SWE-CONN-LLR-508, SWE-CONN-LLR-509

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.
- **BE-02 — History Synchronization:** PASS. Runs deterministic 300-record metadata/chunk/retry/reconstruction/upsert/no-duplicate and time-budget tests.
- **BE-03 — Power-On Recovery:** PASS. Measures fake-device appearance through discovery, connection, current poll, and publication against 10 seconds.
- **BE-04 — Direct Backend Control Command Lifecycle:** PASS. Verifies validated commands use the direct backend/BLE path, are serialized, return terminal results, and do not create a MySQL command queue.

SWE-CONN-MLR-504 — Unavailable-Data Recording

Result PASS
Level / component MLR / Computer BLE Connection Process
Jira [SCRUM-527 — open requirement](#)

Current Jira description

While third-box data is unavailable, the computer software shall preserve the one-second time base and represent missing data distinctly from valid sensor measurements.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-500](#), [SYS-HLR-600](#)

Children: [SWE-CONN-LLR-503](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.

SWE-CONN-MLR-505 — Control Command Relay

Result PASS
Level / component MLR / Computer BLE Connection Process
Jira [SCRUM-528 — open requirement](#)

Current Jira description

The web application and BLE process shall exchange remote third-box control requests through a defined inter-process mechanism that supports the less-than-one-second response requirement.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-510](#)

Children: [SWE-CONN-LLR-510](#), [SWE-CONN-LLR-511](#), [SWE-CONN-LLR-512](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.

SWE-CONN-MLR-506 — Automatic BLE Reconnect

Result PASS
Level / component MLR / Computer BLE Connection Process
Jira [SCRUM-529](#) — open requirement

Current Jira description

The connection process shall automatically detect, reconnect to, and resume polling a compatible third box after it becomes available.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-520](#)

Children: [SWE-CONN-LLR-504](#), [SWE-CONN-LLR-513](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.
- **BE-03 — Power-On Recovery:** PASS. Measures fake-device appearance through discovery, connection, current poll, and publication against 10 seconds.

SWE-DB-MLR-550 — MySQL Persistent Storage

Result PASS
Level / component MLR / MySQL Database
Jira [SCRUM-530 — open requirement](#)

Current Jira description

MySQL shall provide persistent storage for temperature history and shared state used by the computer application.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-110](#), [SYS-HLR-600](#)

Children: [SWE-DB-LLR-553](#), [SWE-DB-LLR-561](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **DB-02 — Samples, Uniqueness, Indexes, and Retention:** PASS. Seeds an isolated database and verifies identity uniqueness, indexed latest/history plans, 300-second minimum retention, and >600-second eligibility.

SWE-DB-MLR-551 — Temperature Sample Schema

Result PASS
Level / component MLR / MySQL Database
Jira [SCRUM-531 — open requirement](#)

Current Jira description

The database shall persist one-second temperature records for three series: probe 1, probe 2, and the computed average of the two, with per-series value and status, third-box device and session identity, and sample identity. Series status shall distinguish valid readings from unplugged-sensor, no-data, provisional, and derived-unavailable conditions.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-500](#), [SYS-HLR-600](#)

Children: [SWE-DB-LLR-550](#), [SWE-DB-LLR-551](#), [SWE-DB-LLR-552](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **DB-02 — Samples, Uniqueness, Indexes, and Retention:** PASS. Seeds an isolated database and verifies identity uniqueness, indexed latest/history plans, 300-second minimum retention, and >600-second eligibility.

SWE-DB-MLR-552 — Configuration and Authorization Storage

Result PASS
Level / component MLR / MySQL Database
Jira [SCRUM-532 — open requirement](#)

Current Jira description

The database shall persist temperature-sample history and email alert recipients/rules required by the web application. Application user roles and third-box control-command state are not persisted because this is a trusted local application with no user accounts and control commands are sent directly to the Python BLE backend.

Revision note

Edited requirement. The command path was revised so MySQL stores history while display commands go directly to the Python BLE backend. Application accounts and role-based access were removed; the released system uses a trusted local deployment boundary. The description above is the newest frozen version used by this run.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: SYS-HLR-510, SYS-HLR-700

Children: SWE-DB-LLR-554, SWE-DB-LLR-555, SWE-DB-LLR-556

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **DB-03 — History-Only Database and Direct Command Boundary:** PASS. Validates that MySQL stores history/email configuration but no command queue/state, while commands use the direct backend API.

SWE-DB-MLR-553 — Control Command Queue

Result PASS
Level / component MLR / MySQL Database
Jira [SCRUM-533 — open requirement](#)

Current Jira description

MySQL shall be the authoritative source for historical temperature samples displayed by the web application. Third-box control commands shall be sent directly from the web application to the Python BLE backend and shall not pass through a MySQL command queue. Verification: issue a display-toggle command via the web UI and confirm the third-box display changes in under 1 second.

Revision note

Edited requirement. The command path was revised so MySQL stores history while display commands go directly to the Python BLE backend. The description above is the newest frozen version used by this run.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-510](#)

Children: [SWE-DB-LLR-557](#), [SWE-DB-LLR-558](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **DB-03 — History-Only Database and Direct Command Boundary:** PASS. Validates that MySQL stores history/email configuration but no command queue/state, while commands use the direct backend API.

SWE-DB-MLR-554 — Canonical Time and Units

Result PASS
Level / component MLR / MySQL Database
Jira [SCRUM-534](#) — open requirement

Current Jira description

Persisted temperature values shall use Celsius as the canonical unit, with conversion to Fahrenheit performed at presentation only. This supports the user-selectable display units required by {Req 5.a} and {Req 5.c.i}, and the third-box display in {Req 4}. Temperatures shall be stored with fixed-point precision sufficient to cover the {Req 8.a} design range of -10 to $+63$ °C without rounding error affecting alert threshold comparisons. Application-created timestamps shall be stored in UTC. This is a team design decision rather than a spec-derived requirement {Req 5.c.iii} labels the graph axis in relative "seconds ago" and no wall-clock time is displayed. UTC is retained because local time is not monotonic across timezone transitions, and both the history index (**SWE-DB-LLR-552**) and the time-range replacement in **SWE-CONN-LLR-509** assume ordered timestamps.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: SYS-HLR-500, SYS-HLR-600, SYS-HLR-700

Children: SWE-DB-LLR-559, SWE-DB-LLR-560

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **DB-02 — Samples, Uniqueness, Indexes, and Retention:** PASS. Seeds an isolated database and verifies identity uniqueness, indexed latest/history plans, 300-second minimum retention, and >600-second eligibility.

SWE-CONN-LLR-500 — Connection Worker Executable

Result PASS
Level / component LLR / Computer BLE Connection Process
Jira [SCRUM-535](#) — open requirement

Current Jira description

The BLE connection component shall start as an independent process/service with its own main loop and database connection.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-CONN-MLR-501](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.

SWE-CONN-LLR-501 — One-Hertz Poll Scheduler

Result PASS
Level / component LLR / Computer BLE Connection Process
Jira [SCRUM-536](#) — open requirement

Current Jira description

The connection worker shall issue GET_CURRENT transactions at 1 Hz, scheduled against a monotonic clock so that sample spacing is unaffected by system time adjustments. Scheduled poll times shall not deviate by more than ± 100 ms from one-second boundaries. Verification: log poll timestamps over 300 seconds and confirm the count is 300 ± 1 and no interval exceeds the stated tolerance.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-CONN-MLR-502](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.
- **BE-02 — History Synchronization:** PASS. Runs deterministic 300-record metadata/chunk/retry/reconstruction/upsert/no-duplicate and time-budget tests.

SWE-CONN-LLR-502 — Successful Poll Persistence

Result PASS
Level / component LLR / Computer BLE Connection Process
Jira [SCRUM-537 — open requirement](#)

Current Jira description

For each valid GET_CURRENT response, the computer BLE connection worker shall store the received sensor temperatures, statuses, display states, boot identifier, and sample sequence number in the database. The worker shall also store the UTC response-receive time and the average temperature, calculated from the same sample when both sensor readings are valid.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-CONN-MLR-502](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.
- **BE-04 — Direct Backend Control Command Lifecycle:** PASS. Verifies validated commands use the direct backend/BLE path, are serialized, return terminal results, and do not create a MySQL command queue.

SWE-CONN-LLR-503 — Missing Poll Persistence

Result PASS
Level / component LLR / Computer BLE Connection Process
Jira [SCRUM-538 — open requirement](#)

Current Jira description

If a scheduled 1 Hz poll does not yield a valid reading, the worker shall persist a record for that one-second interval rather than repeating the previous value, so gaps remain distinguishable from valid data {Req 5.c.iv} and the graph continues scrolling {Req 5.c.v}. Missed-poll records shall be marked provisional, indicating an interval for which no third-box reading was obtained and which may be superseded if that data is later recovered. The worker cannot determine at poll time whether the box is powered off or merely unreachable, so no distinction is made when the record is written. Resolution of provisional records on reconnection is specified in ****SWE-CONN-LLR-509****. Verification: interrupt the third-box connection during operation and confirm records continue to be written at 1 Hz marked provisional, with no value carried forward from the previous reading.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-CONN-MLR-504](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.
- **BE-03 — Power-On Recovery:** PASS. Measures fake-device appearance through discovery, connection, current poll, and publication against 10 seconds.

SWE-CONN-LLR-504 — Reconnect Retry Interval

Result PASS
Level / component LLR / Computer BLE Connection Process
Jira [SCRUM-539 — open requirement](#)

Current Jira description

After disconnection, the worker shall retry compatible-device discovery/connection frequently enough that a newly powered third box is detected within 10 seconds.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-CONN-MLR-506](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.
- **BE-03 — Power-On Recovery:** PASS. Measures fake-device appearance through discovery, connection, current poll, and publication against 10 seconds.

SWE-CONN-LLR-505 — History Sync Trigger

Result PASS
Level / component LLR / Computer BLE Connection Process
Jira [SCRUM-540 — open requirement](#)

Current Jira description

The worker shall initiate a history synchronization after first connection and after reconnecting to a new boot identifier.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-CONN-MLR-503](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.
- **BE-02 — History Synchronization:** PASS. Runs deterministic 300-record metadata/chunk/retry/reconstruction/upsert/no-duplicate and time-budget tests.
- **BE-03 — Power-On Recovery:** PASS. Measures fake-device appearance through discovery, connection, current poll, and publication against 10 seconds.

SWE-CONN-LLR-506 — History Metadata First

Result PASS
Level / component LLR / Computer BLE Connection Process
Jira [SCRUM-541 — open requirement](#)

Current Jira description

Before requesting history chunk data, the worker shall request history metadata from the third box, comprising: the boot/session identifier, the oldest and newest available sample sequence numbers, and the sample count available per sensor. The worker shall use this to determine which chunks are required rather than fetching blindly. If metadata cannot be obtained within [N] seconds, synchronization shall abort and retry rather than proceeding to chunk requests, so as not to exceed the 10-second startup budget in {Req 5.c}. Verification: initiate synchronization with a populated third-box buffer and confirm a metadata request precedes any chunk request, and that requested chunk ranges fall within the reported available range.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-CONN-MLR-503](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.
- **BE-02 — History Synchronization:** PASS. Runs deterministic 300-record metadata/chunk/retry/reconstruction/upsert/no-duplicate and time-budget tests.

SWE-CONN-LLR-507 — History Chunk Loop

Result PASS
Level / component LLR / Computer BLE Connection Process
Jira [SCRUM-542 — open requirement](#)

Current Jira description

Following metadata retrieval, the worker shall request history chunks until the most recent 300 seconds of available records have been obtained, or until the synchronization time budget is exhausted. The combined metadata and chunk-retrieval budget shall not exceed [N] seconds, preserving the 10-second startup requirement in {Req 5.c}. Individual chunk requests shall be retried up to 2 times on failure. If the budget expires before the full 300 seconds is retrieved, synchronization shall persist the records obtained and leave the remaining interval as a gap rather than discarding partial history; gaps remain distinguishable per {Req 5.c.iv}. Verification: initiate synchronization against a full third-box buffer and confirm 300 seconds of records are retrieved within the budget; repeat with an artificially delayed link and confirm partial history is persisted and the budget is not exceeded.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-CONN-MLR-503](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.
- **BE-02 — History Synchronization:** PASS. Runs deterministic 300-record metadata/chunk/retry/reconstruction/upsert/no-duplicate and time-budget tests.

SWE-CONN-LLR-508 — History Timestamp Reconstruction

Result PASS
Level / component LLR / Computer BLE Connection Process
Jira [SCRUM-543 — open requirement](#)

Current Jira description

For third-box records lacking absolute timestamps, the worker shall reconstruct observation times by anchoring the newest recovered sample to the time its response was received and assigning preceding samples at one-second intervals by descending sample sequence. Reconstructed timestamps are approximate; accumulated error across a 300-second recovery shall not exceed 2 seconds, consistent with the "seconds ago" axis labelling in {Req 5.c.iii}. Where a reconstructed timestamp overlaps an interval already covered by live polling, the live sample shall take precedence. Verification: recover a known-length history and confirm reconstructed timestamps are monotonically decreasing at one-second spacing, that the newest aligns with the first live sample without duplication or gap, and that total span matches the recovered sample count.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-CONN-MLR-503](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.
- **BE-02 — History Synchronization:** PASS. Runs deterministic 300-record metadata/chunk/retry/reconstruction/upsert/no-duplicate and time-budget tests.

SWE-CONN-LLR-509 — History De-Duplication

Result PASS
Level / component LLR / Computer BLE Connection Process
Jira [SCRUM-544 — open requirement](#)

Current Jira description

On history synchronization, the worker shall upsert recovered samples keyed on device, boot, and sample identity so that repeated synchronization does not create duplicate records. Provisional records written by ****SWE-CONN-LLR-503**** carry no third-box sample sequence and shall be resolved by time range rather than by upsert: * If the recovered boot identity matches the session active before the gap, provisional records within the recovered range shall be replaced by the recovered samples. * If the boot identity differs, the third box was power-cycled and no data exists for the gap; provisional records preceding the new session's first sample shall be reclassified as no-data {Req 5.a.ii}. Verification: (a) synchronize history twice in succession and confirm record count is unchanged; (b) drop the connection with the box powered, reconnect, and confirm provisional records are replaced by recovered samples; (c) power-cycle the box and confirm provisional records for the off period become no-data.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-CONN-MLR-503](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.
- **BE-02 — History Synchronization:** PASS. Runs deterministic 300-record metadata/chunk/retry/reconstruction/upsert/no-duplicate and time-budget tests.
- **BE-03 — Power-On Recovery:** PASS. Measures fake-device appearance through discovery, connection, current poll, and publication against 10 seconds.

SWE-CONN-LLR-510 — Pending Command Poll

Result PASS
Level / component LLR / Computer BLE Connection Process
Jira [SCRUM-545 — open requirement](#)

Current Jira description

The connection worker shall accept validated direct control requests from the backend API and schedule them promptly ahead of background history synchronization so the end-to-end display response can remain under 1 second. Verification: issue display toggles from the web UI across ten trials and measure browser-to-third-box response latency.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-CONN-MLR-505](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.
- **BE-04 — Direct Backend Control Command Lifecycle:** PASS. Verifies validated commands use the direct backend/BLE path, are serialized, return terminal results, and do not create a MySQL command queue.

SWE-CONN-LLR-511 — Command Claiming

Result PASS
Level / component LLR / Computer BLE Connection Process
Jira [SCRUM-546 — open requirement](#)

Current Jira description

The backend shall serialize each accepted direct display-control operation, execute it once through the BLE connection worker, and report success or failure to the caller. A connection loss or timeout shall produce an explicit failed result rather than leaving an operation indefinitely active. Verification: submit a command, confirm one BLE execution and a terminal response, then repeat with the third box unavailable and confirm failure is reported.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-CONN-MLR-505](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.
- **BE-04 — Direct Backend Control Command Lifecycle:** PASS. Verifies validated commands use the direct backend/BLE path, are serialized, return terminal results, and do not create a MySQL command queue.

SWE-CONN-LLR-512 — Command Completion

Result PASS
Level / component LLR / Computer BLE Connection Process
Jira [SCRUM-547 — open requirement](#)

Current Jira description

After executing a direct remote display command, the backend shall return command success or failure and the resulting third-box display state through the backend API and diagnostic logs. Command completion is not required to be stored in MySQL.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-CONN-MLR-505](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.
- **BE-04 — Direct Backend Control Command Lifecycle:** PASS. Verifies validated commands use the direct backend/BLE path, are serialized, return terminal results, and do not create a MySQL command queue.

SWE-CONN-LLR-513 — Connection State Publication

Result PASS
Level / component LLR / Computer BLE Connection Process
Jira [SCRUM-548 — open requirement](#)

Current Jira description

The worker shall update a database device-state record with connected/unavailable state and last-seen time whenever BLE availability changes.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-CONN-MLR-506](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.

SWE-CONN-LLR-514 — Polling Failure Isolation

Result PASS
Level / component LLR / Computer BLE Connection Process
Jira [SCRUM-549 — open requirement](#)

Current Jira description

A BLE poll failure or database write failure shall be handled as an error for that cycle and shall not terminate the worker process without a controlled fatal condition.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-CONN-MLR-502](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.

SWE-DB-LLR-550 — temperature_samples Table

Result PASS
Level / component LLR / MySQL Database
Jira [SCRUM-550 — open requirement](#)

Current Jira description

Create a temperature_samples table containing device_id, boot_id, sample_seq, observed_at_utc, sensor1_c, sensor1_status, sensor2_c, sensor2_status, average_c, average_valid, and record_source fields.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-DB-MLR-551](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **DB-01 — Schema, Types, Email Configuration, UTC, and Secrets:** PASS. Inspects isolated MySQL schema types, email-only alert models, deliberate absence of user/command tables, UTC, and external credentials.

SWE-DB-LLR-551 — Sample Uniqueness

Result PASS
Level / component LLR / MySQL Database
Jira [SCRUM-551 — open requirement](#)

Current Jira description

Re-inserting a temperature sample that has already been stored shall not create a duplicate record. Uniqueness shall be enforced by a database constraint on third-box device identity, session identity, and sample sequence extended to include series identity if series are stored as separate rows. Verification: attempt to insert a record matching an existing key and confirm the database rejects it without creating a second row.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-DB-MLR-551](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **DB-02 — Samples, Uniqueness, Indexes, and Retention:** PASS. Seeds an isolated database and verifies identity uniqueness, indexed latest/history plans, 300-second minimum retention, and >600-second eligibility.

SWE-DB-LLR-552 — History Index

Result PASS
Level / component LLR / MySQL Database
Jira [SCRUM-552 — open requirement](#)

Current Jira description

The temperature_samples table shall carry an index on (device_id, observed_at_utc) supporting the two required access patterns: retrieval of the most recent sample per series and retrieval of all samples within a bounded time window of up to 300 seconds. If series are stored as separate rows, series identity shall be included in the index. Verification: EXPLAIN on both query forms shows the index in use rather than a full table scan.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-DB-MLR-551](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **DB-02 — Samples, Uniqueness, Indexes, and Retention:** PASS. Seeds an isolated database and verifies identity uniqueness, indexed latest/history plans, 300-second minimum retention, and >600-second eligibility.

SWE-DB-LLR-553 — Persistent Retention

Result PASS
Level / component LLR / MySQL Database
Jira [SCRUM-553 — open requirement](#)

Current Jira description

Temperature samples shall be retained for a minimum of 300 seconds after the observed timestamp. Samples older than 600 seconds shall be eligible for automatic deletion. The retention floor shall exceed the maximum user-selectable graph window (300s) with margin sufficient to avoid contention between deletion and history queries.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-DB-MLR-550](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **DB-02 — Samples, Uniqueness, Indexes, and Retention:** PASS. Seeds an isolated database and verifies identity uniqueness, indexed latest/history plans, 300-second minimum retention, and >600-second eligibility.

SWE-DB-LLR-554 — users Table

Result PASS
Level / component LLR / MySQL Database
Jira [SCRUM-554 — open requirement](#)

Current Jira description

The database schema shall not store application user accounts, passwords, password hashes, or USER/ADMIN roles. The application is operated as a trusted local tool and does not implement user accounts. Verification: inspect the schema and confirm no users or credential table is present.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-DB-MLR-552](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **DB-01 — Schema, Types, Email Configuration, UTC, and Secrets:** PASS. Inspects isolated MySQL schema types, email-only alert models, deliberate absence of user/command tables, UTC, and external credentials.

SWE-DB-LLR-555 — alert_recipients Table

Result PASS
Level / component LLR / MySQL Database
Jira [SCRUM-555 — open requirement](#)

Current Jira description

Create an alert_recipients table storing an email destination address and enabled/configuration audit fields needed by the alerting implementation. The table shall support multiple email recipients. SMS recipients are out of scope. Verification: insert multiple email recipients and confirm all persist and are retrievable by alerting logic.

Revision note

Edited requirement. Notification scope was narrowed to email only; SMS delivery is not part of this release. The description above is the newest frozen version used by this run.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-DB-MLR-552](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **DB-01 — Schema, Types, Email Configuration, UTC, and Secrets:** PASS. Inspects isolated MySQL schema types, email-only alert models, deliberate absence of user/command tables, UTC, and external credentials.

SWE-DB-LLR-556 — alert_rules Table

Result PASS
Level / component LLR / MySQL Database
Jira [SCRUM-556 — open requirement](#)

Current Jira description

Create an alert_rules table storing a minimum threshold, maximum threshold, high-temperature message, low-temperature message, and the monitored series, linked to one or more alert_recipients. Verification: create a rule with both thresholds and messages, confirm all five editable fields persist and are retrievable.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-DB-MLR-552](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **DB-01 — Schema, Types, Email Configuration, UTC, and Secrets:** PASS. Inspects isolated MySQL schema types, email-only alert models, deliberate absence of user/command tables, UTC, and external credentials.

SWE-DB-LLR-557 — control_commands Table

Result PASS
Level / component LLR / MySQL Database
Jira [SCRUM-557 — open requirement](#)

Current Jira description

The database schema shall not require or create a control_commands table. Third-box display commands shall be validated and sent directly to the Python BLE backend. Verification: confirm the command table is absent and exercise a display request through the direct backend endpoint.

Revision note

Edited requirement. The command path was revised so MySQL stores history while display commands go directly to the Python BLE backend. The description above is the newest frozen version used by this run.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-DB-MLR-553](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **DB-03 — History-Only Database and Direct Command Boundary:** PASS. Validates that MySQL stores history/email configuration but no command queue/state, while commands use the direct backend API.

SWE-DB-LLR-558 — Command Status Values

Result PASS
Level / component LLR / MySQL Database
Jira [SCRUM-558 — open requirement](#)

Current Jira description

Direct control operations shall return an explicit accepted/completed or failed result through the backend API, including timeout or unavailable-device failures. Database command states and expiry are not required because commands are not queued in MySQL. Verification: submit a command with the third box unavailable and confirm the API terminates with an actionable failure.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-DB-MLR-553](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **DB-03 — History-Only Database and Direct Command Boundary:** PASS. Validates that MySQL stores history/email configuration but no command queue/state, while commands use the direct backend API.

SWE-DB-LLR-559 — Celsius Storage

Result PASS
Level / component LLR / MySQL Database
Jira [SCRUM-559 — open requirement](#)

Current Jira description

All numeric threshold and temperature fields stored in MySQL shall use Celsius regardless of the user-selected display unit.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-DB-MLR-554](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **DB-01 — Schema, Types, Email Configuration, UTC, and Secrets:** PASS. Inspects isolated MySQL schema types, email-only alert models, deliberate absence of user/command tables, UTC, and external credentials.

SWE-DB-LLR-560 — UTC Timestamp Storage

Result PASS
Level / component LLR / MySQL Database
Jira [SCRUM-560 — open requirement](#)

Current Jira description

All timestamps stored in the database shall use UTC, including sample observation times and alert-configuration audit timestamps. This applies to application-generated timestamps and database defaults; the database server timezone shall be UTC. Conversion to local time is presentation-only.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-DB-MLR-554](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **DB-03 — History-Only Database and Direct Command Boundary:** PASS. Validates that MySQL stores history/email configuration but no command queue/state, while commands use the direct backend API.

SWE-DB-LLR-561 — Database Credentials

Result PASS
Level / component LLR / MySQL Database
Jira [SCRUM-561 — open requirement](#)

Current Jira description

Database credentials and secrets shall be supplied through environment/configuration outside version-controlled source files.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-DB-MLR-550](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **DB-01 — Schema, Types, Email Configuration, UTC, and Secrets:** PASS. Inspects isolated MySQL schema types, email-only alert models, deliberate absence of user/command tables, UTC, and external credentials.

SWE-WEB-MLR-600 — Computer-Hosted Web Application

Result PASS
Level / component MLR / Web Application
Jira [SCRUM-562 — open requirement](#)

Current Jira description

The computer shall host a web application/server that provides the browser UI and mediates access to database-backed temperature, configuration, and control functions.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-110](#), [SYS-HLR-500](#)

Children: [SWE-WEB-LLR-600](#), [SWE-WEB-LLR-601](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-07 — Remote Display Control UI:** PASS. Tests independent controls, pending/completed/failed states, authoritative returned state, and no optimistic success.

SWE-WEB-MLR-601 — One-Hertz UI Data Refresh

Result PASS
Level / component MLR / Web Application
Jira [SCRUM-563](#) — open requirement

Current Jira description

The web UI shall obtain current temperature/status data from MySQL at a nominal 1 Hz rate.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-500](#)

Children: [SWE-WEB-LLR-604](#), [SWE-WEB-LLR-605](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-01 — Latest and History API Contract:** PASS. Verifies latest/history response fields, bounds, ordering, statuses, missing markers, average, availability, and display state.

SWE-WEB-MLR-602 — Responsive User/Admin Modes

Result PASS
Level / component MLR / Web Application
Jira [SCRUM-564 — open requirement](#)

Current Jira description

The web application shall be usable on desktop and mobile displays. It is a trusted local application and shall not implement application user accounts or USER/ADMIN roles. Deployment shall restrict exposed services to the intended local host/network boundary.

Revision note

Edited requirement. Application accounts and role-based access were removed; the released system uses a trusted local deployment boundary. The description above is the newest frozen version used by this run.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-500](#), [SYS-HLR-510](#), [SYS-HLR-700](#)

Children: [SWE-WEB-LLR-606](#), [SWE-WEB-LLR-607](#), [SWE-WEB-LLR-608](#), [SWE-WEB-LLR-609](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-05 — Trusted-Local Access Boundary and Validation:** PASS. Checks deliberate absence of application accounts/credentials, loopback-scoped deployment, CORS restrictions, and server-side command/email validation.
- **FE-07 — Remote Display Control UI:** PASS. Tests independent controls, pending/completed/failed states, authoritative returned state, and no optimistic success.

SWE-WEB-MLR-603 — Settings Page

Result PASS
Level / component MLR / Web Application
Jira [SCRUM-565 — open requirement](#)

Current Jira description

The UI shall provide settings for configuring multiple email recipients and alert thresholds/messages. SMS configuration is out of scope.

Revision note

Edited requirement. Notification scope was narrowed to email only; SMS delivery is not part of this release. The description above is the newest frozen version used by this run.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-700](#)

Children: [SWE-WEB-LLR-610](#), [SWE-WEB-LLR-611](#), [SWE-WEB-LLR-612](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-06 — Email Alert Configuration and Delivery:** PASS. Checks email-only recipients/rules, email validation, configurable thresholds/messages, Gmail-compatible SMTP delivery, and persistence schema.

SWE-WEB-MLR-604 — Main Graph Page and Series Selection

Result PASS
Level / component MLR / Web Application
Jira [SCRUM-566 — open requirement](#)

Current Jira description

The main view shall be a graph page supporting simultaneous selection of Sensor 1, Sensor 2, and Average series, with current values for selected series.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-500](#), [SYS-HLR-600](#)

Children: [SWE-WEB-LLR-613](#), [SWE-WEB-LLR-614](#), [SWE-WEB-LLR-615](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.

SWE-WEB-MLR-605 — Celsius/Fahrenheit Presentation

Result PASS
Level / component MLR / Web Application
Jira [SCRUM-567 — open requirement](#)

Current Jira description

The web UI shall allow the user to display real-time and historical temperatures in Celsius or Fahrenheit.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-500](#), [SYS-HLR-600](#)

Children: [SWE-WEB-LLR-616](#), [SWE-WEB-LLR-617](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.
- **FE-05 — Trusted-Local Access Boundary and Validation:** PASS. Checks deliberate absence of application accounts/credentials, loopback-scoped deployment, CORS restrictions, and server-side command/email validation.

SWE-WEB-MLR-606 — Missing/Unavailable Presentation

Result PASS
Level / component MLR / Web Application
Jira [SCRUM-568 — open requirement](#)

Current Jira description

The UI shall distinguish disconnected sensors, unavailable third-box data, missing graph data, and valid off-scale graph data.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-500](#), [SYS-HLR-600](#)

Children: [SWE-WEB-LLR-618](#), [SWE-WEB-LLR-619](#), [SWE-WEB-LLR-620](#), [SWE-WEB-LLR-621](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.

SWE-WEB-MLR-607 — Control Page

Result PASS
Level / component MLR / Web Application
Jira [SCRUM-569 — open requirement](#)

Current Jira description

The UI shall provide a local control page or section for remotely selecting the temperature information displayed on the third-box LCD. No application ADMIN account is required.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-510](#)

Children: [SWE-WEB-LLR-602](#), [SWE-WEB-LLR-603](#), [SWE-WEB-LLR-622](#), [SWE-WEB-LLR-623](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-05 — Trusted-Local Access Boundary and Validation:** PASS. Checks deliberate absence of application accounts/credentials, loopback-scoped deployment, CORS restrictions, and server-side command/email validation.
- **FE-07 — Remote Display Control UI:** PASS. Tests independent controls, pending/completed/failed states, authoritative returned state, and no optimistic success.

SWE-WEB-MLR-608 — Compliance Graph Y-Axis

Result PASS
Level / component MLR / Web Application
Jira [SCRUM-570 — open requirement](#)

Current Jira description

The stakeholder-required graph shall use fixed limits of 10 to 50 deg C or 50 to 122 deg F.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-600](#)

Children: [SWE-WEB-LLR-624](#), [SWE-WEB-LLR-625](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.

SWE-WEB-MLR-610 — Compliance Graph Time Window

Result PASS
Level / component MLR / Web Application
Jira [SCRUM-572 — open requirement](#)

Current Jira description

The stakeholder-required graph shall display a total time record of 300 seconds.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-600](#)

Children: [SWE-WEB-LLR-626](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.

SWE-WEB-LLR-600 — Current Data API

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-574 — open requirement](#)

Current Jira description

The web server shall provide a read-only API endpoint that returns the latest database-backed Sensor 1, Sensor 2, average, sensor-status, box-availability, and display-state data.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-600](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-01 — Latest and History API Contract:** PASS. Verifies latest/history response fields, bounds, ordering, statuses, missing markers, average, availability, and display state.
- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.

SWE-WEB-LLR-601 — History API

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-575 — open requirement](#)

Current Jira description

The web server shall provide a read-only history endpoint accepting a requested duration up to the permitted limit and returning timestamped sensor/average values and status/missing markers.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-600](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-01 — Latest and History API Contract:** PASS. Verifies latest/history response fields, bounds, ordering, statuses, missing markers, average, availability, and display state.
- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.

SWE-WEB-LLR-602 — Display Control API

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-576 — open requirement](#)

Current Jira description

The web/backend service shall provide an endpoint that validates a sensor display request and sends the corresponding command directly to the Python BLE backend. It shall not insert commands into MySQL and does not require application-user authentication in the trusted local deployment.

Revision note

Edited requirement. The command path was revised so MySQL stores history while display commands go directly to the Python BLE backend. The description above is the newest frozen version used by this run.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-607](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-05 — Trusted-Local Access Boundary and Validation:** PASS. Checks deliberate absence of application accounts/credentials, loopback-scoped deployment, CORS restrictions, and server-side command/email validation.
- **FE-07 — Remote Display Control UI:** PASS. Tests independent controls, pending/completed/failed states, authoritative returned state, and no optimistic success.

SWE-WEB-LLR-603 — Command Status API

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-577 — open requirement](#)

Current Jira description

The web/backend service shall expose the success, failure, timeout, and resulting display state of a submitted direct third-box control command through its API. No ADMIN authorization or database command-status record is required in the trusted local deployment.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-607](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-05 — Trusted-Local Access Boundary and Validation:** PASS. Checks deliberate absence of application accounts/credentials, loopback-scoped deployment, CORS restrictions, and server-side command/email validation.
- **FE-07 — Remote Display Control UI:** PASS. Tests independent controls, pending/completed/failed states, authoritative returned state, and no optimistic success.

SWE-WEB-LLR-604 — Browser Poll Timer

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-578 — open requirement](#)

Current Jira description

The graph/current-value client shall request current data from the web server every 1000 ms while the page is active.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-601](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-02 — Live Readout and Stale-Data Behavior:** PASS. Exercises valid/change/disconnected/box-off scenarios, 1-second refresh, stale-number removal, exact unavailable text, and prominent values.
- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.

SWE-WEB-LLR-605 — No Stale Current Value

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-579 — open requirement](#)

Current Jira description

When the current API reports box unavailable or sensor invalid, the browser shall replace the prior numeric current value with the appropriate status text instead of retaining it as current.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-601](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-01 — Latest and History API Contract:** PASS. Verifies latest/history response fields, bounds, ordering, statuses, missing markers, average, availability, and display state.

SWE-WEB-LLR-606 — Responsive Layout

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-580 — open requirement](#)

Current Jira description

The graph, current-value cards, settings forms, and control page shall use responsive layout rules that avoid horizontal overflow at common phone and desktop viewport widths.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-602](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-02 — Live Readout and Stale-Data Behavior:** PASS. Exercises valid/change/disconnected/box-off scenarios, 1-second refresh, stale-number removal, exact unavailable text, and prominent values.
- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.
- **FE-04 — Responsive Layout:** PASS. Checks required controls and horizontal overflow expectations for desktop, laptop, and two mobile viewport definitions.
- **FE-06 — Email Alert Configuration and Delivery:** PASS. Checks email-only recipients/rules, email validation, configurable thresholds/messages, Gmail-compatible SMTP delivery, and persistence schema.
- **FE-07 — Remote Display Control UI:** PASS. Tests independent controls, pending/completed/failed states, authoritative returned state, and no optimistic success.

SWE-WEB-LLR-607 — Role Model

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-581 — open requirement](#)

Current Jira description

The application shall operate without application user accounts or USER/ADMIN roles. The operator is trusted by the local deployment boundary. Verification: confirm no login, user-role model, or account database is required for normal operation.

Revision note

Edited requirement. Application accounts and role-based access were removed; the released system uses a trusted local deployment boundary. The description above is the newest frozen version used by this run.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-602](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-05 — Trusted-Local Access Boundary and Validation:** PASS. Checks deliberate absence of application accounts/credentials, loopback-scoped deployment, CORS restrictions, and server-side command/email validation.

SWE-WEB-LLR-608 — User Permissions

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-582 — open requirement](#)

Current Jira description

A local operator shall be permitted to view current temperatures and graphs, control the third-box displays, and configure email alerts. The application does not distinguish USER and ADMIN roles.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-602](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.
- **FE-05 — Trusted-Local Access Boundary and Validation:** PASS. Checks deliberate absence of application accounts/credentials, loopback-scoped deployment, CORS restrictions, and server-side command/email validation.
- **FE-06 — Email Alert Configuration and Delivery:** PASS. Checks email-only recipients/rules, email validation, configurable thresholds/messages, Gmail-compatible SMTP delivery, and persistence schema.
- **FE-07 — Remote Display Control UI:** PASS. Tests independent controls, pending/completed/failed states, authoritative returned state, and no optimistic success.

SWE-WEB-LLR-609 — Admin Permissions

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-583](#) — open requirement

Current Jira description

A local operator shall be permitted to use third-box control and email notification/settings functions without an application ADMIN account.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-602](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-05 — Trusted-Local Access Boundary and Validation:** PASS. Checks deliberate absence of application accounts/credentials, loopback-scoped deployment, CORS restrictions, and server-side command/email validation.
- **FE-06 — Email Alert Configuration and Delivery:** PASS. Checks email-only recipients/rules, email validation, configurable thresholds/messages, Gmail-compatible SMTP delivery, and persistence schema.
- **FE-07 — Remote Display Control UI:** PASS. Tests independent controls, pending/completed/failed states, authoritative returned state, and no optimistic success.

SWE-WEB-LLR-610 — Recipient List UI

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-584 — open requirement](#)

Current Jira description

The settings page shall display all configured email recipients and allow the local operator to add, edit, enable/disable, and remove recipients. SMS recipients and ADMIN authorization are out of scope.

Revision note

Edited requirement. Notification scope was narrowed to email only; SMS delivery is not part of this release. The description above is the newest frozen version used by this run.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-603](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-05 — Trusted-Local Access Boundary and Validation:** PASS. Checks deliberate absence of application accounts/credentials, loopback-scoped deployment, CORS restrictions, and server-side command/email validation.
- **FE-06 — Email Alert Configuration and Delivery:** PASS. Checks email-only recipients/rules, email validation, configurable thresholds/messages, Gmail-compatible SMTP delivery, and persistence schema.

SWE-WEB-LLR-611 — Per-Recipient Threshold UI

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-585 — open requirement](#)

Current Jira description

The settings page shall allow the local operator to configure minimum and maximum Celsius-equivalent thresholds independently for each email alert recipient/rule.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-603](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.
- **FE-05 — Trusted-Local Access Boundary and Validation:** PASS. Checks deliberate absence of application accounts/credentials, loopback-scoped deployment, CORS restrictions, and server-side command/email validation.
- **FE-06 — Email Alert Configuration and Delivery:** PASS. Checks email-only recipients/rules, email validation, configurable thresholds/messages, Gmail-compatible SMTP delivery, and persistence schema.

SWE-WEB-LLR-612 — Message Configuration UI

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-586 — open requirement](#)

Current Jira description

The settings page shall allow the local operator to edit the high-temperature and low-temperature message text associated with an email alert rule.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-603](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-05 — Trusted-Local Access Boundary and Validation:** PASS. Checks deliberate absence of application accounts/credentials, loopback-scoped deployment, CORS restrictions, and server-side command/email validation.
- **FE-06 — Email Alert Configuration and Delivery:** PASS. Checks email-only recipients/rules, email validation, configurable thresholds/messages, Gmail-compatible SMTP delivery, and persistence schema.

SWE-WEB-LLR-613 — Series Toggle Controls

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-587 — open requirement](#)

Current Jira description

The graph page shall provide independent controls for displaying Sensor 1, Sensor 2, and Average, and shall permit any combination of the three.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-604](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.
- **FE-07 — Remote Display Control UI:** PASS. Tests independent controls, pending/completed/failed states, authoritative returned state, and no optimistic success.

SWE-WEB-LLR-614 — Selected Current Values

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-588 — open requirement](#)

Current Jira description

The graph page shall show a current-value indicator for each selected series and suppress or mark the average unavailable when one or both sensor readings are invalid.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-604](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-02 — Live Readout and Stale-Data Behavior:** PASS. Exercises valid/change/disconnected/box-off scenarios, 1-second refresh, stale-number removal, exact unavailable text, and prominent values.
- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.

SWE-WEB-LLR-615 — Average Calculation Source

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-589 — open requirement](#)

Current Jira description

The web application shall calculate or read Average only from valid Sensor 1 and Sensor 2 values from the same sample record.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-604](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.

SWE-WEB-LLR-616 — Canonical Conversion

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-590 — open requirement](#)

Current Jira description

The web layer shall treat Celsius database values as canonical and convert to Fahrenheit using $F = C * 9/5 + 32$ for presentation.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-605](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.

SWE-WEB-LLR-617 — Unit Switch Refresh

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-591](#) — open requirement

Current Jira description

Changing the unit selector shall update current-value labels, graph values, graph-axis labels, and configured threshold presentation without altering stored Celsius data.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-605](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-02 — Live Readout and Stale-Data Behavior:** PASS. Exercises valid/change/disconnected/box-off scenarios, 1-second refresh, stale-number removal, exact unavailable text, and prominent values.
- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.
- **FE-06 — Email Alert Configuration and Delivery:** PASS. Checks email-only recipients/rules, email validation, configurable thresholds/messages, Gmail-compatible SMTP delivery, and persistence schema.

SWE-WEB-LLR-618 — Unplugged Sensor Text

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-592 — open requirement](#)

Current Jira description

For a disconnected sensor, the current-value UI shall display a textual indication of an unplugged sensor instead of a numeric temperature. I.E: “Sensor X Unplugged”

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-606](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-02 — Live Readout and Stale-Data Behavior:** PASS. Exercises valid/change/disconnected/box-off scenarios, 1-second refresh, stale-number removal, exact unavailable text, and prominent values.

SWE-WEB-LLR-619 — No Data Available Text

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-593](#) — open requirement

Current Jira description

When the third box is OFF/unavailable, the current-value UI shall display the exact text "No data available" instead of a numeric temperature.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-606](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-02 — Live Readout and Stale-Data Behavior:** PASS. Exercises valid/change/disconnected/box-off scenarios, 1-second refresh, stale-number removal, exact unavailable text, and prominent values.

SWE-WEB-LLR-620 — Graph Missing Gap

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-594 — open requirement](#)

Current Jira description

A missing/unavailable sample shall be rendered as a visible gap and/or dedicated missing marker and shall not be connected as if it were a valid temperature sample.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-606](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.

SWE-WEB-LLR-621 — Off-Scale Marker

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-595 — open requirement](#)

Current Jira description

A valid sample below/above the visible Y-axis shall be represented differently from a missing sample, for example by a boundary/clipped indicator while preserving its valid status.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-606](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.

SWE-WEB-LLR-622 — Sensor Display Controls

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-596 — open requirement](#)

Current Jira description

The control page shall provide independent local-operator controls for Sensor 1 and Sensor 2 LCD display-enable state.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-607](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-05 — Trusted-Local Access Boundary and Validation:** PASS. Checks deliberate absence of application accounts/credentials, loopback-scoped deployment, CORS restrictions, and server-side command/email validation.
- **FE-07 — Remote Display Control UI:** PASS. Tests independent controls, pending/completed/failed states, authoritative returned state, and no optimistic success.

SWE-WEB-LLR-623 — Control State Confirmation

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-597 — open requirement](#)

Current Jira description

After submitting a display-control command, the control page shall update from the command result/current state rather than assuming the command succeeded.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-607](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-07 — Remote Display Control UI:** PASS. Tests independent controls, pending/completed/failed states, authoritative returned state, and no optimistic success.

SWE-WEB-LLR-624 — Fixed Celsius Axis

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-598 — open requirement](#)

Current Jira description

In the stakeholder compliance graph, Celsius Y-axis minimum shall be 10 and maximum shall be 50.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-608](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.

SWE-WEB-LLR-625 — Fixed Fahrenheit Axis

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-599](#) — open requirement

Current Jira description

In the stakeholder compliance graph, Fahrenheit Y-axis minimum shall be 50 and maximum shall be 122.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-608](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.

SWE-WEB-LLR-626 — 300-Second X Axis

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-600 — open requirement](#)

Current Jira description

The stakeholder compliance graph shall request/render 300 seconds of history and label the X-axis in seconds ago from 300 on the left to 0 on the right.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-610](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.

SWE-ALR-MLR-700 — Email and SMS Delivery

Result PASS
Level / component MLR / Alerting/Security
Jira [SCRUM-603 — open requirement](#)

Current Jira description

The computer software shall support email alerts and shall allow multiple configurable email destinations. SMS alerts are out of scope.

Revision note

Edited requirement. Notification scope was narrowed to email only; SMS delivery is not part of this release. The description above is the newest frozen version used by this run.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-700](#)

Children: [SWE-ALR-LLR-700](#), [SWE-ALR-LLR-701](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **ALR-01 — Alert Rule Matrix:** PASS. Table-drives HIGH/LOW/normal/missing/disconnected/invalid, both sensors/average, multiple recipients, and independent thresholds.
- **ALR-03 — Notification Provider Adapters and Failure Isolation:** PASS. Checks email destinations/messages, external SMTP credentials, and failure isolation from subsequent polling.

SWE-ALR-MLR-701 — Per-Recipient Alert Rules

Result PASS
Level / component MLR / Alerting/Security
Jira [SCRUM-604 — open requirement](#)

Current Jira description

Each configured recipient shall support independently configurable high and low temperature thresholds.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-700](#)

Children: [SWE-ALR-LLR-702](#), [SWE-ALR-LLR-703](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **ALR-01 — Alert Rule Matrix:** PASS. Table-drives HIGH/LOW/normal/missing/disconnected/invalid, both sensors/average, multiple recipients, and independent thresholds.

SWE-ALR-MLR-702 — Configurable Alert Messages

Result PASS
Level / component MLR / Alerting/Security
Jira [SCRUM-605 — open requirement](#)

Current Jira description

The high-temperature and low-temperature alert message text shall be configurable from the computer UI.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-700](#)

Children: [SWE-ALR-LLR-704](#), [SWE-ALR-LLR-705](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **ALR-01 — Alert Rule Matrix:** PASS. Table-drives HIGH/LOW/normal/missing/disconnected/invalid, both sensors/average, multiple recipients, and independent thresholds.

SWE-ALR-MLR-703 — Third-Party Notification Adapters

Result PASS
Level / component MLR / Alerting/Security
Jira [SCRUM-606 — open requirement](#)

Current Jira description

Notification delivery shall be performed by the computer through a configured third-party email service API or SMTP service rather than by the third box. SMS delivery is out of scope.

Revision note

Edited requirement. Notification scope was narrowed to email only; SMS delivery is not part of this release. The description above is the newest frozen version used by this run.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-700](#)

Children: [SWE-ALR-LLR-706](#), [SWE-ALR-LLR-707](#), [SWE-ALR-LLR-708](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **ALR-03 — Notification Provider Adapters and Failure Isolation:** PASS. Checks email destinations/messages, external SMTP credentials, and failure isolation from subsequent polling.

SWE-ALR-MLR-704 — Valid-Data Alert Evaluation

Result PASS
Level / component MLR / Alerting/Security
Jira [SCRUM-607 — open requirement](#)

Current Jira description

Only valid current temperature measurements shall be evaluated against alert thresholds.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-700](#)

Children: [SWE-ALR-LLR-709](#), [SWE-ALR-LLR-710](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **ALR-01 — Alert Rule Matrix:** PASS. Table-drives HIGH/LOW/normal/missing/disconnected/invalid, both sensors/average, multiple recipients, and independent thresholds.

SWE-ALR-MLR-705 — Alert Repetition Policy

Result PASS
Level / component MLR / Alerting/Security
Jira [SCRUM-608 — open requirement](#)

Current Jira description

The software shall avoid sending an alert every second while a temperature remains continuously out of range; an edge/episode-based notification policy shall be used unless the stakeholder specifies otherwise.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-700](#)

Children: [SWE-ALR-LLR-711](#), [SWE-ALR-LLR-712](#), [SWE-ALR-LLR-713](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **ALR-02 — Alert Episode State Machine:** PASS. Checks edge-triggered HIGH/LOW episodes, suppression while out of range, NORMAL reset, and re-alert.

SWE-SEC-MLR-750 — Web Authentication and Authorization

Result PASS
Level / component MLR / Alerting/Security
Jira [SCRUM-609](#) — open requirement

Current Jira description

The web application shall not implement application user accounts, login, or role-based authorization. It shall be deployed as a trusted local tool with services bound to the documented local access boundary, while server-side validation remains mandatory for control and email-alert configuration inputs.

Revision note

Edited requirement. Application accounts and role-based access were removed; the released system uses a trusted local deployment boundary. The description above is the newest frozen version used by this run.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: SYS-HLR-510, SYS-HLR-700

Children: SWE-SEC-LLR-750, SWE-SEC-LLR-751, SWE-SEC-LLR-752, SWE-SEC-LLR-753, SWE-SEC-LLR-754

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-05 — Trusted-Local Access Boundary and Validation:** PASS. Checks deliberate absence of application accounts/credentials, loopback-scoped deployment, CORS restrictions, and server-side command/email validation.

SWE-ALR-LLR-700 — Recipient Types

Result PASS
Level / component LLR / Alerting/Security
Jira [SCRUM-610 — open requirement](#)

Current Jira description

Each alert recipient record shall identify an email destination address. EMAIL is the only supported delivery type; SMS is out of scope.

Revision note

Edited requirement. Notification scope was narrowed to email only; SMS delivery is not part of this release. The description above is the newest frozen version used by this run.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-ALR-MLR-700](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **ALR-01 — Alert Rule Matrix:** PASS. Table-drives HIGH/LOW/normal/missing/disconnected/invalid, both sensors/average, multiple recipients, and independent thresholds.
- **ALR-03 — Notification Provider Adapters and Failure Isolation:** PASS. Checks email destinations/messages, external SMTP credentials, and failure isolation from subsequent polling.

SWE-ALR-LLR-701 — Multiple Recipients

Result PASS
Level / component LLR / Alerting/Security
Jira [SCRUM-611 — open requirement](#)

Current Jira description

The alert evaluator shall process all enabled recipient/rule records applicable to a valid sample rather than stopping after the first matching recipient.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-ALR-MLR-700](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **ALR-01 — Alert Rule Matrix:** PASS. Table-drives HIGH/LOW/normal/missing/disconnected/invalid, both sensors/average, multiple recipients, and independent thresholds.

SWE-ALR-LLR-702 — Threshold Rule Fields

Result PASS
Level / component LLR / Alerting/Security
Jira [SCRUM-612 — open requirement](#)

Current Jira description

Each alert rule shall contain minimum and maximum Celsius thresholds and a monitored source identifier.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-ALR-MLR-701](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **ALR-01 — Alert Rule Matrix:** PASS. Table-drives HIGH/LOW/normal/missing/disconnected/invalid, both sensors/average, multiple recipients, and independent thresholds.

SWE-ALR-LLR-703 — Per-Recipient Evaluation

Result PASS
Level / component LLR / Alerting/Security
Jira [SCRUM-613 — open requirement](#)

Current Jira description

The alert evaluator shall compare each enabled recipient rule using that rule's own thresholds; one recipient's thresholds shall not alter another recipient's evaluation.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-ALR-MLR-701](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **ALR-01 — Alert Rule Matrix:** PASS. Table-drives HIGH/LOW/normal/missing/disconnected/invalid, both sensors/average, multiple recipients, and independent thresholds.

SWE-ALR-LLR-704 — High Message Selection

Result PASS
Level / component LLR / Alerting/Security
Jira [SCRUM-614 — open requirement](#)

Current Jira description

A high-threshold notification shall use the configured high-message text for the matching rule/recipient.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-ALR-MLR-702](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **ALR-01 — Alert Rule Matrix:** PASS. Table-drives HIGH/LOW/normal/missing/disconnected/invalid, both sensors/average, multiple recipients, and independent thresholds.

SWE-ALR-LLR-705 — Low Message Selection

Result PASS
Level / component LLR / Alerting/Security
Jira [SCRUM-615 — open requirement](#)

Current Jira description

A low-threshold notification shall use the configured low-message text for the matching rule/recipient.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-ALR-MLR-702](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **ALR-01 — Alert Rule Matrix:** PASS. Table-drives HIGH/LOW/normal/missing/disconnected/invalid, both sensors/average, multiple recipients, and independent thresholds.

SWE-ALR-LLR-706 — SMS Provider Adapter

Result PASS
Level / component LLR / Alerting/Security
Jira [SCRUM-616 — open requirement](#)

Current Jira description

Implement an email provider adapter that accepts destination and message text and invokes the configured third-party email API or SMTP service using credentials supplied outside source control. The adapter shall support Gmail-compatible SMTP configuration for integration testing.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-ALR-MLR-703](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **ALR-01 — Alert Rule Matrix:** PASS. Table-drives HIGH/LOW/normal/missing/disconnected/invalid, both sensors/average, multiple recipients, and independent thresholds.
- **ALR-03 — Notification Provider Adapters and Failure Isolation:** PASS. Checks email destinations/messages, external SMTP credentials, and failure isolation from subsequent polling.

SWE-ALR-LLR-707 — Email Provider Adapter

Result PASS
Level / component LLR / Alerting/Security
Jira [SCRUM-617 — open requirement](#)

Current Jira description

Implement an email provider adapter that accepts destination and message text and invokes the configured third-party email API using credentials supplied outside source control.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-ALR-MLR-703](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **ALR-01 — Alert Rule Matrix:** PASS. Table-drives HIGH/LOW/normal/missing/disconnected/invalid, both sensors/average, multiple recipients, and independent thresholds.
- **ALR-03 — Notification Provider Adapters and Failure Isolation:** PASS. Checks email destinations/messages, external SMTP credentials, and failure isolation from subsequent polling.

SWE-ALR-LLR-708 — Provider Failure Handling

Result PASS
Level / component LLR / Alerting/Security
Jira [SCRUM-618 — open requirement](#)

Current Jira description

A notification-provider failure shall be logged/recorded and shall not block subsequent one-second temperature polling.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-ALR-MLR-703](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **ALR-03 — Notification Provider Adapters and Failure Isolation:** PASS. Checks email destinations/messages, external SMTP credentials, and failure isolation from subsequent polling.

SWE-ALR-LLR-709 — Invalid Sample Skip

Result PASS
Level / component LLR / Alerting/Security
Jira [SCRUM-619 — open requirement](#)

Current Jira description

The alert evaluator shall skip a sensor/rule evaluation when the underlying sample status is unavailable/disconnected/invalid.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-ALR-MLR-704](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **ALR-01 — Alert Rule Matrix:** PASS. Table-drives HIGH/LOW/normal/missing/disconnected/invalid, both sensors/average, multiple recipients, and independent thresholds.

SWE-ALR-LLR-710 — Average Alert Validity

Result PASS
Level / component LLR / Alerting/Security
Jira [SCRUM-620 — open requirement](#)

Current Jira description

If Average is supported as an alert source, it shall be evaluated only when both Sensor 1 and Sensor 2 values in the same sample are valid.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-ALR-MLR-704](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **ALR-01 — Alert Rule Matrix:** PASS. Table-drives HIGH/LOW/normal/missing/disconnected/invalid, both sensors/average, multiple recipients, and independent thresholds.

SWE-ALR-LLR-711 — Alert Episode State

Result PASS
Level / component LLR / Alerting/Security
Jira [SCRUM-621 — open requirement](#)

Current Jira description

For each enabled recipient/rule/source combination, maintain alert state NORMAL, HIGH, or LOW.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-ALR-MLR-705](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **ALR-01 — Alert Rule Matrix:** PASS. Table-drives HIGH/LOW/normal/missing/disconnected/invalid, both sensors/average, multiple recipients, and independent thresholds.
- **ALR-02 — Alert Episode State Machine:** PASS. Checks edge-triggered HIGH/LOW episodes, suppression while out of range, NORMAL reset, and re-alert.

SWE-ALR-LLR-712 — High Edge Trigger

Result PASS
Level / component LLR / Alerting/Security
Jira [SCRUM-622 — open requirement](#)

Current Jira description

Send the high alert only when the rule state transitions from NORMAL/LOW into HIGH; do not resend each one-second sample while it remains HIGH.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-ALR-MLR-705](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **ALR-01 — Alert Rule Matrix:** PASS. Table-drives HIGH/LOW/normal/missing/disconnected/invalid, both sensors/average, multiple recipients, and independent thresholds.
- **ALR-02 — Alert Episode State Machine:** PASS. Checks edge-triggered HIGH/LOW episodes, suppression while out of range, NORMAL reset, and re-alert.

SWE-ALR-LLR-713 — Low Edge Trigger

Result PASS
Level / component LLR / Alerting/Security
Jira [SCRUM-623 — open requirement](#)

Current Jira description

Send the low alert only when the rule state transitions from NORMAL/HIGH into LOW; do not resend each one-second sample while it remains LOW.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-ALR-MLR-705](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **ALR-01 — Alert Rule Matrix:** PASS. Table-drives HIGH/LOW/normal/missing/disconnected/invalid, both sensors/average, multiple recipients, and independent thresholds.
- **ALR-02 — Alert Episode State Machine:** PASS. Checks edge-triggered HIGH/LOW episodes, suppression while out of range, NORMAL reset, and re-alert.

SWE-SEC-LLR-750 — Password Hashing

Result PASS
Level / component LLR / Alerting/Security
Jira [SCRUM-625 — open requirement](#)

Current Jira description

The application shall not store application-user credentials, plaintext passwords, or password hashes because application user accounts are out of scope. Service credentials, including email-provider credentials, shall be supplied outside source control.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-SEC-MLR-750](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-05 — Trusted-Local Access Boundary and Validation:** PASS. Checks deliberate absence of application accounts/credentials, loopback-scoped deployment, CORS restrictions, and server-side command/email validation.

SWE-SEC-LLR-751 — Authenticated Session

Result PASS
Level / component LLR / Alerting/Security
Jira [SCRUM-626 — open requirement](#)

Current Jira description

The trusted local deployment shall not require an application authenticated session or token for control/settings operations. Backend and frontend services shall be limited to the documented local access boundary.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-SEC-MLR-750](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-05 — Trusted-Local Access Boundary and Validation:** PASS. Checks deliberate absence of application accounts/credentials, loopback-scoped deployment, CORS restrictions, and server-side command/email validation.

SWE-SEC-LLR-752 — Server-Side Authorization

Result PASS
Level / component LLR / Alerting/Security
Jira [SCRUM-627 — open requirement](#)

Current Jira description

Every third-box control and email-alert configuration API endpoint shall enforce server-side input validation independent of UI controls. Application ADMIN authorization is not required in the trusted local deployment.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-SEC-MLR-750](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-05 — Trusted-Local Access Boundary and Validation:** PASS. Checks deliberate absence of application accounts/credentials, loopback-scoped deployment, CORS restrictions, and server-side command/email validation.

SWE-SEC-LLR-753 — Session Cookie Protection

Result PASS
Level / component LLR / Alerting/Security
Jira [SCRUM-628 — open requirement](#)

Current Jira description

The application shall not create authentication cookies because application login is out of scope. Any non-sensitive preference or session cookie that is used shall apply an appropriate SameSite setting and shall not contain credentials.

Revision note

Edited requirement. Application accounts and role-based access were removed; the released system uses a trusted local deployment boundary. The description above is the newest frozen version used by this run.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-SEC-MLR-750](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-05 — Trusted-Local Access Boundary and Validation:** PASS. Checks deliberate absence of application accounts/credentials, loopback-scoped deployment, CORS restrictions, and server-side command/email validation.

SWE-SEC-LLR-754 — Input Validation

Result PASS
Level / component LLR / Alerting/Security
Jira [SCRUM-629 — open requirement](#)

Current Jira description

Server-side validation shall reject malformed email addresses, invalid threshold ordering, unsupported sensor identifiers, and out-of-range control values before persistence or command execution. Phone-number validation is out of scope.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-SEC-MLR-750](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-05 — Trusted-Local Access Boundary and Validation:** PASS. Checks deliberate absence of application accounts/credentials, loopback-scoped deployment, CORS restrictions, and server-side command/email validation.

HWE-MLR-213 — Indoor LCD Readability

Result PASS
Level / component MLR / Third Box Hardware
Jira [SCRUM-634 — open requirement](#)

Current Jira description

The third-box LCD shall be clearly readable under normal indoor lighting conditions.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-410](#)

Children: [HWE-LLR-222](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

HWE-MLR-214 — Display Size and Resolution Unconstrained

Result PASS
Level / component MLR / Third Box Hardware
Jira [SCRUM-635 — open requirement](#)

Current Jira description

No stakeholder-mandated physical display size or pixel/character resolution applies beyond the requirements to render all required information correctly and readably.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-410](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

HWE-MLR-215 — Drop Disconnect Allowance

Result PASS
Level / component MLR / Third Box Hardware
Jira [SCRUM-636 — open requirement](#)

Current Jira description

During the required drop with cables attached, a cable may become disconnected, but the cable and connector shall not break.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-200](#)

Children: [HWE-LLR-223](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-02 — Mechanical Robustness:** PASS. Automatically captures pre/post full-system smoke evidence around guided drop, upside-down, connected-cable, handling, and immersion actions.
- **HW-03 — Power Switch and Hot-Plug:** PASS. Guides sensor hot-plug and power cycling while capturing independent operation, no-reset behavior, loss of current data, and recovery.

HWE-MLR-216 — Sensor Thermal Response Capability

Result PASS
Level / component MLR / Third Box Hardware
Jira [SCRUM-637 — open requirement](#)

Current Jira description

The selected sensor assembly shall respond observably after a few seconds of hand heating and shall respond more quickly to nearby/brief soldering-iron heating.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-310](#)

Children: [HWE-LLR-224](#), [HWE-LLR-225](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-04 — Temperature Accuracy and Thermal Response:** PASS. Captures room, ice-water, hand-heating, and soldering-iron response measurements for both sensors.

HWE-LLR-222 — Indoor Readability Verification

Result PASS
Level / component LLR / Third Box Hardware
Jira [SCRUM-638 — open requirement](#)

Current Jira description

Verify the selected LCD/backlight combination can be read at normal laboratory viewing distance under normal indoor lighting at the intended enclosure orientation.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [HWE-MLR-213](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **FW-04 — LCD and Buttons:** PASS. Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.

HWE-LLR-223 — Connected-Cable Drop Acceptance

Result PASS
Level / component LLR / Third Box Hardware
Jira [SCRUM-639 — open requirement](#)

Current Jira description

A connected-cable drop test passes if the cables/connectors remain unbroken and functional after reconnection even when one or more cables detach during impact.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [HWE-MLR-215](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-02 — Mechanical Robustness:** PASS. Automatically captures pre/post full-system smoke evidence around guided drop, upside-down, connected-cable, handling, and immersion actions.

HWE-LLR-224 — Hand-Heating Verification

Result PASS
Level / component LLR / Third Box Hardware
Jira [SCRUM-640 — open requirement](#)

Current Jira description

Verify that holding each sensor in a hand produces an increasing reported temperature after a few seconds.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [HWE-MLR-216](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **HW-04 — Temperature Accuracy and Thermal Response:** PASS. Captures room, ice-water, hand-heating, and soldering-iron response measurements for both sensors.

HWE-LLR-225 — Soldering-Iron Response Verification

Result PASS
Level / component LLR / Third Box Hardware
Jira [SCRUM-641 — open requirement](#)

Current Jira description

Verify that bringing a soldering iron near or briefly against each sensor produces an increasing reported temperature more rapidly than the hand-heating test.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [HWE-MLR-216](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **[HW-01 — Physical Configuration and Design Inspection](#):** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- **[HW-04 — Temperature Accuracy and Thermal Response](#):** PASS. Captures room, ice-water, hand-heating, and soldering-iron response measurements for both sensors.

SWE-CONN-MLR-507 — Startup History Availability Timing

Result PASS
Level / component MLR / Computer BLE Connection Process
Jira [SCRUM-642 — open requirement](#)

Current Jira description

When the third box is available, the connection/database path shall make the previous 300 seconds of available history usable by the web application within 10 seconds of computer-software startup.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-600](#)

Children: [SWE-CONN-LLR-515](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.
- **BE-02 — History Synchronization:** PASS. Runs deterministic 300-record metadata/chunk/retry/reconstruction/upsert/no-duplicate and time-budget tests.

SWE-CONN-MLR-508 — Third-Box Power-On Recovery Timing

Result PASS
Level / component MLR / Computer BLE Connection Process
Jira [SCRUM-643 — open requirement](#)

Current Jira description

When the computer software is already running, real-time and graph data shall become available within 10 seconds after the third box is turned ON.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-520](#)

Children: [SWE-CONN-LLR-516](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-03 — Power-On Recovery:** PASS. Measures fake-device appearance through discovery, connection, current poll, and publication against 10 seconds.

SWE-CONN-LLR-515 — Ten-Second Startup Sync Budget

Result PASS
Level / component LLR / Computer BLE Connection Process
Jira [SCRUM-644 — open requirement](#)

Current Jira description

The startup connection, history retrieval, history persistence, and first current-data publication path shall be implemented and tested so the required 300-second history can be served within 10 seconds when the third box is already available.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-CONN-MLR-507](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.
- **BE-02 — History Synchronization:** PASS. Runs deterministic 300-record metadata/chunk/retry/reconstruction/upsert/no-duplicate and time-budget tests.
- **SYS-03 — Startup, Power-On, and History Recovery:** PASS. Separately measures box-OFF power recovery and 300-second history recovery after a host-radio outage while the box remains powered.

SWE-CONN-LLR-516 — Ten-Second Power-On Recovery Budget

Result PASS
Level / component LLR / Computer BLE Connection Process
Jira [SCRUM-645 — open requirement](#)

Current Jira description

The reconnect scan interval, BLE connection setup, first current poll, and database publication shall collectively satisfy the 10-second third-box power-on recovery requirement.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-CONN-MLR-508](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BE-01 — Worker Isolation, 1-Hz Polling, and Persistence:** PASS. Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.
- **BE-03 — Power-On Recovery:** PASS. Measures fake-device appearance through discovery, connection, current poll, and publication against 10 seconds.
- **SYS-03 — Startup, Power-On, and History Recovery:** PASS. Separately measures box-OFF power recovery and 300-second history recovery after a host-radio outage while the box remains powered.

INT-LLR-416 — Remote Control End-to-End Timing

Result PASS
Level / component LLR / BLE Interface
Jira [SCRUM-646 — open requirement](#)

Current Jira description

The direct backend API request, BLE SET_DISPLAY transaction, firmware state update, and LCD rendering path shall be tested end-to-end to complete the physical display response in less than 1 second after the computer user action.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [INT-MLR-403](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **BLE-02 — Authenticated Display Control:** PASS. Confirms authenticated SET_DISPLAY and rejection of unauthorized/malformed state-changing requests.
- **SYS-04 — Remote Display Response Latency:** PASS. Measures 10 ON/OFF changes per sensor from user action to physical LCD render; every sample must be below 1000 ms.

SWE-WEB-MLR-613 — Prominent Current Temperature Display

Result PASS
Level / component MLR / Web Application
Jira [SCRUM-647 — open requirement](#)

Current Jira description

The web interface shall display current Sensor 1 and Sensor 2 temperatures prominently using a large-font presentation when the values are valid.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-500](#)

Children: [SWE-WEB-LLR-630](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-02 — Live Readout and Stale-Data Behavior:** PASS. Exercises valid/change/disconnected/box-off scenarios, 1-second refresh, stale-number removal, exact unavailable text, and prominent values.

SWE-WEB-MLR-614 — Chart-Recorder Scrolling Behavior

Result PASS
Level / component MLR / Web Application
Jira [SCRUM-648 — open requirement](#)

Current Jira description

The graph shall place newest data on the right, older data toward the left, add a new value once per second, and remove values that age beyond the displayed 300-second record.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SYS-HLR-600](#)

Children: [SWE-WEB-LLR-631](#), [SWE-WEB-LLR-632](#), [SWE-WEB-LLR-633](#)

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.

SWE-WEB-LLR-630 — Large Current-Value Styling

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-649 — open requirement](#)

Current Jira description

The graph/main page shall render valid current temperature values in dedicated current-value elements visually larger than normal body text and immediately visible without opening settings/control views.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-613](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-02 — Live Readout and Stale-Data Behavior:** PASS. Exercises valid/change/disconnected/box-off scenarios, 1-second refresh, stale-number removal, exact unavailable text, and prominent values.
- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.
- **FE-06 — Email Alert Configuration and Delivery:** PASS. Checks email-only recipients/rules, email validation, configurable thresholds/messages, Gmail-compatible SMTP delivery, and persistence schema.
- **FE-07 — Remote Display Control UI:** PASS. Tests independent controls, pending/completed/failed states, authoritative returned state, and no optimistic success.

SWE-WEB-LLR-631 — Newest Sample Placement

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-650 — open requirement](#)

Current Jira description

The graph rendering data order shall map the newest sample to the rightmost X-axis position.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-614](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.

SWE-WEB-LLR-632 — Leftward Aging

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-651 — open requirement](#)

Current Jira description

On each one-second graph update, existing samples shall move logically one second farther from the right edge so older data appears to scroll left.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-614](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.

SWE-WEB-LLR-633 — Expired Sample Removal

Result PASS
Level / component LLR / Web Application
Jira [SCRUM-652 — open requirement](#)

Current Jira description

The graph data model shall exclude samples older than 300 seconds from the stakeholder compliance view after each update.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: [SWE-WEB-MLR-614](#)

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **FE-03 — Chart Recorder Compliance:** PASS. Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.

PRJ-REQ-005 — Full-Range Design Verification

Result PASS
Level / component Project / System
Jira [SCRUM-653 — open requirement](#)

Current Jira description

Physical testing of the complete -10 through +63 deg C range is not required, but the design documentation shall demonstrate that the chosen components and representations support that range.

Derivation and relationships

Source: Atlassian Jira SCRUM read-only snapshot (snapshot 2026-09-22)

Parents: None — top-level project or stakeholder requirement.

Children: None.

Passing criteria

This requirement passes only when every linked applicable test passes and the observed behavior satisfies the complete Jira description above.

Tests that verify this requirement

- **HW-01 — Physical Configuration and Design Inspection:** PASS. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.

3 Test results

4 FW-01 — Firmware Build, Startup, and Configuration

Outcome: PASS

Method: automated

Subsystem: firmware

4.1 What this test proves and why

This test verifies the following behavior: Builds or statically validates real and simulated firmware configuration, protocol generation, app_main initialization, Celsius, and boot identity. The inputs and expected results are controlled, so the check is repeatable and does not rely on operator judgment.

4.2 Passing criteria

- The observed result must match this test objective: Builds or statically validates real and simulated firmware configuration, protocol generation, app_main initialization, Celsius, and boot identity.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

4.3 Required setup

Software, simulation, and container checks (software)

- No ESP32 or physical sensors are required for this section.
- Docker must be running for integration-profile MySQL checks.
- The runner may start local backend/frontend/database services, but it will not use the physical BLE device.

4.4 Procedure

1. Run the automated FW-01 qualification check. The runner applies the defined inputs, records the observations, and evaluates every required assertion.

4.5 Requirements covered

- **SWE-EMB-LLR-300** — PASS. FW-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-EMB-LLR-323** — PASS. FW-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-EMB-MLR-300** — PASS. FW-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-300** — PASS. FW-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

4.6 Recorded results and limitations

No structured measurements were recorded. **Limitations:** A successful build does not prove operation on physical ESP32/LCD/sensors.

4.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

5 FW-02 — Sensor Acquisition, State Machine, and Recovery

Outcome: PASS

Method: automated

Subsystem: firmware

5.1 What this test proves and why

This test verifies the following behavior: Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic. The inputs and expected results are controlled, so the check is repeatable and does not rely on operator judgment.

5.2 Passing criteria

- The observed result must match this test objective: Exercises both-valid, individual/both failure, preservation of pre-disconnection ON/OFF state, independent acquisition, and no-restart logic.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

5.3 Required setup

Software, simulation, and container checks (software)

- No ESP32 or physical sensors are required for this section.
- Docker must be running for integration-profile MySQL checks.
- The runner may start local backend/frontend/database services, but it will not use the physical BLE device.

5.4 Procedure

1. Run the automated FW-02 qualification check. The runner applies the defined inputs, records the observations, and evaluates every required assertion.

5.5 Requirements covered

- **SWE-EMB-LLR-300** — PASS. FW-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-EMB-LLR-301** — PASS. FW-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-EMB-LLR-302** — PASS. FW-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-EMB-LLR-303** — PASS. FW-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-EMB-LLR-304** — PASS. FW-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-EMB-LLR-305** — PASS. FW-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-EMB-LLR-306** — PASS. FW-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

- [illegible]

It passes only when every required observation and limit is satisfied.

5.6 Recorded results and limitations

No structured measurements were recorded. **Limitations:** Host logic/static evidence does not prove real DS18B20 electrical behavior.

5.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

6 FW-03 — History, Average, and Atomic Snapshot

Outcome: PASS

Method: automated

Subsystem: firmware

6.1 What this test proves and why

This test verifies the following behavior: Verifies 1 Hz logical sampling, 300-record wrap/order, record fields, averages, synchronization, and boot identity. The inputs and expected results are controlled, so the check is repeatable and does not rely on operator judgment.

6.2 Passing criteria

- The observed result must match this test objective: Verifies 1 Hz logical sampling, 300-record wrap/order, record fields, averages, synchronization, and boot identity.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

6.3 Required setup

Software, simulation, and container checks (software)

- No ESP32 or physical sensors are required for this section.
- Docker must be running for integration-profile MySQL checks.
- The runner may start local backend/frontend/database services, but it will not use the physical BLE device.

6.4 Procedure

1. Run the automated FW-03 qualification check. The runner applies the defined inputs, records the observations, and evaluates every required assertion.

6.5 Requirements covered

- **SWE-EMB-LLR-300** — PASS. FW-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-EMB-LLR-304** — PASS. FW-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-EMB-LLR-305** — PASS. FW-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-EMB-LLR-306** — PASS. FW-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-EMB-LLR-307** — PASS. FW-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-EMB-LLR-314** — PASS. FW-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-EMB-LLR-315** — PASS. FW-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

- **SWE-EMB-LLR-324** — PASS. FW-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-EMB-LLR-325** — PASS. FW-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-EMB-MLR-306** — PASS. FW-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-EMB-MLR-311** — PASS. FW-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-600** — PASS. FW-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

6.6 Recorded results and limitations

No structured measurements were recorded. **Limitations:** Host tests do not measure real BLE transfer timing.

6.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

7 BLE-01 — GATT and Protocol Contract

Outcome: PASS

Method: automated

Subsystem: ble

7.1 What this test proves and why

This test verifies the following behavior: Checks UUIDs, layouts, request IDs, GET_CURRENT, statuses/version rejection, authentication framing, and requester-only behavior. The inputs and expected results are controlled, so the check is repeatable and does not rely on operator judgment.

7.2 Passing criteria

- The observed result must match this test objective: Checks UUIDs, layouts, request IDs, GET_CURRENT, statuses/version rejection, authentication framing, and requester-only behavior.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

7.3 Required setup

Software, simulation, and container checks (software)

- No ESP32 or physical sensors are required for this section.
- Docker must be running for integration-profile MySQL checks.
- The runner may start local backend/frontend/database services, but it will not use the physical BLE device.

7.4 Procedure

1. Run the automated BLE-01 qualification check. The runner applies the defined inputs, records the observations, and evaluates every required assertion.

7.5 Requirements covered

- **INT-LLR-400** — PASS. BLE-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **INT-LLR-402** — PASS. BLE-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **INT-LLR-403** — PASS. BLE-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **INT-LLR-404** — PASS. BLE-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **INT-LLR-412** — PASS. BLE-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **INT-MLR-400** — PASS. BLE-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **INT-MLR-401** — PASS. BLE-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

- **INT-MLR-402** — PASS. BLE-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **INT-MLR-403** — PASS. BLE-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **INT-MLR-407** — PASS. BLE-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

7.6 Recorded results and limitations

No structured measurements were recorded. **Limitations:** Fake/client contract evidence does not prove radio interoperability.

7.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

8 BE-02 — History Synchronization

Outcome: PASS

Method: automated

Subsystem: backend

8.1 What this test proves and why

This test verifies the following behavior: Runs deterministic 300-record metadata/chunk/retry/reconstruction/upsert/no-duplicate and time-budget tests. The inputs and expected results are controlled, so the check is repeatable and does not rely on operator judgment.

8.2 Passing criteria

- The observed result must match this test objective: Runs deterministic 300-record metadata/chunk/retry/reconstruction/upsert/no-duplicate and time-budget tests.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

8.3 Required setup

Software, simulation, and container checks (software)

- No ESP32 or physical sensors are required for this section.
- Docker must be running for integration-profile MySQL checks.
- The runner may start local backend/frontend/database services, but it will not use the physical BLE device.

8.4 Procedure

1. Run the automated BE-02 qualification check. The runner applies the defined inputs, records the observations, and evaluates every required assertion.

8.5 Requirements covered

- **SWE-CONN-LLR-501** — PASS. BE-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-CONN-LLR-505** — PASS. BE-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-CONN-LLR-506** — PASS. BE-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-CONN-LLR-507** — PASS. BE-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-CONN-LLR-508** — PASS. BE-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-CONN-LLR-509** — PASS. BE-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-CONN-LLR-515** — PASS. BE-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

- **SWE-CONN-MLR-503** — PASS. BE-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-CONN-MLR-507** — PASS. BE-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-600** — PASS. BE-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

8.6 Recorded results and limitations

No structured measurements were recorded. **Limitations:** The 10-second requirement remains incomplete until SYS-03 measures real hardware.

8.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

9 BE-03 — Power-On Recovery

Outcome: PASS

Method: automated

Subsystem: backend

9.1 What this test proves and why

This test verifies the following behavior: Measures fake-device appearance through discovery, connection, current poll, and publication against 10 seconds. The inputs and expected results are controlled, so the check is repeatable and does not rely on operator judgment.

9.2 Passing criteria

- The observed result must match this test objective: Measures fake-device appearance through discovery, connection, current poll, and publication against 10 seconds.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

9.3 Required setup

Software, simulation, and container checks (software)

- No ESP32 or physical sensors are required for this section.
- Docker must be running for integration-profile MySQL checks.
- The runner may start local backend/frontend/database services, but it will not use the physical BLE device.

9.4 Procedure

1. Run the automated BE-03 qualification check. The runner applies the defined inputs, records the observations, and evaluates every required assertion.

9.5 Requirements covered

- **SWE-CONN-LLR-503** — PASS. BE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-CONN-LLR-504** — PASS. BE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-CONN-LLR-505** — PASS. BE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-CONN-LLR-509** — PASS. BE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-CONN-LLR-516** — PASS. BE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-CONN-MLR-503** — PASS. BE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-CONN-MLR-506** — PASS. BE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

- **SWE-CONN-MLR-508** — PASS. BE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-520** — PASS. BE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

9.6 Recorded results and limitations

No structured measurements were recorded. **Limitations:** Real Windows/BlueZ advertising and power-on latency require SYS-03.

9.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

10 BE-04 — Direct Backend Control Command Lifecycle

Outcome: PASS

Method: automated

Subsystem: backend

10.1 What this test proves and why

This test verifies the following behavior: Verifies validated commands use the direct backend/BLE path, are serialized, return terminal results, and do not create a MySQL command queue. The inputs and expected results are controlled, so the check is repeatable and does not rely on operator judgment.

10.2 Passing criteria

- The observed result must match this test objective: Verifies validated commands use the direct backend/BLE path, are serialized, return terminal results, and do not create a MySQL command queue.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

10.3 Required setup

Software, simulation, and container checks (software)

- No ESP32 or physical sensors are required for this section.
- Docker must be running for integration-profile MySQL checks.
- The runner may start local backend/frontend/database services, but it will not use the physical BLE device.

10.4 Procedure

1. Run the automated BE-04 qualification check. The runner applies the defined inputs, records the observations, and evaluates every required assertion.

10.5 Requirements covered

- **SWE-CONN-LLR-502** — PASS. BE-04 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-CONN-LLR-510** — PASS. BE-04 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-CONN-LLR-511** — PASS. BE-04 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-CONN-LLR-512** — PASS. BE-04 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-CONN-MLR-503** — PASS. BE-04 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

10.6 Recorded results and limitations

No structured measurements were recorded. **Limitations:** Physical under-one-second response is covered by SYS-04.

10.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

11 DB-01 — Schema, Types, Email Configuration, UTC, and Secrets

Outcome: PASS

Method: automated

Subsystem: database

11.1 What this test proves and why

This test verifies the following behavior: Inspects isolated MySQL schema types, email-only alert models, deliberate absence of user/command tables, UTC, and external credentials. The inputs and expected results are controlled, so the check is repeatable and does not rely on operator judgment.

11.2 Passing criteria

- The observed result must match this test objective: Inspects isolated MySQL schema types, email-only alert models, deliberate absence of user/command tables, UTC, and external credentials.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

11.3 Required setup

Software, simulation, and container checks (software)

- No ESP32 or physical sensors are required for this section.
- Docker must be running for integration-profile MySQL checks.
- The runner may start local backend/frontend/database services, but it will not use the physical BLE device.

11.4 Procedure

1. Run the automated DB-01 qualification check. The runner applies the defined inputs, records the observations, and evaluates every required assertion.

11.5 Requirements covered

- **SWE-DB-LLR-550** — PASS. DB-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-DB-LLR-554** — PASS. DB-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-DB-LLR-555** — PASS. DB-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-DB-LLR-556** — PASS. DB-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-DB-LLR-559** — PASS. DB-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-DB-LLR-561** — PASS. DB-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

11.6 Recorded results and limitations

No structured measurements were recorded. **Limitations:** Static schema parsing alone is explicitly insufficient; unavailable isolated MySQL blocks this test.

11.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

12 DB-02 — Samples, Uniqueness, Indexes, and Retention

Outcome: PASS

Method: automated

Subsystem: database

12.1 What this test proves and why

This test verifies the following behavior: Seeds an isolated database and verifies identity uniqueness, indexed latest/history plans, 300-second minimum retention, and >600-second eligibility. The inputs and expected results are controlled, so the check is repeatable and does not rely on operator judgment.

12.2 Passing criteria

- The observed result must match this test objective: Seeds an isolated database and verifies identity uniqueness, indexed latest/history plans, 300-second minimum retention, and >600-second eligibility.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

12.3 Required setup

Software, simulation, and container checks (software)

- No ESP32 or physical sensors are required for this section.
- Docker must be running for integration-profile MySQL checks.
- The runner may start local backend/frontend/database services, but it will not use the physical BLE device.

12.4 Procedure

1. Run the automated DB-02 qualification check. The runner applies the defined inputs, records the observations, and evaluates every required assertion.

12.5 Requirements covered

- **SWE-DB-LLR-551** — PASS. DB-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-DB-LLR-552** — PASS. DB-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-DB-LLR-553** — PASS. DB-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-DB-MLR-550** — PASS. DB-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-DB-MLR-551** — PASS. DB-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-DB-MLR-554** — PASS. DB-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

12.6 Recorded results and limitations

No structured measurements were recorded. **Limitations:** Does not run against the development database.

12.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

13 DB-03 — History-Only Database and Direct Command Boundary

Outcome: PASS

Method: automated

Subsystem: database

13.1 What this test proves and why

This test verifies the following behavior: Validates that MySQL stores history/email configuration but no command queue/state, while commands use the direct backend API. The inputs and expected results are controlled, so the check is repeatable and does not rely on operator judgment.

13.2 Passing criteria

- The observed result must match this test objective: Validates that MySQL stores history/email configuration but no command queue/state, while commands use the direct backend API.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

13.3 Required setup

Software, simulation, and container checks (software)

- No ESP32 or physical sensors are required for this section.
- Docker must be running for integration-profile MySQL checks.
- The runner may start local backend/frontend/database services, but it will not use the physical BLE device.

13.4 Procedure

1. Run the automated DB-03 qualification check. The runner applies the defined inputs, records the observations, and evaluates every required assertion.

13.5 Requirements covered

- **SWE-DB-LLR-557** — PASS. DB-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-DB-LLR-558** — PASS. DB-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-DB-LLR-560** — PASS. DB-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-DB-MLR-552** — PASS. DB-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-DB-MLR-553** — PASS. DB-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

13.6 Recorded results and limitations

No structured measurements were recorded. **Limitations:** Direct command execution is verified separately by BE-04 and SYS-04.

13.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

14 FE-01 — Latest and History API Contract

Outcome: PASS

Method: automated

Subsystem: frontend

14.1 What this test proves and why

This test verifies the following behavior: Verifies latest/history response fields, bounds, ordering, statuses, missing markers, average, availability, and display state. The inputs and expected results are controlled, so the check is repeatable and does not rely on operator judgment.

14.2 Passing criteria

- The observed result must match this test objective: Verifies latest/history response fields, bounds, ordering, statuses, missing markers, average, availability, and display state.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

14.3 Required setup

Software, simulation, and container checks (software)

- No ESP32 or physical sensors are required for this section.
- Docker must be running for integration-profile MySQL checks.
- The runner may start local backend/frontend/database services, but it will not use the physical BLE device.

14.4 Procedure

1. Run the automated FE-01 qualification check. The runner applies the defined inputs, records the observations, and evaluates every required assertion.

14.5 Requirements covered

- **SWE-WEB-LLR-600** — PASS. FE-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-601** — PASS. FE-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-605** — PASS. FE-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-MLR-601** — PASS. FE-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

14.6 Recorded results and limitations

No structured measurements were recorded. **Limitations:** Unit/Supertest coverage is partial if the production endpoint omits Jira-required fields.

14.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

15 FE-02 — Live Readout and Stale-Data Behavior

Outcome: PASS

Method: automated

Subsystem: frontend

15.1 What this test proves and why

This test verifies the following behavior: Exercises valid/change/disconnected/box-off scenarios, 1-second refresh, stale-number removal, exact unavailable text, and prominent values. The inputs and expected results are controlled, so the check is repeatable and does not rely on operator judgment.

15.2 Passing criteria

- The observed result must match this test objective: Exercises valid/change/disconnected/box-off scenarios, 1-second refresh, stale-number removal, exact unavailable text, and prominent values.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

15.3 Required setup

Software, simulation, and container checks (software)

- No ESP32 or physical sensors are required for this section.
- Docker must be running for integration-profile MySQL checks.
- The runner may start local backend/frontend/database services, but it will not use the physical BLE device.

15.4 Procedure

1. Run the automated FE-02 qualification check. The runner applies the defined inputs, records the observations, and evaluates every required assertion.

15.5 Requirements covered

- **SWE-WEB-LLR-604** — PASS. FE-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-606** — PASS. FE-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-614** — PASS. FE-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-617** — PASS. FE-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-618** — PASS. FE-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-619** — PASS. FE-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-630** — PASS. FE-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

- **SWE-WEB-MLR-613** — PASS. FE-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-500** — PASS. FE-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

15.6 Recorded results and limitations

No structured measurements were recorded. **Limitations:** DOM unit evidence is partial; the full profile also requires operator/browser evidence through SYS tests.

15.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

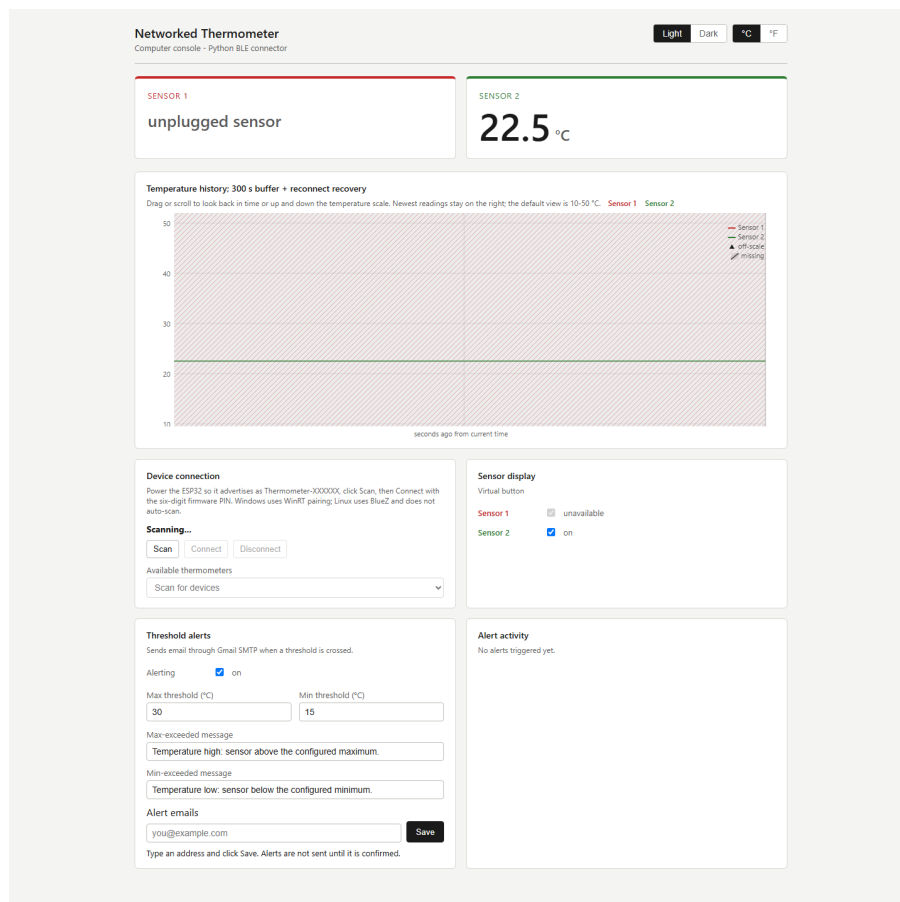


Figure 1: Selected visual evidence for FE-02

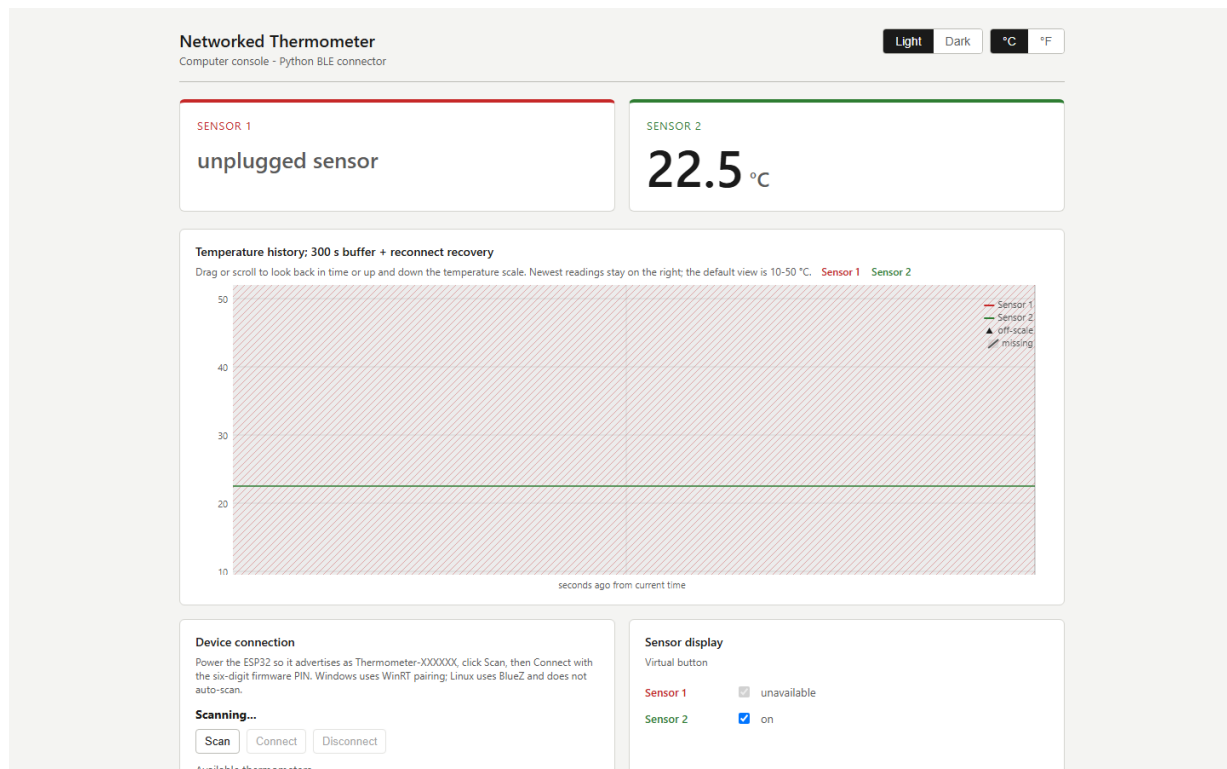


Figure 2: Selected visual evidence for FE-02

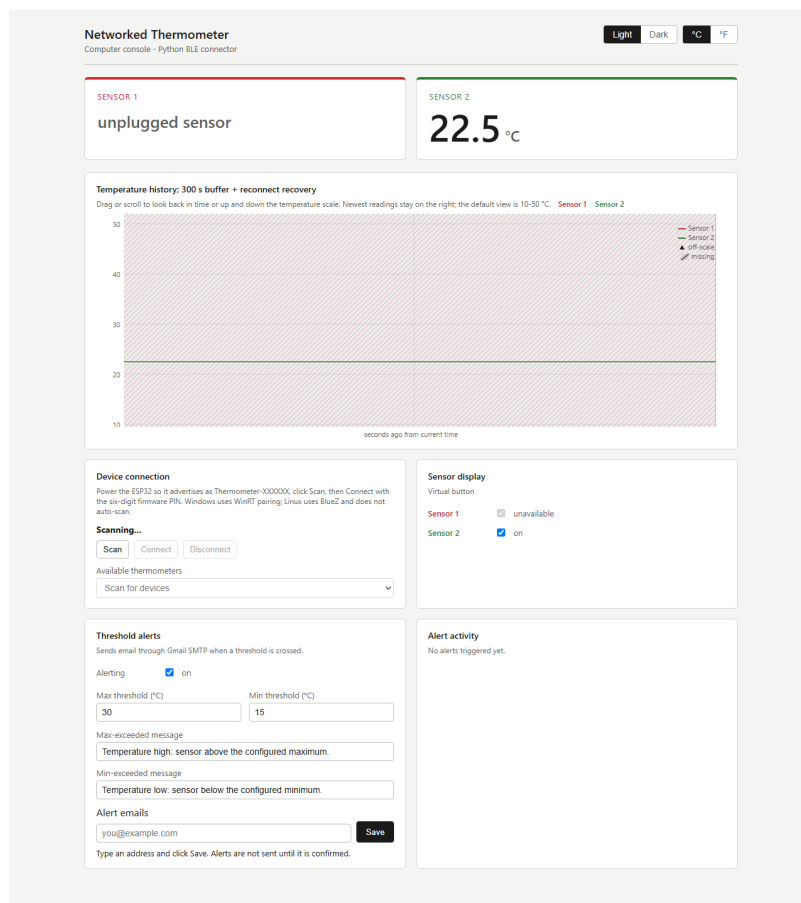


Figure 3: Selected visual evidence for FE-02

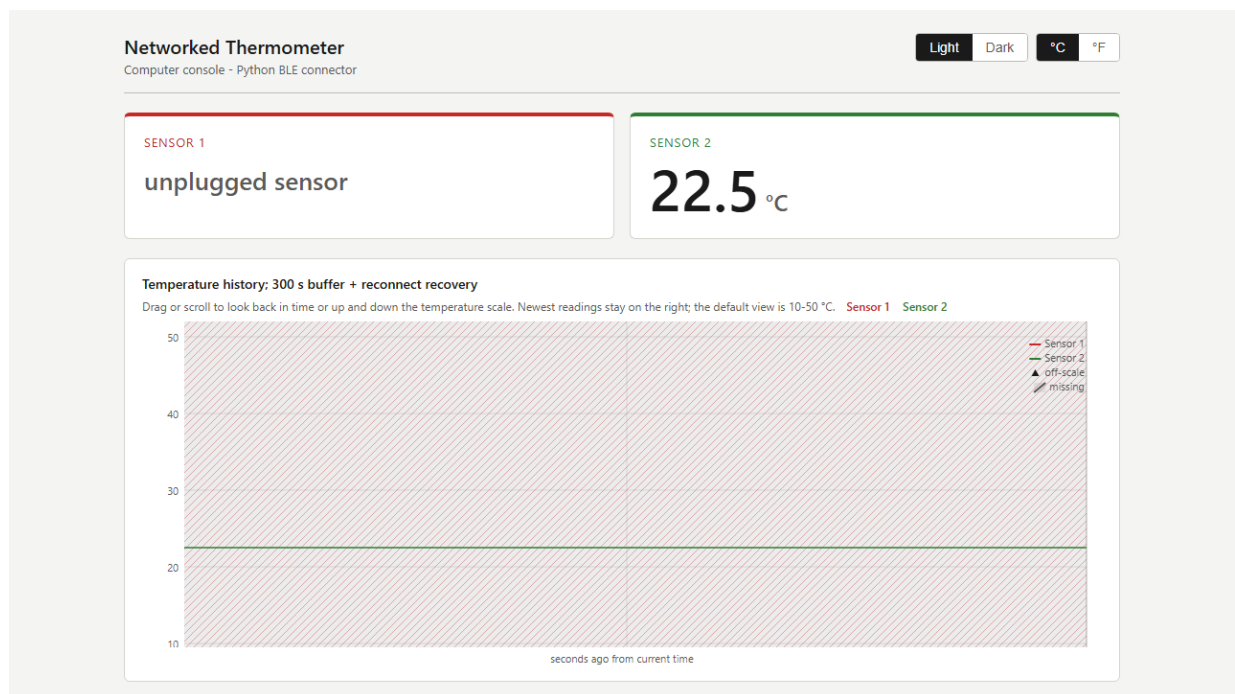


Figure 4: Selected visual evidence for FE-02

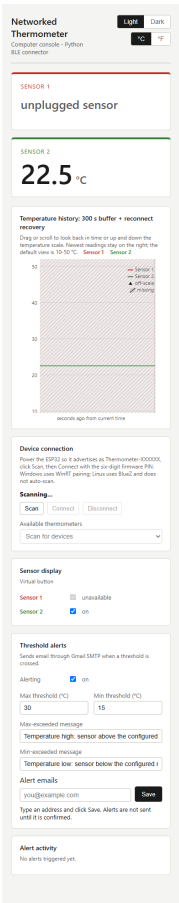


Figure 5: Selected visual evidence for FE-02

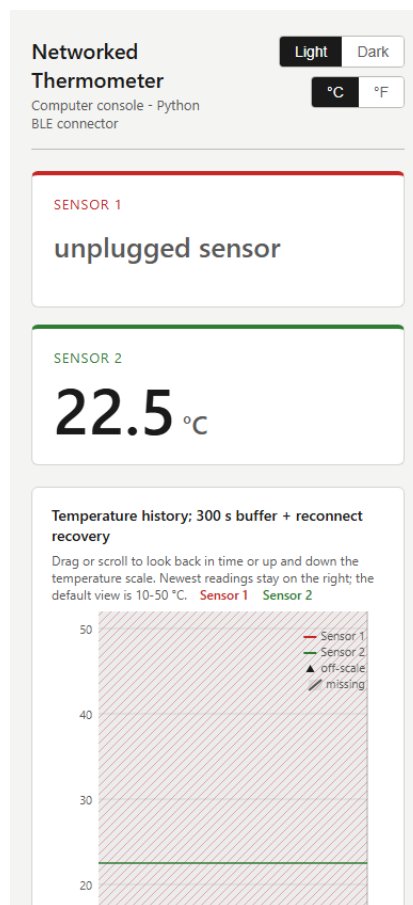


Figure 6: Selected visual evidence for FE-02

16 FE-03 — Chart Recorder Compliance

Outcome: PASS

Method: automated

Subsystem: frontend

16.1 What this test proves and why

This test verifies the following behavior: Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average. The inputs and expected results are controlled, so the check is repeatable and does not rely on operator judgment.

16.2 Passing criteria

- The observed result must match this test objective: Checks all series combinations, unit conversion, fixed limits, 300-to-0 axis, order/scroll/pruning, gaps/off-scale distinction, and same-sample average.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

16.3 Required setup

Software, simulation, and container checks (software)

- No ESP32 or physical sensors are required for this section.
- Docker must be running for integration-profile MySQL checks.
- The runner may start local backend/frontend/database services, but it will not use the physical BLE device.

16.4 Procedure

1. Run the automated FE-03 qualification check. The runner applies the defined inputs, records the observations, and evaluates every required assertion.

16.5 Requirements covered

- **SWE-WEB-LLR-600** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-601** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-604** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-606** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-608** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-611** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-613** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

- **SWE-WEB-LLR-614** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-615** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-616** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-617** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-620** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-621** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-624** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-625** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-626** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-630** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-631** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-632** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-633** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-MLR-604** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-MLR-605** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-MLR-606** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-MLR-608** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-MLR-610** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-MLR-614** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-600** — PASS. FE-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

16.6 Recorded results and limitations

No structured measurements were recorded. **Limitations:** Canvas pixels and cross-browser presentation require the optional Playwright/HIL evidence when available.

16.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

17 FE-04 — Responsive Layout

Outcome: PASS

Method: automated

Subsystem: frontend

17.1 What this test proves and why

This test verifies the following behavior: Checks required controls and horizontal overflow expectations for desktop, laptop, and two mobile viewport definitions. The inputs and expected results are controlled, so the check is repeatable and does not rely on operator judgment.

17.2 Passing criteria

- The observed result must match this test objective: Checks required controls and horizontal overflow expectations for desktop, laptop, and two mobile viewport definitions.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

17.3 Required setup

Software, simulation, and container checks (software)

- No ESP32 or physical sensors are required for this section.
- Docker must be running for integration-profile MySQL checks.
- The runner may start local backend/frontend/database services, but it will not use the physical BLE device.

17.4 Procedure

1. Run the automated FE-04 qualification check. The runner applies the defined inputs, records the observations, and evaluates every required assertion.

17.5 Requirements covered

- **SWE-WEB-LLR-606** — PASS. FE-04 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

17.6 Recorded results and limitations

No structured measurements were recorded. **Limitations:** JSDOM cannot prove rendered geometry; absent Playwright browser evidence causes this check to report BLOCKED.

17.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

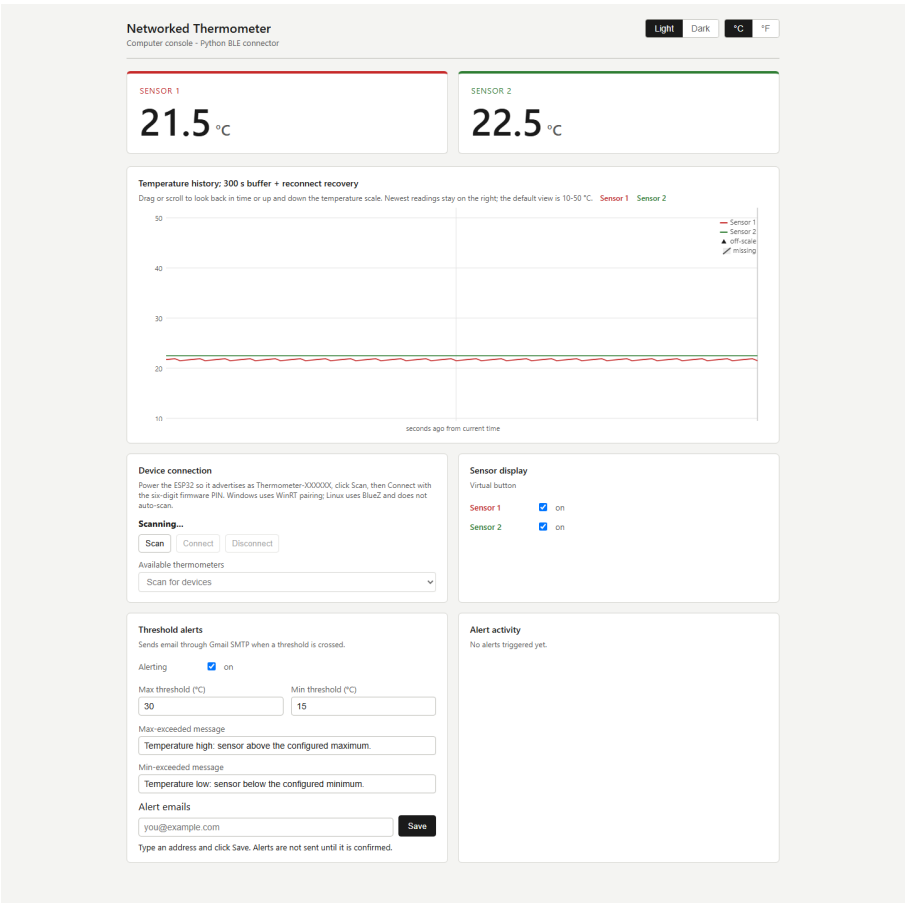


Figure 7: Selected visual evidence for FE-04

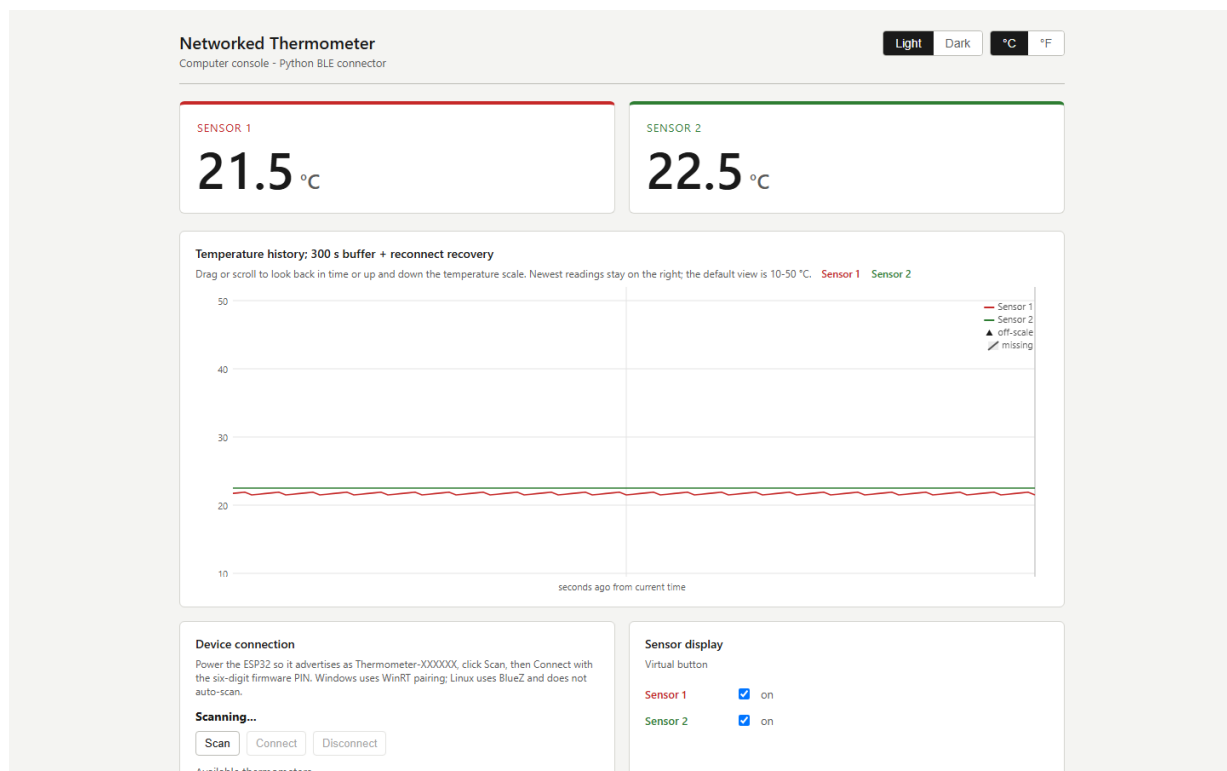


Figure 8: Selected visual evidence for FE-04

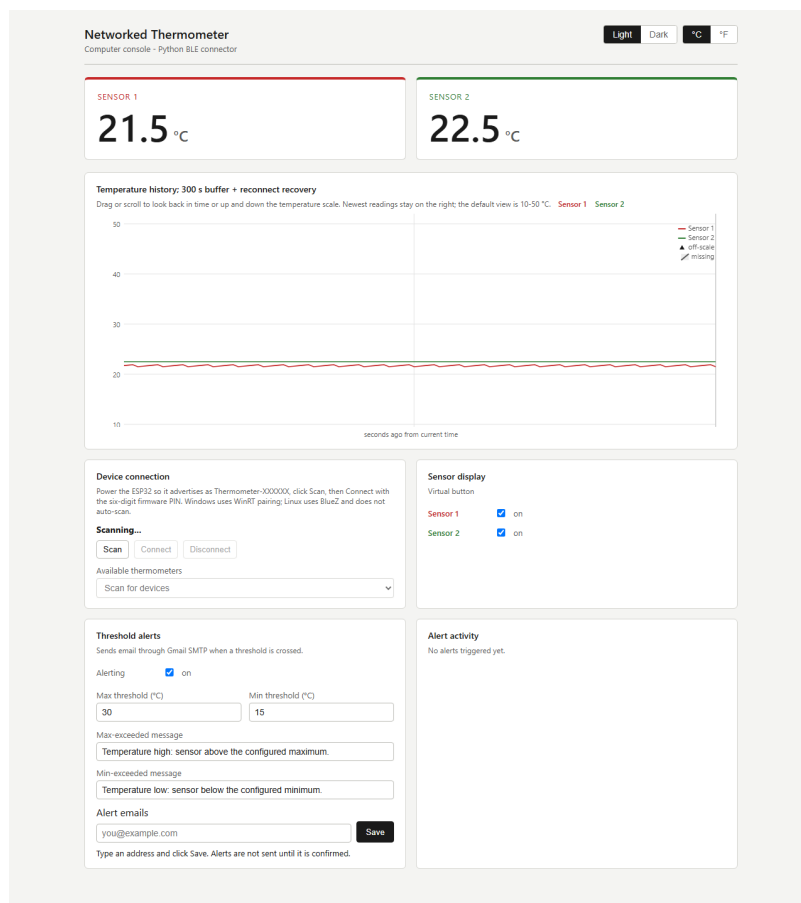


Figure 9: Selected visual evidence for FE-04

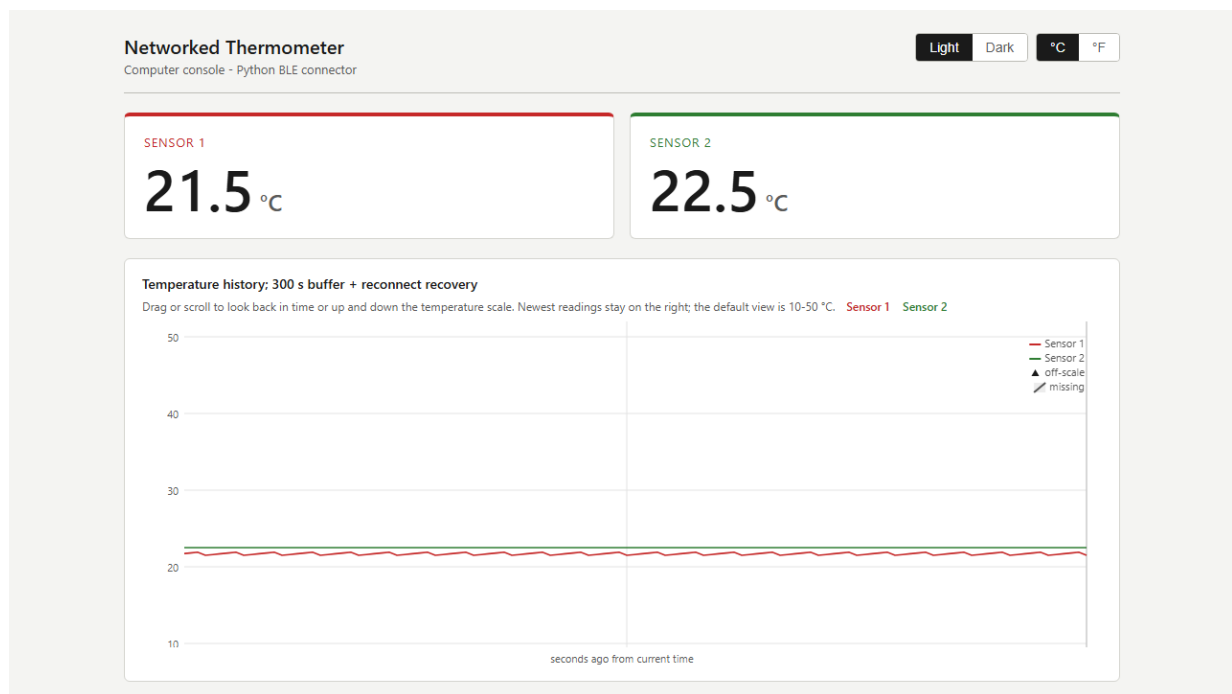


Figure 10: Selected visual evidence for FE-04

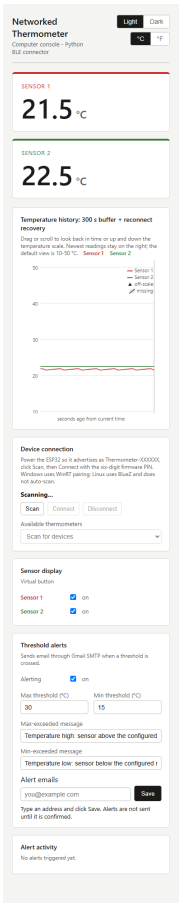


Figure 11: Selected visual evidence for FE-04

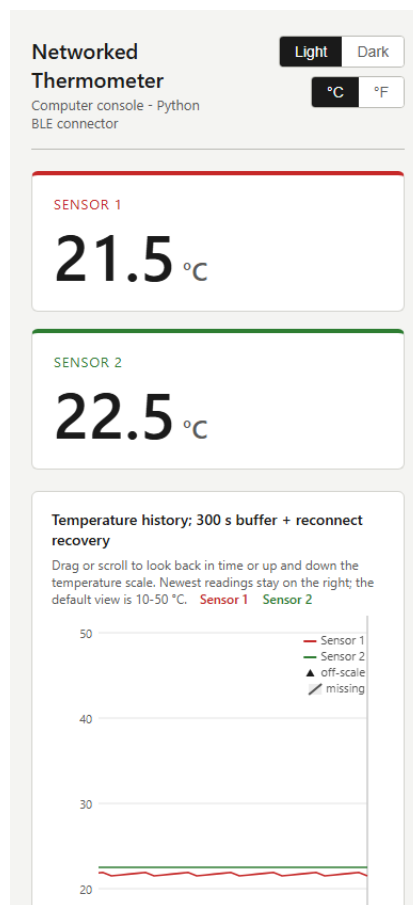


Figure 12: Selected visual evidence for FE-04

18 FE-05 — Trusted-Local Access Boundary and Validation

Outcome: PASS

Method: automated

Subsystem: frontend

18.1 What this test proves and why

This test verifies the following behavior: Checks deliberate absence of application accounts/credentials, loopback-scoped deployment, CORS restrictions, and server-side command/email validation. The inputs and expected results are controlled, so the check is repeatable and does not rely on operator judgment.

18.2 Passing criteria

- The observed result must match this test objective: Checks deliberate absence of application accounts/credentials, loopback-scoped deployment, CORS restrictions, and server-side command/email validation.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

18.3 Required setup

Software, simulation, and container checks (software)

- No ESP32 or physical sensors are required for this section.
- Docker must be running for integration-profile MySQL checks.
- The runner may start local backend/frontend/database services, but it will not use the physical BLE device.

18.4 Procedure

1. Run the automated FE-05 qualification check. The runner applies the defined inputs, records the observations, and evaluates every required assertion.

18.5 Requirements covered

- **SWE-SEC-LLR-750** — PASS. FE-05 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-SEC-LLR-751** — PASS. FE-05 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-SEC-LLR-752** — PASS. FE-05 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-SEC-LLR-753** — PASS. FE-05 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-SEC-LLR-754** — PASS. FE-05 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-SEC-MLR-750** — PASS. FE-05 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-602** — PASS. FE-05 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

- **SWE-WEB-LLR-603** — PASS. FE-05 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-607** — PASS. FE-05 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-608** — PASS. FE-05 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-609** — PASS. FE-05 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-610** — PASS. FE-05 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-611** — PASS. FE-05 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-612** — PASS. FE-05 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-622** — PASS. FE-05 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-MLR-602** — PASS. FE-05 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-MLR-605** — PASS. FE-05 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-MLR-607** — PASS. FE-05 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

18.6 Recorded results and limitations

No structured measurements were recorded. **Limitations:** Network/firewall policy beyond loopback binding remains a deployment responsibility.

18.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

19 FE-06 — Email Alert Configuration and Delivery

Outcome: PASS

Method: automated

Subsystem: frontend

19.1 What this test proves and why

This test verifies the following behavior: Checks email-only recipients/rules, email validation, configurable thresholds/messages, Gmail-compatible SMTP delivery, and persistence schema. The inputs and expected results are controlled, so the check is repeatable and does not rely on operator judgment.

19.2 Passing criteria

- The observed result must match this test objective: Checks email-only recipients/rules, email validation, configurable thresholds/messages, Gmail-compatible SMTP delivery, and persistence schema.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

19.3 Required setup

Software, simulation, and container checks (software)

- No ESP32 or physical sensors are required for this section.
- Docker must be running for integration-profile MySQL checks.
- The runner may start local backend/frontend/database services, but it will not use the physical BLE device.

19.4 Procedure

1. Run the automated FE-06 qualification check. The runner applies the defined inputs, records the observations, and evaluates every required assertion.

19.5 Requirements covered

- **SWE-WEB-LLR-606** — PASS. FE-06 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-608** — PASS. FE-06 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-609** — PASS. FE-06 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-610** — PASS. FE-06 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-611** — PASS. FE-06 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-612** — PASS. FE-06 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-617** — PASS. FE-06 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

- **SWE-WEB-LLR-630** — PASS. FE-06 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-MLR-603** — PASS. FE-06 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-700** — PASS. FE-06 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

19.6 Recorded results and limitations

No structured measurements were recorded. **Limitations:** A real Gmail inbox confirmation remains part of SYS-05 because automated tests use a capture provider.

19.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

20 FE-07 — Remote Display Control UI

Outcome: PASS

Method: automated

Subsystem: frontend

20.1 What this test proves and why

This test verifies the following behavior: Tests independent controls, pending/completed/failed states, authoritative returned state, and no optimistic success. The inputs and expected results are controlled, so the check is repeatable and does not rely on operator judgment.

20.2 Passing criteria

- The observed result must match this test objective: Tests independent controls, pending/completed/failed states, authoritative returned state, and no optimistic success.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

20.3 Required setup

Software, simulation, and container checks (software)

- No ESP32 or physical sensors are required for this section.
- Docker must be running for integration-profile MySQL checks.
- The runner may start local backend/frontend/database services, but it will not use the physical BLE device.

20.4 Procedure

1. Run the automated FE-07 qualification check. The runner applies the defined inputs, records the observations, and evaluates every required assertion.

20.5 Requirements covered

- **SWE-WEB-LLR-602** — PASS. FE-07 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-603** — PASS. FE-07 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-606** — PASS. FE-07 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-608** — PASS. FE-07 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-609** — PASS. FE-07 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-613** — PASS. FE-07 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-622** — PASS. FE-07 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

- **SWE-WEB-LLR-623** — PASS. FE-07 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-LLR-630** — PASS. FE-07 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-MLR-600** — PASS. FE-07 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-MLR-602** — PASS. FE-07 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-WEB-MLR-607** — PASS. FE-07 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-510** — PASS. FE-07 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

20.6 Recorded results and limitations

No structured measurements were recorded. **Limitations:** UI behavior alone does not prove the physical <1 second response; SYS-04 is mandatory.

20.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

21 ALR-01 — Alert Rule Matrix

Outcome: PASS

Method: automated

Subsystem: alerting

21.1 What this test proves and why

This test verifies the following behavior: Table-drives HIGH/LOW/normal/missing/disconnected/invalid, both sensors/average, multiple recipients, and independent thresholds. The inputs and expected results are controlled, so the check is repeatable and does not rely on operator judgment.

21.2 Passing criteria

- The observed result must match this test objective: Table-drives HIGH/LOW/normal/missing/disconnected/invalid, both sensors/average, multiple recipients, and independent thresholds.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

21.3 Required setup

Software, simulation, and container checks (software)

- No ESP32 or physical sensors are required for this section.
- Docker must be running for integration-profile MySQL checks.
- The runner may start local backend/frontend/database services, but it will not use the physical BLE device.

21.4 Procedure

1. Run the automated ALR-01 qualification check. The runner applies the defined inputs, records the observations, and evaluates every required assertion.

21.5 Requirements covered

- **SWE-ALR-LLR-700** — PASS. ALR-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-ALR-LLR-701** — PASS. ALR-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-ALR-LLR-702** — PASS. ALR-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-ALR-LLR-703** — PASS. ALR-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-ALR-LLR-704** — PASS. ALR-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-ALR-LLR-705** — PASS. ALR-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-ALR-LLR-706** — PASS. ALR-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

- **SWE-ALR-LLR-707** — PASS. ALR-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-ALR-LLR-709** — PASS. ALR-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-ALR-LLR-710** — PASS. ALR-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-ALR-LLR-711** — PASS. ALR-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-ALR-LLR-712** — PASS. ALR-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-ALR-LLR-713** — PASS. ALR-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-ALR-MLR-700** — PASS. ALR-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-ALR-MLR-701** — PASS. ALR-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-ALR-MLR-702** — PASS. ALR-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-ALR-MLR-704** — PASS. ALR-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-700** — PASS. ALR-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

21.6 Recorded results and limitations

No structured measurements were recorded. **Limitations:** Does not prove delivery by a real third-party provider.

21.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

22 ALR-02 — Alert Episode State Machine

Outcome: PASS

Method: automated

Subsystem: alerting

22.1 What this test proves and why

This test verifies the following behavior: Checks edge-triggered HIGH/LOW episodes, suppression while out of range, NORMAL reset, and re-alert. The inputs and expected results are controlled, so the check is repeatable and does not rely on operator judgment.

22.2 Passing criteria

- The observed result must match this test objective: Checks edge-triggered HIGH/LOW episodes, suppression while out of range, NORMAL reset, and re-alert.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

22.3 Required setup

Software, simulation, and container checks (software)

- No ESP32 or physical sensors are required for this section.
- Docker must be running for integration-profile MySQL checks.
- The runner may start local backend/frontend/database services, but it will not use the physical BLE device.

22.4 Procedure

1. Run the automated ALR-02 qualification check. The runner applies the defined inputs, records the observations, and evaluates every required assertion.

22.5 Requirements covered

- **SWE-ALR-LLR-711** — PASS. ALR-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-ALR-LLR-712** — PASS. ALR-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-ALR-LLR-713** — PASS. ALR-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-ALR-MLR-705** — PASS. ALR-02 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

22.6 Recorded results and limitations

No structured measurements were recorded.

22.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

23 ALR-03 — Notification Provider Adapters and Failure Isolation

Outcome: PASS

Method: automated

Subsystem: alerting

23.1 What this test proves and why

This test verifies the following behavior: Checks email destinations/messages, external SMTP credentials, and failure isolation from subsequent polling. The inputs and expected results are controlled, so the check is repeatable and does not rely on operator judgment.

23.2 Passing criteria

- The observed result must match this test objective: Checks email destinations/messages, external SMTP credentials, and failure isolation from subsequent polling.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

23.3 Required setup

Software, simulation, and container checks (software)

- No ESP32 or physical sensors are required for this section.
- Docker must be running for integration-profile MySQL checks.
- The runner may start local backend/frontend/database services, but it will not use the physical BLE device.

23.4 Procedure

1. Run the automated ALR-03 qualification check. The runner applies the defined inputs, records the observations, and evaluates every required assertion.

23.5 Requirements covered

- **SWE-ALR-LLR-700** — PASS. ALR-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-ALR-LLR-706** — PASS. ALR-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-ALR-LLR-707** — PASS. ALR-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-ALR-LLR-708** — PASS. ALR-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-ALR-MLR-700** — PASS. ALR-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SWE-ALR-MLR-703** — PASS. ALR-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-700** — PASS. ALR-03 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

23.6 Recorded results and limitations

No structured measurements were recorded. **Limitations:** Fake providers do not prove delivery to a real email inbox; SYS-05 provides controlled end-to-end evidence.

23.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

24 DOC-01 — Requirement Coverage Integrity

Outcome: PASS

Method: automated

Subsystem: documentation

24.1 What this test proves and why

This test verifies the following behavior: Fails on unmapped/unknown requirements, missing tests, invalid physical claims, or mishandled TBD/conflict requirements. The inputs and expected results are controlled, so the check is repeatable and does not rely on operator judgment.

24.2 Passing criteria

- The observed result must match this test objective: Fails on unmapped/unknown requirements, missing tests, invalid physical claims, or mishandled TBD/conflict requirements.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

24.3 Required setup

Software, simulation, and container checks (software)

- No ESP32 or physical sensors are required for this section.
- Docker must be running for integration-profile MySQL checks.
- The runner may start local backend/frontend/database services, but it will not use the physical BLE device.

24.4 Procedure

1. Run the automated DOC-01 qualification check. The runner applies the defined inputs, records the observations, and evaluates every required assertion.

24.5 Requirements covered

- **PRJ-REQ-001** — PASS. DOC-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **PRJ-REQ-003** — PASS. DOC-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.
- **PRJ-REQ-004** — PASS. DOC-01 checks this requirement using repeatable software checks and controlled inputs. It passes only when every required observation and limit is satisfied.

24.6 Recorded results and limitations

No structured measurements were recorded. **Limitations:** Checks traceability integrity, not product behavior.

24.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

25 HW-01 — Physical Configuration and Design Inspection

Outcome: PASS

Method: semi-automated

Subsystem: hardware

25.1 What this test proves and why

This test verifies the following behavior: Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range. A person performs the physical action; the runner records the resulting states or timing and applies the pass limits.

25.2 Passing criteria

- The observed result must match this test objective: Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

25.3 Required setup

Battery-powered BLE and physical-sensor checks (UART disconnected) (ble_no_uart)

- Disconnect the ESP32 micro-USB/UART cable from the PC. It can back-power the board and invalidate later power-off observations.
- Power the enclosure from its battery or intended non-PC supply and switch it ON.
- Connect both physical temperature sensors and confirm the built-in PC Bluetooth radio is ON.
- Close ESP-IDF Monitor and connect the thermometer from the real web UI when prompted.

25.4 Procedure

1. Human evidence is mandatory. Measure both sensor cables (pass: 0.9–1.1 m); inspect strain relief, permanent port identification, connector support/protection, battery/regulator/switch, ESP32/BLE capability, sensor range/accuracy datasheets, LCD character capacity/readability, signed values, and normal indoor readability. Record model/part numbers, numerical measurements, datasheet revisions, and photo paths. Adjustable back-light control is outside project scope and is not inspected. This inspection supports design/construction claims but does not replace HW-02 drop testing or HW-04 accuracy testing.
2. Prepare every fixture and instrumentation item listed for HW-01. Expected evidence: yes.
3. Measures cable lengths and inspects construction, parts, power, labels, LCD capacity/readability, and supported design range. Expected evidence: yes/no.
4. Record measurement/evidence file paths in comments and retain them under this test's evidence directory. Expected evidence: yes.

25.5 Requirements covered

- **HWE-LLR-200** — PASS. HW-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-LLR-201** — PASS. HW-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.

- [illegible]

and timing capture. It passes only when every required observation and limit is satisfied.

- **HWE-MLR-205** — PASS. HW-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-MLR-206** — PASS. HW-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-MLR-209** — PASS. HW-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-MLR-210** — PASS. HW-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-MLR-211** — PASS. HW-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-MLR-212** — PASS. HW-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-MLR-213** — PASS. HW-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-MLR-214** — PASS. HW-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-MLR-215** — PASS. HW-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-MLR-216** — PASS. HW-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **PRJ-REQ-005** — PASS. HW-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-100** — PASS. HW-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-200** — PASS. HW-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-300** — PASS. HW-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-410** — PASS. HW-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.

25.6 Recorded results and limitations

component_models_and_revisions: N/A

construction_inspection: Pass

datasheet_range: Pass

initial_hardware_check: BLE backend reports connected and ready

lcd_readability: Pass

sensor_1_cable_length_m: 1.4

sensor_2_cable_length_m: 1.4

Operator decision: Works as intended **Limitations:** Physical dimensions, construction and datasheet capabilities require an operator and photographs.

25.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

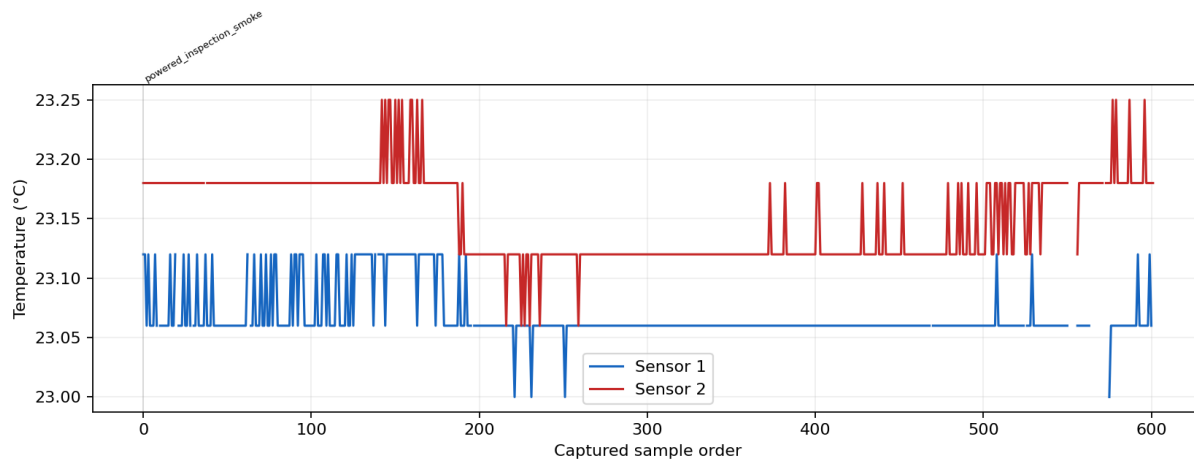


Figure 13: Selected visual evidence for HW-01

26 HW-04 — Temperature Accuracy and Thermal Response

Outcome: PASS

Method: semi-automated

Subsystem: hardware

26.1 What this test proves and why

This test verifies the following behavior: Captures room, ice-water, hand-heating, and soldering-iron response measurements for both sensors. A person performs the physical action; the runner records the resulting states or timing and applies the pass limits.

26.2 Passing criteria

- The observed result must match this test objective: Captures room, ice-water, hand-heating, and soldering-iron response measurements for both sensors.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

26.3 Required setup

Battery-powered BLE and physical-sensor checks (UART disconnected) (ble_no_uart)

- Disconnect the ESP32 micro-USB/UART cable from the PC. It can back-power the board and invalidate later power-off observations.
- Power the enclosure from its battery or intended non-PC supply and switch it ON.
- Connect both physical temperature sensors and confirm the built-in PC Bluetooth radio is ON.
- Close ESP-IDF Monitor and connect the thermometer from the real web UI when prompted.

26.4 Procedure

1. Capture both sensors and a reference at room temperature (pass target $22\text{ }^{\circ}\text{C} \pm 4\text{ }^{\circ}\text{C}$) and in a properly prepared stirred ice-water bath ($0\text{ }^{\circ}\text{C} \pm 2\text{ }^{\circ}\text{C}$). For each sensor, time an observable rise after hand heating and a faster rise from safe nearby/brief soldering-iron heating. Preserve the complete CSV, reference identity/calibration information, setup photos, timestamps, and safety notes. Code/simulator results do not verify physical accuracy.
2. Prepare every fixture and instrumentation item listed for HW-04. Expected evidence: yes.
3. Captures room, ice-water, hand-heating, and soldering-iron response measurements for both sensors. Expected evidence: yes/no.
4. Record measurement/evidence file paths in comments and retain them under this test's evidence directory. Expected evidence: yes.

26.5 Requirements covered

- **HWE-LLR-201** — PASS. HW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-LLR-208** — PASS. HW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-LLR-209** — PASS. HW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.

- **HWE-LLR-218** — PASS. HW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-LLR-219** — PASS. HW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-LLR-220** — PASS. HW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-LLR-224** — PASS. HW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-LLR-225** — PASS. HW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-MLR-200** — PASS. HW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-MLR-201** — PASS. HW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-MLR-210** — PASS. HW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-MLR-211** — PASS. HW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-MLR-212** — PASS. HW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-MLR-216** — PASS. HW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-300** — PASS. HW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-310** — PASS. HW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-320** — PASS. HW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-321** — PASS. HW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-322** — PASS. HW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.

26.6 Recorded results and limitations

initial_hardware_check: BLE backend reports connected and ready

sensor_1_hand_rise_c: 1.3100000000000023

sensor_1_ice_c: 11.43

sensor_1_iron_response_faster: True

sensor_1_iron_rise_c: 4.0

sensor_1_room_c: 23.12

sensor_2_hand_rise_c: 2.5

sensor_2_ice_c: 12.25

sensor_2_iron_response_faster: True

sensor_2_iron_rise_c: 25.25

sensor_2_room_c: 23.25

Operator decision: Previously adjudicated passing behavior is accepted as a standard PASS for this finalized run; original observations and adjudication history remain retained. **Limitations:** Physical accuracy and thermal response require calibrated references and operator handling.

26.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

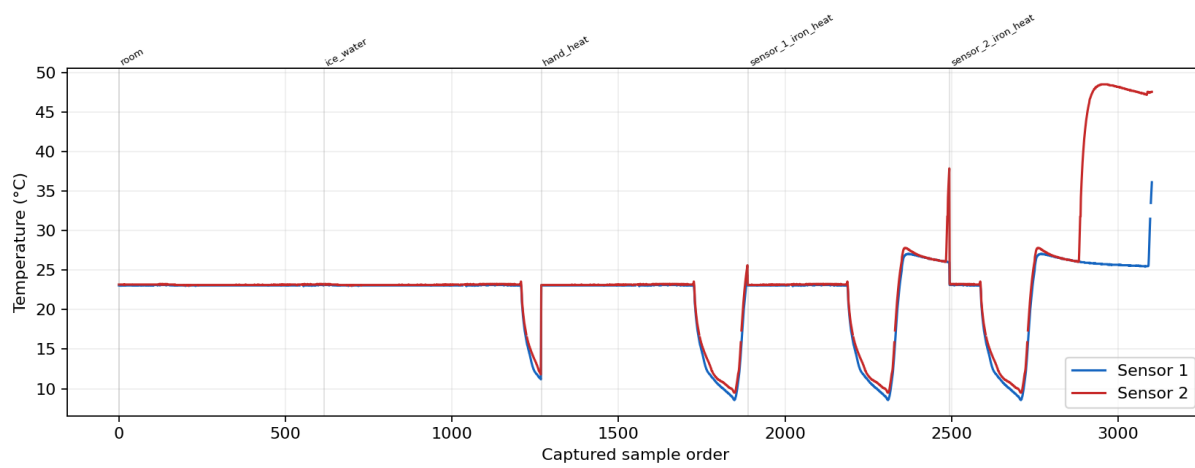


Figure 14: Selected visual evidence for HW-04

27 FW-05 — Shared Local/Remote Display State

Outcome: PASS

Method: semi-automated

Subsystem: firmware

27.1 What this test proves and why

This test verifies the following behavior: Confirms local and remote commands update one authoritative display state and malformed commands do not change it. A person performs the physical action; the runner records the resulting states or timing and applies the pass limits.

27.2 Passing criteria

- The observed result must match this test objective: Confirms local and remote commands update one authoritative display state and malformed commands do not change it.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

27.3 Required setup

Battery-powered BLE and physical-sensor checks (UART disconnected) (ble_no_uart)

- Disconnect the ESP32 micro-USB/UART cable from the PC. It can back-power the board and invalidate later power-off observations.
- Power the enclosure from its battery or intended non-PC supply and switch it ON.
- Connect both physical temperature sensors and confirm the built-in PC Bluetooth radio is ON.
- Close ESP-IDF Monitor and connect the thermometer from the real web UI when prompted.

27.4 Procedure

1. Capture serial and BLE transactions. Alternate local button and remote SET_DISPLAY commands for both sensors and verify both paths mutate the same authoritative state shown by the LCD and current snapshot. Send invalid sensor/value/length commands and confirm rejection with no state change. Physical controls and display observation are mandatory.
2. Prepare every fixture and instrumentation item listed for FW-05. Expected evidence: yes.
3. Confirms local and remote commands update one authoritative display state and malformed commands do not change it. Expected evidence: yes/no.
4. Record measurement/evidence file paths in comments and retain them under this test's evidence directory. Expected evidence: yes.

27.5 Requirements covered

- **SWE-EMB-MLR-313** — PASS. FW-05 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-400** — PASS. FW-05 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.

27.6 Recorded results and limitations

initial_hardware_check: BLE backend reports connected and ready

invalid_sensor_rejected: True

sensor_1_physical_toggle_observed: True

sensor_1_remote_command_ms: 25.8

sensor_1_remote_confirmed: True

sensor_2_physical_toggle_observed: True

sensor_2_remote_command_ms: 22.0

sensor_2_remote_confirmed: True

Limitations: Physical buttons and LCD require operator observation.

27.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

28 BLE-02 — Authenticated Display Control

Outcome: PASS

Method: semi-automated

Subsystem: ble

28.1 What this test proves and why

This test verifies the following behavior: Confirms authenticated SET_DISPLAY and rejection of unauthorized/malformed state-changing requests. A person performs the physical action; the runner records the resulting states or timing and applies the pass limits.

28.2 Passing criteria

- The observed result must match this test objective: Confirms authenticated SET_DISPLAY and rejection of unauthorized/malformed state-changing requests.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

28.3 Required setup

Battery-powered BLE and physical-sensor checks (UART disconnected) (ble_no_uart)

- Disconnect the ESP32 micro-USB/UART cable from the PC. It can back-power the board and invalidate later power-off observations.
- Power the enclosure from its battery or intended non-PC supply and switch it ON.
- Connect both physical temperature sensors and confirm the built-in PC Bluetooth radio is ON.
- Close ESP-IDF Monitor and connect the thermometer from the real web UI when prompted.

28.4 Procedure

1. Using a real ESP32, run an authorized SET_DISPLAY command and verify the physical LCD. From an unbonded/unauthenticated client, attempt state-changing access and verify rejection. Send malformed sensor IDs, values, lengths, and protocol versions and verify defined error status with no LCD state change. Preserve BLE/serial traces; a backend acknowledgement alone is insufficient.
2. Prepare every fixture and instrumentation item listed for BLE-02. Expected evidence: yes.
3. Confirms authenticated SET_DISPLAY and rejection of unauthorized/malformed state-changing requests. Expected evidence: yes/no.
4. Record measurement/evidence file paths in comments and retain them under this test's evidence directory. Expected evidence: yes.

28.5 Requirements covered

- **INT-LLR-405** — PASS. BLE-02 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **INT-LLR-406** — PASS. BLE-02 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **INT-LLR-413** — PASS. BLE-02 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.

- **INT-LLR-414** — PASS. BLE-02 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **INT-LLR-416** — PASS. BLE-02 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **INT-MLR-406** — PASS. BLE-02 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-510** — PASS. BLE-02 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.

28.6 Recorded results and limitations

authorized_command_ms: 1.4

authorized_confirmed: True

initial_hardware_check: BLE backend reports connected and ready

malformed_inputs_rejected: True

physical_lcd: LCD changed correctly

unauthorized_client: Write rejected; state unchanged

Limitations: Requires real bonded and untrusted BLE clients.

28.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

29 BLE-03 — History Transfer and MTU Adaptation

Outcome: PASS

Method: semi-automated

Subsystem: ble

29.1 What this test proves and why

This test verifies the following behavior: Validates metadata-first chunks, boundaries, retries, complete reconstruction, small MTU behavior, and negotiated real MTU. A person performs the physical action; the runner records the resulting states or timing and applies the pass limits.

29.2 Passing criteria

- The observed result must match this test objective: Validates metadata-first chunks, boundaries, retries, complete reconstruction, small MTU behavior, and negotiated real MTU.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

29.3 Required setup

Battery-powered BLE and physical-sensor checks (UART disconnected) (ble_no_uart)

- Disconnect the ESP32 micro-USB/UART cable from the PC. It can back-power the board and invalidate later power-off observations.
- Power the enclosure from its battery or intended non-PC supply and switch it ON.
- Connect both physical temperature sensors and confirm the built-in PC Bluetooth radio is ON.
- Close ESP-IDF Monitor and connect the thermometer from the real web UI when prompted.

29.4 Procedure

1. Populate at least 300 seconds. Capture real negotiated MTU and request metadata first, then reconstruct all required oldest-to-newest chunks. Exercise/review retry evidence and a small-MTU case; verify only complete records and exact boot/sequence/status/value reconstruction. The simulator can cover algorithms, but real negotiated-MTU evidence is required here.
2. Prepare every fixture and instrumentation item listed for BLE-03. Expected evidence: yes.
3. Validates metadata-first chunks, boundaries, retries, complete reconstruction, small MTU behavior, and negotiated real MTU. Expected evidence: yes/no.
4. Record measurement/evidence file paths in comments and retain them under this test's evidence directory. Expected evidence: yes.

29.5 Requirements covered

- **INT-LLR-401** — PASS. BLE-03 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **INT-LLR-407** — PASS. BLE-03 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **INT-LLR-408** — PASS. BLE-03 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.

- **INT-LLR-409** — PASS. BLE-03 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **INT-LLR-410** — PASS. BLE-03 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **INT-LLR-411** — PASS. BLE-03 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **INT-MLR-404** — PASS. BLE-03 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-600** — PASS. BLE-03 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.

29.6 Recorded results and limitations

be-02_precheck: True

ble-01_precheck: True

graph_confirmation: Graph shows recovered history

history_operation_id: 470ac0f7-b2bb-4f84-b185-cd289dc32721

history_rows: 604

history_sync:

history_to_database_ms: 10155.2

initial_hardware_check: BLE backend reports connected and ready

sensor_1_history_records: 604

sensor_2_history_records: 604

Operator decision: Previously adjudicated passing behavior is accepted as a standard PASS for this finalized run; original observations and adjudication history remain retained. **Limitations:** Simulated MTU checks are partial until a negotiated hardware MTU is captured.

29.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

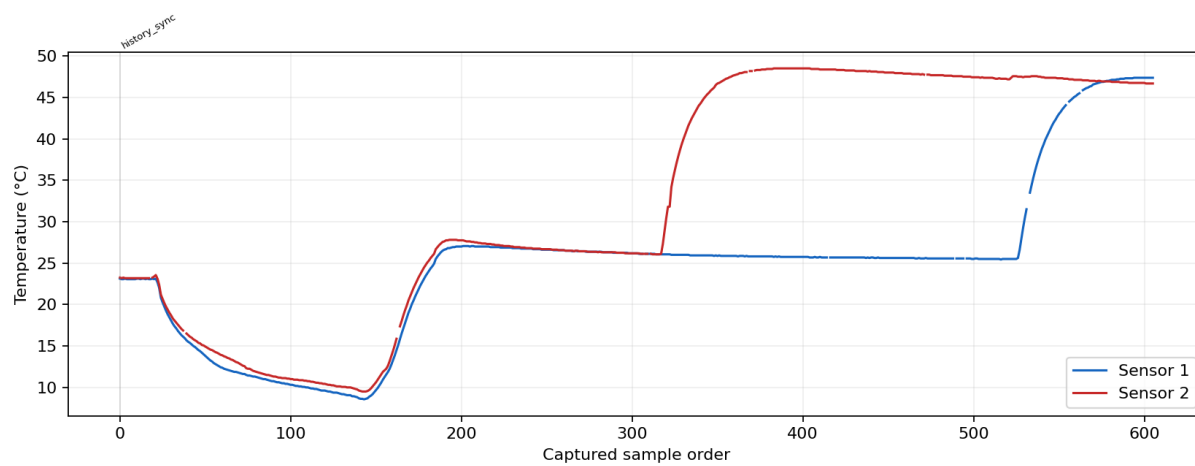


Figure 15: Selected visual evidence for BLE-03

30 BE-01 — Worker Isolation, 1-Hz Polling, and Persistence

Outcome: PASS

Method: semi-automated

Subsystem: backend

30.1 What this test proves and why

This test verifies the following behavior: Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states. A person performs the physical action; the runner records the resulting states or timing and applies the pass limits.

30.2 Passing criteria

- The observed result must match this test objective: Runs the backend worker unit precheck, then guides Sensor 1 and Sensor 2 disconnect/reconnect actions while measuring persisted 1 Hz fault, independent-channel, and recovery states.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

30.3 Required setup

Battery-powered BLE and physical-sensor checks (UART disconnected) (ble_no_uart)

- Disconnect the ESP32 micro-USB/UART cable from the PC. It can back-power the board and invalidate later power-off observations.
- Power the enclosure from its battery or intended non-PC supply and switch it ON.
- Connect both physical temperature sensors and confirm the built-in PC Bluetooth radio is ON.
- Close ESP-IDF Monitor and connect the thermometer from the real web UI when prompted.

30.4 Procedure

1. The runner first executes the deterministic backend worker tests. It then confirms the real third box is connected, captures a baseline, prompts the operator to disconnect and reconnect each sensor in turn, and records backend status plus MySQL-backed `/api/samples` data throughout. Pass/fail is computed from fault visibility, continued valid samples from the unaffected sensor, persistence, and automatic recovery. The operator only performs the physical cable actions.
2. Connect the real third box and both sensors; the runner verifies backend readiness. Expected evidence: yes.

30.5 Requirements covered

- **SWE-CONN-LLR-500** — PASS. BE-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SWE-CONN-LLR-501** — PASS. BE-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SWE-CONN-LLR-502** — PASS. BE-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SWE-CONN-LLR-503** — PASS. BE-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.

- [illegible]

30.6 Recorded results and limitations

```
backend unit precheck: True
```

baseline boot id: 1965552206

initial_hardware_check: BLE backend reports connected and ready

sensor_1_continued_during_sensor_2_on_fault: True

sensor_1_on_automatic_recovery: True

sensor_1_on_boot_id_unchanged: True

sensor_1_on_fault_observed: True

sensor_1_on_state_preserved_during_fault: True

sensor_1_on_state_restored: True

sensor_2_continued_during_sensor_1_on_fault: True

sensor_2_on_automatic_recovery: True

sensor_2_on_boot_id_unchanged: True

sensor_2_on_fault_observed: True

sensor_2_on_state_preserved_during_fault: True

sensor_2_on_state_restored: True

Limitations: The physical phase is operator-triggered; the 300-second scheduler tolerance remains established by the automated precheck.

30.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

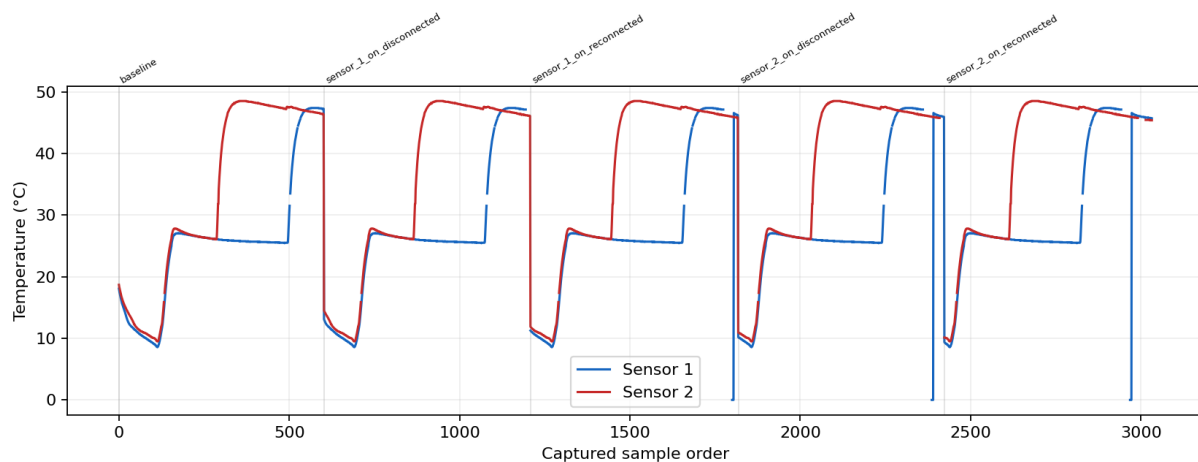


Figure 16: Selected visual evidence for BE-01

31 SYS-01 — Complete Data Path

Outcome: PASS

Method: semi-automated

Subsystem: system

31.1 What this test proves and why

This test verifies the following behavior: Verifies both physical sensor values/status through ESP32, BLE, backend, MySQL, server, and browser. A person performs the physical action; the runner records the resulting states or timing and applies the pass limits.

31.2 Passing criteria

- The observed result must match this test objective: Verifies both physical sensor values/status through ESP32, BLE, backend, MySQL, server, and browser.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

31.3 Required setup

Battery-powered BLE and physical-sensor checks (UART disconnected) (ble_no_uart)

- Disconnect the ESP32 micro-USB/UART cable from the PC. It can back-power the board and invalidate later power-off observations.
- Power the enclosure from its battery or intended non-PC supply and switch it ON.
- Connect both physical temperature sensors and confirm the built-in PC Bluetooth radio is ON.
- Close ESP-IDF Monitor and connect the thermometer from the real web UI when prompted.

31.4 Procedure

1. For both real sensors, record a distinguishable physical stimulus and compare the ESP32/LCD value and status with BLE trace, backend record, MySQL row, sample API, and browser readout/graph. Preserve one correlated evidence set per sensor. Every hop—sensor → ESP32 → BLE → backend → MySQL → Node API → browser—must be observed; simulated input does not pass the physical system requirement.
2. Prepare every fixture and instrumentation item listed for SYS-01. Expected evidence: yes.
3. Verifies both physical sensor values/status through ESP32, BLE, backend, MySQL, server, and browser. Expected evidence: yes/no.
4. Record measurement/evidence file paths in comments and retain them under this test's evidence directory. Expected evidence: yes.

31.5 Requirements covered

- **PRJ-REQ-002** — PASS. SYS-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **PRJ-REQ-003** — PASS. SYS-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-100** — PASS. SYS-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.

- **SYS-HLR-110** — PASS. SYS-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-500** — PASS. SYS-01 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.

31.6 Recorded results and limitations

initial_hardware_check: BLE backend reports connected and ready

operator_correlation: Both sensors match at all visible hops

sensor_1_backend_database_match: True

sensor_2_backend_database_match: True

Limitations: Requires operator comparison of physical LCD/sensors and browser values.

31.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

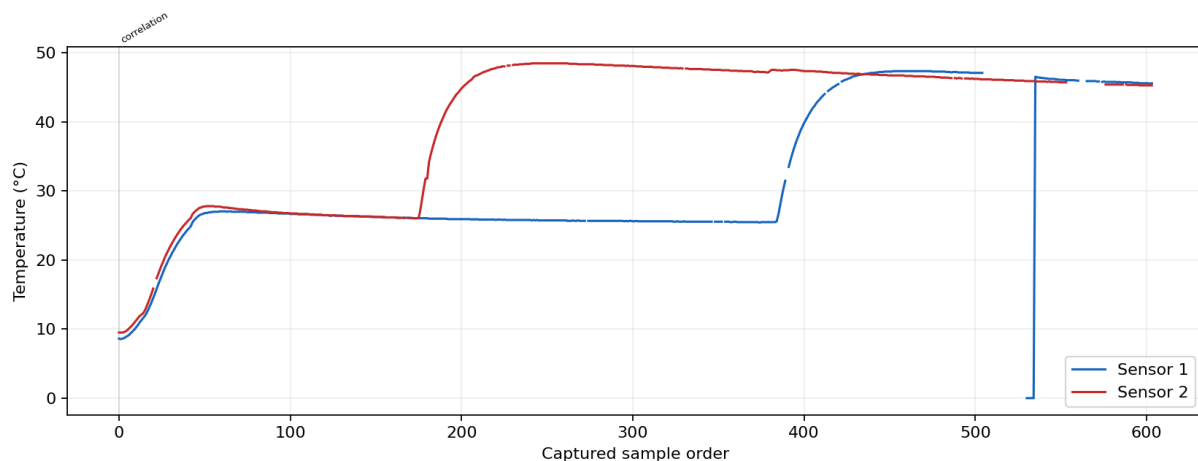


Figure 17: Selected visual evidence for SYS-01

32 SYS-02 — Full-System Sensor Disconnect and Recovery

Outcome: PASS

Method: semi-automated

Subsystem: system

32.1 What this test proves and why

This test verifies the following behavior: Disconnects each real sensor in both ON and OFF display states and verifies other-sensor continuity, local/remote fault, graph gap, automatic recovery, and preservation of the pre-disconnection state. A person performs the physical action; the runner records the resulting states or timing and applies the pass limits.

32.2 Passing criteria

- The observed result must match this test objective: Disconnects each real sensor in both ON and OFF display states and verifies other-sensor continuity, local/remote fault, graph gap, automatic recovery, and preservation of the pre-disconnection state.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

32.3 Required setup

Battery-powered BLE and physical-sensor checks (UART disconnected) (ble_no_uart)

- Disconnect the ESP32 micro-USB/UART cable from the PC. It can back-power the board and invalidate later power-off observations.
- Power the enclosure from its battery or intended non-PC supply and switch it ON.
- Connect both physical temperature sensors and confirm the built-in PC Bluetooth radio is ON.
- Close ESP-IDF Monitor and connect the thermometer from the real web UI when prompted.

32.4 Procedure

1. For each sensor, test both pre-disconnection display states (ON and OFF). The runner first sets and confirms the state, then actively disconnects the computer's BLE link. While the PC is disconnected, unplug and reconnect that physical sensor without restarting the ESP32. Confirm on the physical LCD that the other sensor continues and that the recovered sensor independently returns to its pre-disconnection state. The runner then reconnects the backend and confirms the firmware-reported state. Repeat all four sensor/state combinations.
2. This separation is essential: backend observations alone cannot prove that state restoration is implemented locally in firmware. Any unperformed action or unavailable visual observation is BLOCKED, while an observed wrong state is FAIL.
3. Prepare every fixture and instrumentation item listed for SYS-02. Expected evidence: yes.
4. Disconnects each real sensor and verifies other-sensor continuity, local/remote fault, graph gap, automatic recovery, and initial OFF. Expected evidence: yes/no.
5. Record measurement/evidence file paths in comments and retain them under this test's evidence directory. Expected evidence: yes.

32.5 Requirements covered

- **SWE-EMB-LLR-311** — PASS. SYS-02 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-210** — PASS. SYS-02 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-420** — PASS. SYS-02 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.

32.6 Recorded results and limitations

initial_hardware_check: BLE backend reports connected and ready

sensor_1_on_backend_reconfirmed: False

sensor_1_on_local_restore_without_pc: True

Operator decision: Previously adjudicated passing behavior is accepted as a standard PASS for this finalized run; original observations and adjudication history remain retained. **Limitations:** Requires physical cable manipulation.

32.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

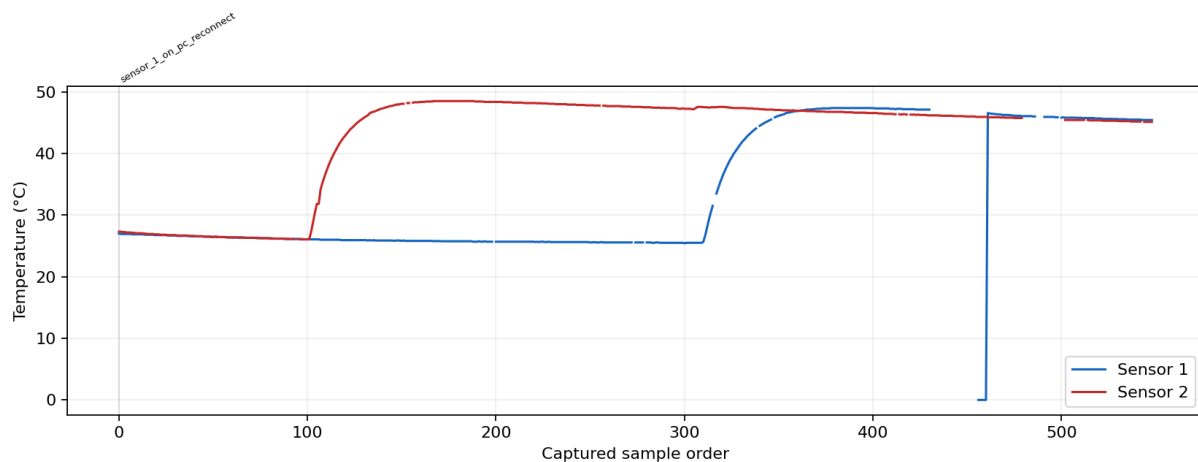


Figure 18: Selected visual evidence for SYS-02

33 SYS-05 — End-to-End Notifications

Outcome: PASS

Method: semi-automated

Subsystem: system

33.1 What this test proves and why

This test verifies the following behavior: Configures actual UI rules, triggers controlled crossings, and verifies storage, evaluation, episode transition, provider, destination, and message. A person performs the physical action; the runner records the resulting states or timing and applies the pass limits.

33.2 Passing criteria

- The observed result must match this test objective: Configures actual UI rules, triggers controlled crossings, and verifies storage, evaluation, episode transition, provider, destination, and message.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

33.3 Required setup

Battery-powered BLE and physical-sensor checks (UART disconnected) (ble_no_uart)

- Disconnect the ESP32 micro-USB/UART cable from the PC. It can back-power the board and invalidate later power-off observations.
- Power the enclosure from its battery or intended non-PC supply and switch it ON.
- Connect both physical temperature sensors and confirm the built-in PC Bluetooth radio is ON.
- Close ESP-IDF Monitor and connect the thermometer from the real web UI when prompted.

33.4 Procedure

1. Before starting the frontend on Windows, run 'frontend\configure-smtp.ps1'. Enter the Gmail sending address and a Google App Password when prompted; the password input is hidden and is stored only in the gitignored 'backend/.env'. Restart 'frontend/run.ps1', then confirm '/api/health' reports 'mode: live'.
2. Through the actual UI configure controlled email recipients, monitored source, independent thresholds, and high/low messages. Trigger controlled high and low crossings. Correlate the database sample, rule evaluation, episode transition, provider call, destination, and exact message. Confirm both controlled end-to-end emails in the destination inbox without placing credentials in evidence. SMS delivery is out of scope.
3. Prepare every fixture and instrumentation item listed for SYS-05. Expected evidence: yes.
4. Configures actual UI rules, triggers controlled crossings, and verifies storage, evaluation, episode transition, provider, destination, and message. Expected evidence: yes/no.
5. Record measurement/evidence file paths in comments and retain them under this test's evidence directory. Expected evidence: yes.

33.5 Requirements covered

- **SYS-HLR-700** — PASS. SYS-05 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.

33.6 Recorded results and limitations

email_mode: console

high_low_requests: 2

inbox_confirmation: Both HIGH and LOW emails received

initial_hardware_check: BLE backend reports connected and ready

Operator decision: Previously adjudicated passing behavior is accepted as a standard PASS for this finalized run; original observations and adjudication history remain retained. **Limitations:** Controlled provider evidence is qualification evidence; optional real delivery is a separate smoke test.

33.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

34 HW-02 — Mechanical Robustness

Outcome: PASS

Method: semi-automated

Subsystem: hardware

34.1 What this test proves and why

This test verifies the following behavior: Automatically captures pre/post full-system smoke evidence around guided drop, upside-down, connected-cable, handling, and immersion actions. A person performs the physical action; the runner records the resulting states or timing and applies the pass limits.

34.2 Passing criteria

- The observed result must match this test objective: Automatically captures pre/post full-system smoke evidence around guided drop, upside-down, connected-cable, handling, and immersion actions.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

34.3 Required setup

Power-switch, restart, and radio-loss checks (all UART/USB power removed) (power_cycle_no_uart)

- Physically disconnect every micro-USB/UART cable from the ESP32 before continuing; closing the serial monitor is not enough.
- Power the enclosure only from its battery or the intended switched supply so OFF truly removes ESP32 power.
- Keep the PC Bluetooth radio ON unless a test explicitly asks you to disable only that radio.
- Do not reconnect USB/UART during this section. These tests measure real restart and advertising behavior.

34.4 Procedure

1. Human evidence is mandatory. Photograph and functionally test the baseline. Perform the approved workbench-to-floor enclosure drop, upside-down operation, connected-cable drop, ordinary handling, and temporary sensor immersion. Photograph after-test condition, inspect all retention/strain relief/connectors, reconnect any permitted detached cable, and repeat the functional smoke test. Record height, surface, cable detachments, damage, and operator initials. A simulated or visual-only inspection cannot pass this test.
2. Prepare every fixture and instrumentation item listed for HW-02. Expected evidence: yes.
3. Records baseline, drop/upside-down/connected-cable/handling/immersion tests, post-test inspection, and functional smoke test. Expected evidence: yes/no.
4. Record measurement/evidence file paths in comments and retain them under this test's evidence directory. Expected evidence: yes.

34.5 Requirements covered

- **HWE-LLR-201** — PASS. HW-02 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-LLR-202** — PASS. HW-02 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.

- **HWE-LLR-203** — PASS. HW-02 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-LLR-204** — PASS. HW-02 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-LLR-205** — PASS. HW-02 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-LLR-223** — PASS. HW-02 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-MLR-201** — PASS. HW-02 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-MLR-202** — PASS. HW-02 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-MLR-203** — PASS. HW-02 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-MLR-215** — PASS. HW-02 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-200** — PASS. HW-02 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.

34.6 Recorded results and limitations

connected_cable_drop: Completed with no disallowed damage

enclosure_drop: Completed with no disallowed damage

initial_hardware_check: BLE backend reports connected and ready

ordinary_handling: Completed with no disallowed damage

post_test_smoke_pass: False

pre_test_smoke_pass: False

sensor_immersion: Completed with no disallowed damage

upside_down: Completed with no disallowed damage

Operator decision: Previously adjudicated passing behavior is accepted as a standard PASS for this finalized run; original observations and adjudication history remain retained. **Limitations:** No simulation or software-only evidence can satisfy physical robustness.

34.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

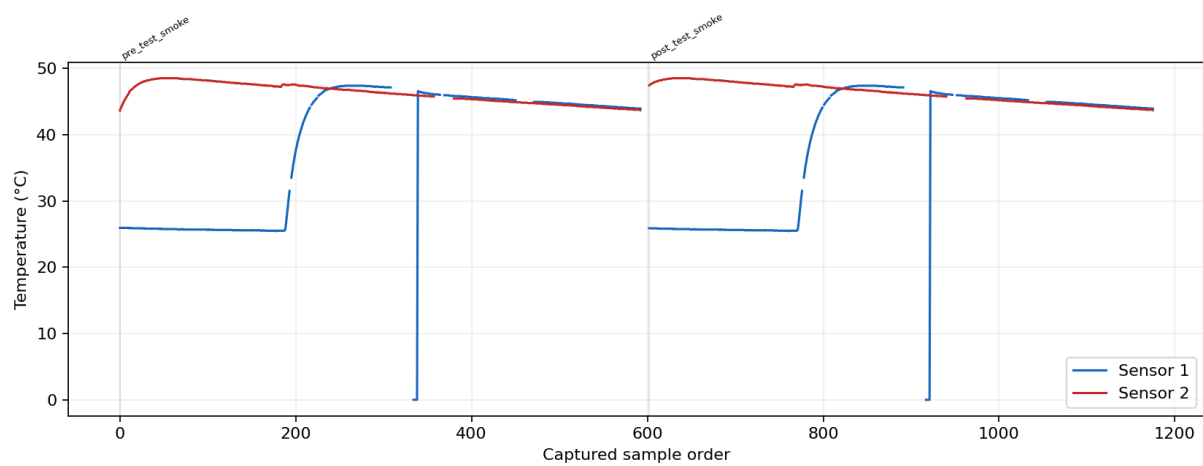


Figure 19: Selected visual evidence for HW-02

35 HW-03 — Power Switch and Hot-Plug

Outcome: PASS

Method: semi-automated

Subsystem: hardware

35.1 What this test proves and why

This test verifies the following behavior: Guides sensor hot-plug and power cycling while capturing independent operation, no-reset behavior, loss of current data, and recovery. A person performs the physical action; the runner records the resulting states or timing and applies the pass limits.

35.2 Passing criteria

- The observed result must match this test objective: Guides sensor hot-plug and power cycling while capturing independent operation, no-reset behavior, loss of current data, and recovery.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

35.3 Required setup

Power-switch, restart, and radio-loss checks (all UART/USB power removed) (power_cycle_no_uart)

- Physically disconnect every micro-USB/UART cable from the ESP32 before continuing; closing the serial monitor is not enough.
- Power the enclosure only from its battery or the intended switched supply so OFF truly removes ESP32 power.
- Keep the PC Bluetooth radio ON unless a test explicitly asks you to disable only that radio.
- Do not reconnect USB/UART during this section. These tests measure real restart and advertising behavior.

35.4 Procedure

1. Disconnect the ESP32 micro-USB/UART cable before starting and power the box only through its battery or intended switched supply. Disconnect/reconnect Sensor 1 and then Sensor 2 while powered; verify the other channel continues, the fault is reported, and each sensor restores its pre-disconnection ON/OFF state. Switch the third box OFF and verify local operation/current remote data cease; switch ON and verify automatic recovery. Backend, database, and current-data captures provide the machine evidence. Human cable/switch actions are required; UART evidence is deliberately excluded because USB power would invalidate the OFF state.
2. Prepare every fixture and instrumentation item listed for HW-03. Expected evidence: yes.
3. Guides sensor hot-plug and power cycling while capturing independent operation, no-reset behavior, loss of current data, and recovery. Expected evidence: yes/no.
4. Record measurement/evidence file paths in comments and retain them under this test's evidence directory. Expected evidence: yes.

35.5 Requirements covered

- **HWE-LLR-206** — PASS. HW-03 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-LLR-208** — PASS. HW-03 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.

- **HWE-MLR-204** — PASS. HW-03 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-MLR-205** — PASS. HW-03 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-MLR-211** — PASS. HW-03 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **HWE-MLR-215** — PASS. HW-03 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-210** — PASS. HW-03 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-220** — PASS. HW-03 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.

35.6 Recorded results and limitations

automatic_reconnect: True

baseline_boot_id: 1203119999

boot_id_after_power_on: 68608410

boot_id_before_power_off: 1203119999

initial_hardware_check: BLE backend reports connected and ready

link_loss_observed: True

physical_restart_confirmed: True

reconnect_ms: 1354.3

sensor_1_continued_during_sensor_2_off_fault: True

sensor_1_off_automatic_recovery: True

sensor_1_off_boot_id_unchanged: True

sensor_1_off_fault_observed: True

sensor_1_off_state_preserved_during_fault: True

sensor_1_off_state_restored: True

sensor_2_continued_during_sensor_1_off_fault: True

sensor_2_off_automatic_recovery: True

sensor_2_off_boot_id_unchanged: True

sensor_2_off_fault_observed: True

sensor_2_off_state_preserved_during_fault: True

sensor_2_off_state_restored: True

Limitations: An operator must physically disconnect sensors and operate the switch.

35.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

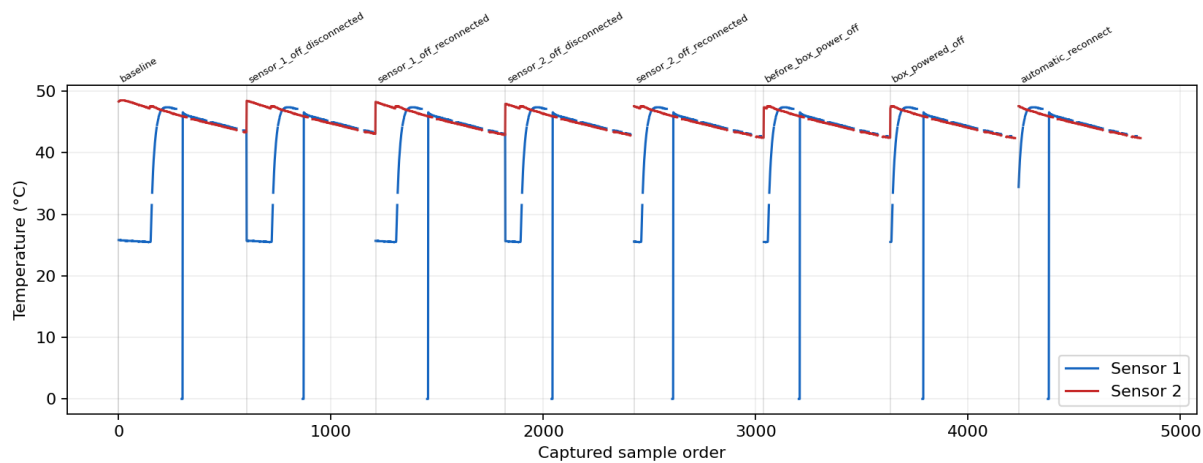


Figure 20: Selected visual evidence for HW-03

36 BLE-04 — BLE Link-Loss Recovery

Outcome: PASS

Method: semi-automated

Subsystem: ble

36.1 What this test proves and why

This test verifies the following behavior: Forces physical link loss and verifies automatic rediscovery/reconnection without web restart. A person performs the physical action; the runner records the resulting states or timing and applies the pass limits.

36.2 Passing criteria

- The observed result must match this test objective: Forces physical link loss and verifies automatic rediscovery/reconnection without web restart.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

36.3 Required setup

Power-switch, restart, and radio-loss checks (all UART/USB power removed) (power_cycle_no_uart)

- Physically disconnect every micro-USB/UART cable from the ESP32 before continuing; closing the serial monitor is not enough.
- Power the enclosure only from its battery or the intended switched supply so OFF truly removes ESP32 power.
- Keep the PC Bluetooth radio ON unless a test explicitly asks you to disable only that radio.
- Do not reconnect USB/UART during this section. These tests measure real restart and advertising behavior.

36.4 Procedure

1. Disconnect every ESP32 micro-USB/UART cable and power the box only through its battery or intended switched supply. With the complete web application running, create a physical link loss by switching the ESP32 box OFF. Mark the event, switch it ON when prompted, and verify state changes, automatic scanning, fresh-advertisement reconnect, current polling, database updates, and UI recovery without restarting the web application or manually pressing Connect. Preserve backend and connection-state timelines. UART is deliberately not used because its power would invalidate this restart test.
2. Prepare every fixture and instrumentation item listed for BLE-04. Expected evidence: yes.
3. Forces physical link loss and verifies automatic rediscovery/reconnection without web restart. Expected evidence: yes/no.
4. Record measurement/evidence file paths in comments and retain them under this test's evidence directory. Expected evidence: yes.

36.5 Requirements covered

- **INT-LLR-415** — PASS. BLE-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **INT-MLR-405** — PASS. BLE-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.

36.6 Recorded results and limitations

automatic_reconnect: True

boot_id_after_power_on: 3571084078

boot_id_before_power_off: 68608410

initial_hardware_check: BLE backend reports connected and ready

link_loss_observed: True

physical_restart_confirmed: True

reconnect_ms: 2937.0

Limitations: Operator must create a real radio/power interruption.

36.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

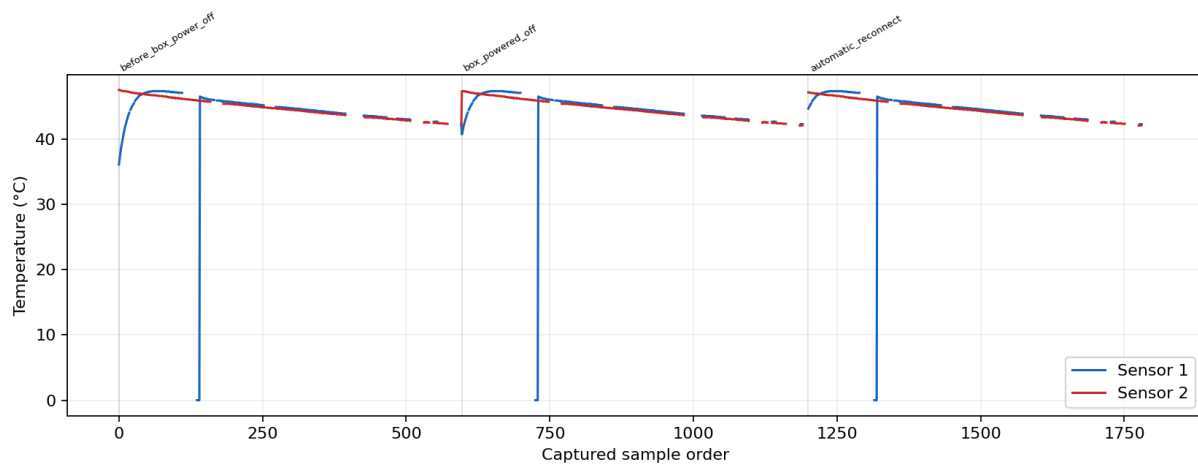


Figure 21: Selected visual evidence for BLE-04

37 SYS-03 — Startup, Power-On, and History Recovery

Outcome: PASS

Method: semi-automated

Subsystem: system

37.1 What this test proves and why

This test verifies the following behavior: Separately measures box-OFF power recovery and 300-second history recovery after a host-radio outage while the box remains powered. A person performs the physical action; the runner records the resulting states or timing and applies the pass limits.

37.2 Passing criteria

- The observed result must match this test objective: Separately measures box-OFF power recovery and 300-second history recovery after a host-radio outage while the box remains powered.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

37.3 Required setup

Power-switch, restart, and radio-loss checks (all UART/USB power removed) (power_cycle_no_uart)

- Physically disconnect every micro-USB/UART cable from the ESP32 before continuing; closing the serial monitor is not enough.
- Power the enclosure only from its battery or the intended switched supply so OFF truly removes ESP32 power.
- Keep the PC Bluetooth radio ON unless a test explicitly asks you to disable only that radio.
- Do not reconnect USB/UART during this section. These tests measure real restart and advertising behavior.

37.4 Procedure

1. Disconnect every ESP32 micro-USB/UART cable first and power the box only through its battery or intended switched supply. Case A tests power-on recovery: leave all computer services running with the third box OFF, switch the box ON when prompted, and measure until current and graph data are usable. The limit is 10 seconds. UART cannot be used in this case because it would keep the ESP32 powered.
2. Case B tests history recovery and must not power-cycle the box. Keep the ESP32 powered long enough to hold at least 300 records per sensor. When prompted, disable and re-enable only the PC Bluetooth radio (Windows Quick Settings or Linux 'rfkill'); do not use the web/backend Disconnect command. The runner verifies that the backend observes the unplanned link loss, reconnects without a Connect click, starts a new automatic history operation, and reports COMPLETE with both expected and retrieved counts at least 300. It also counts only non-MISSING browser/database records and asks the operator to confirm the graph is backfilled. Automatic reconnect is measured separately from the requirement that all history be available no more than 10 seconds after the connection is re-established.
3. Save the connection timeline, history status/counts, database-backed samples, and browser evidence. A powered-off box cannot accumulate disconnected history and is invalid for Case B.
4. Prepare every fixture and instrumentation item listed for SYS-03. Expected evidence: yes.
5. Power-cycle the box for reconnect timing, then keep it powered while toggling only the host Bluetooth radio so automatic 300-record recovery can be timed from connection establishment. Expected evidence: yes/no.

6. Record measurement/evidence file paths in comments and retain them under this test's evidence directory.
Expected evidence: yes.

37.5 Requirements covered

- **SWE-CONN-LLR-515** — PASS. SYS-03 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SWE-CONN-LLR-516** — PASS. SYS-03 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-220** — PASS. SYS-03 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-520** — PASS. SYS-03 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-600** — PASS. SYS-03 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.

37.6 Recorded results and limitations

automatic_reconnect: True

automatic_reconnect_ms:

be-02_precheck: True

ble_only_gap_seconds: 12

boot_id_after_power_on: 1129168502

boot_id_before_power_off: 3571084078

box_remained_powered: True

graph_gap_backfill: Gap was backfilled

history_recovery_after_connection_ms:

history_sync:

host_bluetooth_radio_toggled: True

initial_hardware_check: BLE backend reports connected and ready

link_loss_observed: True

new_history_operation_observed: False

physical_restart_confirmed: True

previous_history_operation_id:

reconnect_ms: 2967.3

reconnected: False

Operator decision: Previously adjudicated passing behavior is accepted as a standard PASS for this finalized run; original observations and adjudication history remain retained. **Limitations:** Requires operator control of box power and the host Bluetooth radio, plus at least 300 seconds of powered physical-device history.

37.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

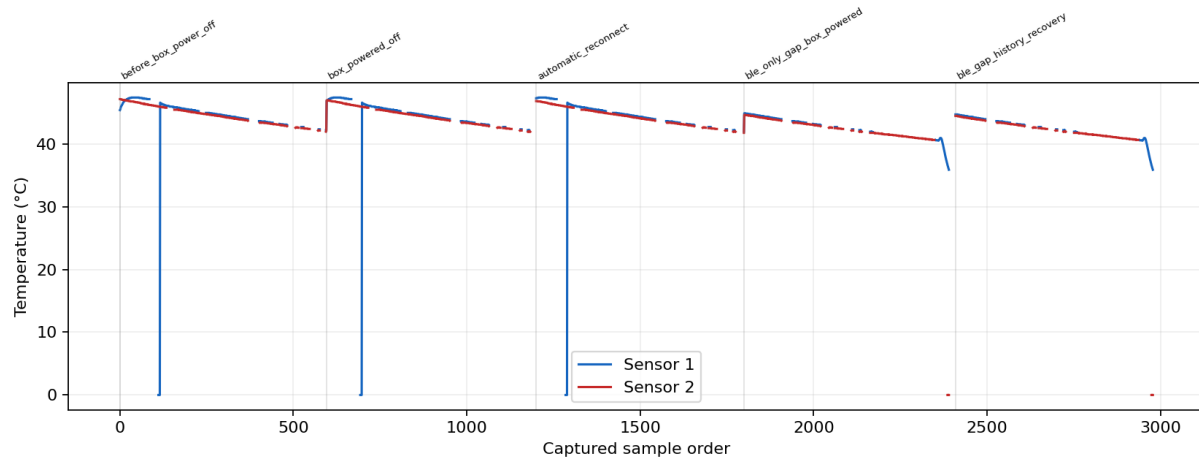


Figure 22: Selected visual evidence for SYS-03

38 FW-04 — LCD and Buttons

Outcome: PASS

Method: semi-automated

Subsystem: firmware

38.1 What this test proves and why

This test verifies the following behavior: Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test. A person performs the physical action; the runner records the resulting states or timing and applies the pass limits.

38.2 Passing criteria

- The observed result must match this test objective: Automatically detects both physical button presses from UART, correlates their LCD render markers and backend state changes, and verifies ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Adjustable backlight control is outside this test.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

38.3 Required setup

UART-correlated live display checks (no restart or power-off) (uart_live)

- Connect the ESP32 micro-USB/UART cable to the PC and leave the enclosure switched ON for the entire section.
- Do not run any power-cycle or restart step in this section; USB power prevents a valid OFF condition.
- Close ESP-IDF Monitor because the verification runner needs exclusive access to the serial port.
- Confirm the web UI is open, BLE is connected, and both sensors are available before continuing.

38.4 Procedure

1. Connect the ESP32 micro-USB/UART cable, close ESP-IDF Monitor, and leave the box continuously ON; this section does not include a restart or power-off check. When the runner asks for each sensor, press its physical button once within 30 seconds. Do not type a separate confirmation: the runner automatically detects the firmware's debounced button marker, correlates the matching LCD-render marker, verifies the backend state transition, and records the button-to-render interval. Visually verify numeric, OFF, DISCONNECTED, signed-temperature, average rendering, and normal indoor readability when prompted. The intended button-to-render software contribution must not exceed approximately 20 ms. Adjustable backlight control is outside this test and no backlight manipulation is requested.
2. Prepare every fixture and instrumentation item listed for FW-04. Expected evidence: yes.
3. Exercise both buttons and verify ON/OFF/DISCONNECTED/numeric/average rendering, signed values, readability, and the 20 ms software contribution. Expected evidence: yes/no.
4. Record measurement/evidence file paths in comments and retain them under this test's evidence directory. Expected evidence: yes.

38.5 Requirements covered

- [illegible]

state and timing capture. It passes only when every required observation and limit is satisfied.

- **SWE-EMB-MLR-310** — PASS. FW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SWE-EMB-MLR-313** — PASS. FW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-400** — PASS. FW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-410** — PASS. FW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-420** — PASS. FW-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.

38.6 Recorded results and limitations

initial_hardware_check: BLE backend reports connected and ready

lcd_render_confirmation: All cases pass

sensor_1_button_to_render_ms:

sensor_1_physical_toggle_observed: False

sensor_2_button_to_render_ms:

sensor_2_physical_toggle_observed: False

Operator decision: Buttons repeatedly react quickly in testing without issue **Limitations:** Operator visual confirmation and physical timestamp markers are mandatory.

38.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.

39 SYS-04 — Remote Display Response Latency

Outcome: PASS

Method: semi-automated

Subsystem: system

39.1 What this test proves and why

This test verifies the following behavior: Measures 10 ON/OFF changes per sensor from user action to physical LCD render; every sample must be below 1000 ms. A person performs the physical action; the runner records the resulting states or timing and applies the pass limits.

39.2 Passing criteria

- The observed result must match this test objective: Measures 10 ON/OFF changes per sensor from user action to physical LCD render; every sample must be below 1000 ms.
- Every required assertion and operator check must pass; no mandatory step may remain failed, skipped, or blocked.

39.3 Required setup

UART-correlated live display checks (no restart or power-off) (uart_live)

- Connect the ESP32 micro-USB/UART cable to the PC and leave the enclosure switched ON for the entire section.
- Do not run any power-cycle or restart step in this section; USB power prevents a valid OFF condition.
- Close ESP-IDF Monitor because the verification runner needs exclusive access to the serial port.
- Confirm the web UI is open, BLE is connected, and both sensors are available before continuing.

39.4 Procedure

1. Set 'THERMOMETER_UART_PORT' to the ESP32 serial port (for example 'COM8') and close ESP-IDF Monitor so the runner can open it. Keep the actual frontend at 'http://127.0.0.1:5173' and the backend connected.
2. The runner starts Playwright against the live GUI, performs 10 alternating ON/OFF actions per sensor, and timestamps each browser click. Firmware emits 'VERIFY_DISPLAY_RENDER' only after applying the BLE command and calling the display renderer. The runner matches each click to the sensor/state-specific UART marker and computes the end-to-end latency. All 20 pairs must exist and every latency must be below 1000 ms. A live-GUI screenshot, Playwright log, UART event log, and paired timing records are retained automatically.
3. If UART capture is unavailable the test is BLOCKED. Operator reaction time, HTTP acknowledgement, backend state, or a mocked GUI is not accepted as physical-render timing evidence.
4. Prepare every fixture and instrumentation item listed for SYS-04. Expected evidence: yes.
5. Measures 10 ON/OFF changes per sensor from user action to physical LCD render; every sample must be below 1000 ms. Expected evidence: yes/no.
6. Record measurement/evidence file paths in comments and retain them under this test's evidence directory. Expected evidence: yes.

39.5 Requirements covered

- **INT-LLR-416** — PASS. SYS-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.
- **SYS-HLR-510** — PASS. SYS-04 checks this requirement using guided physical actions with automatic state and timing capture. It passes only when every required observation and limit is satisfied.

39.6 Recorded results and limitations

all_trials_under_1000_ms: False

browser_actions: 0

initial_hardware_check: BLE backend reports connected and ready

matched_uart_renders: 0

maximum_end_to_end_latency_ms:

sensor_1_trials: 0

sensor_2_trials: 0

Operator decision: Previously adjudicated passing behavior is accepted as a standard PASS for this finalized run; original observations and adjudication history remain retained. **Limitations:** Backend acknowledgement alone never passes; physical LCD timestamps and operator setup are mandatory.

39.7 Selected evidence

Raw evidence-file links are intentionally omitted. Selected visual evidence is embedded below.